Overview

Model Figure 1

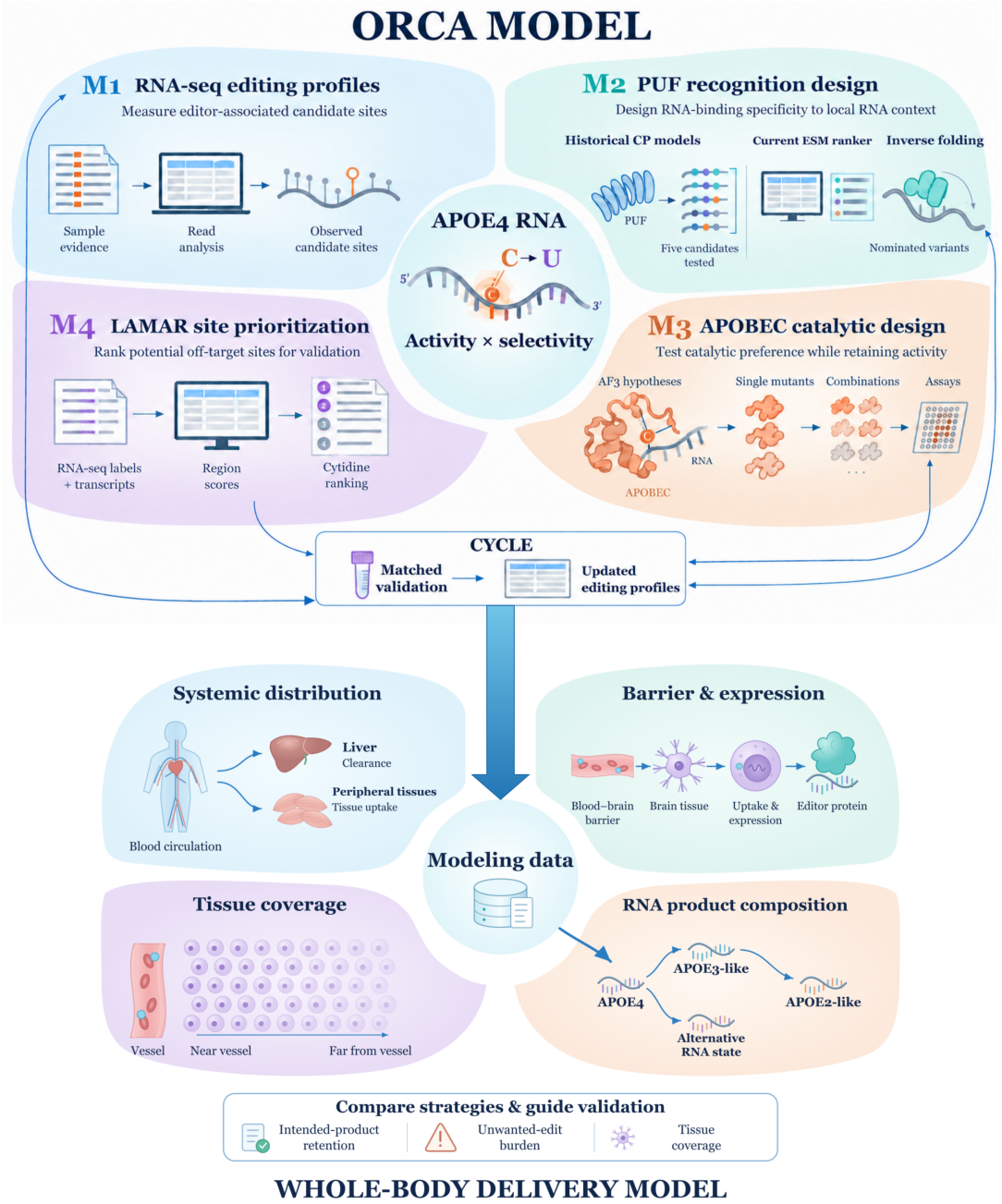


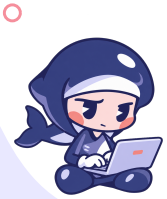
Figure 1. ORCA model overview. RNA-seq maps the editing landscape; protein foundation and structural models guide recognition and catalytic design; generative models propose PUF sequences; LAMAR ranks RNA sites; and delivery simulations connect editor exposure to RNA products. Experimental measurements connect the sequence-level models to observed editing outcomes.

Module 1: Transcriptomic Landscape

Module 1 Table 1

1.2 Experimental design

Component	Configuration
Editor treatments	PUF10, PUF12, 132D, E72A, GVE and SNE
Background controls	Control/mock, APOBEC-only and PUF-only
Reference	Human GRCh38 primary assembly with matching GENCODE v50 annotation
Candidate substitutions	Genomic C>T on the positive transcript strand and G>A on the negative transcript strand
Comparison scope	Positions with REF + target ALT depth ⩾20 in all 36 libraries



Model Figure 2

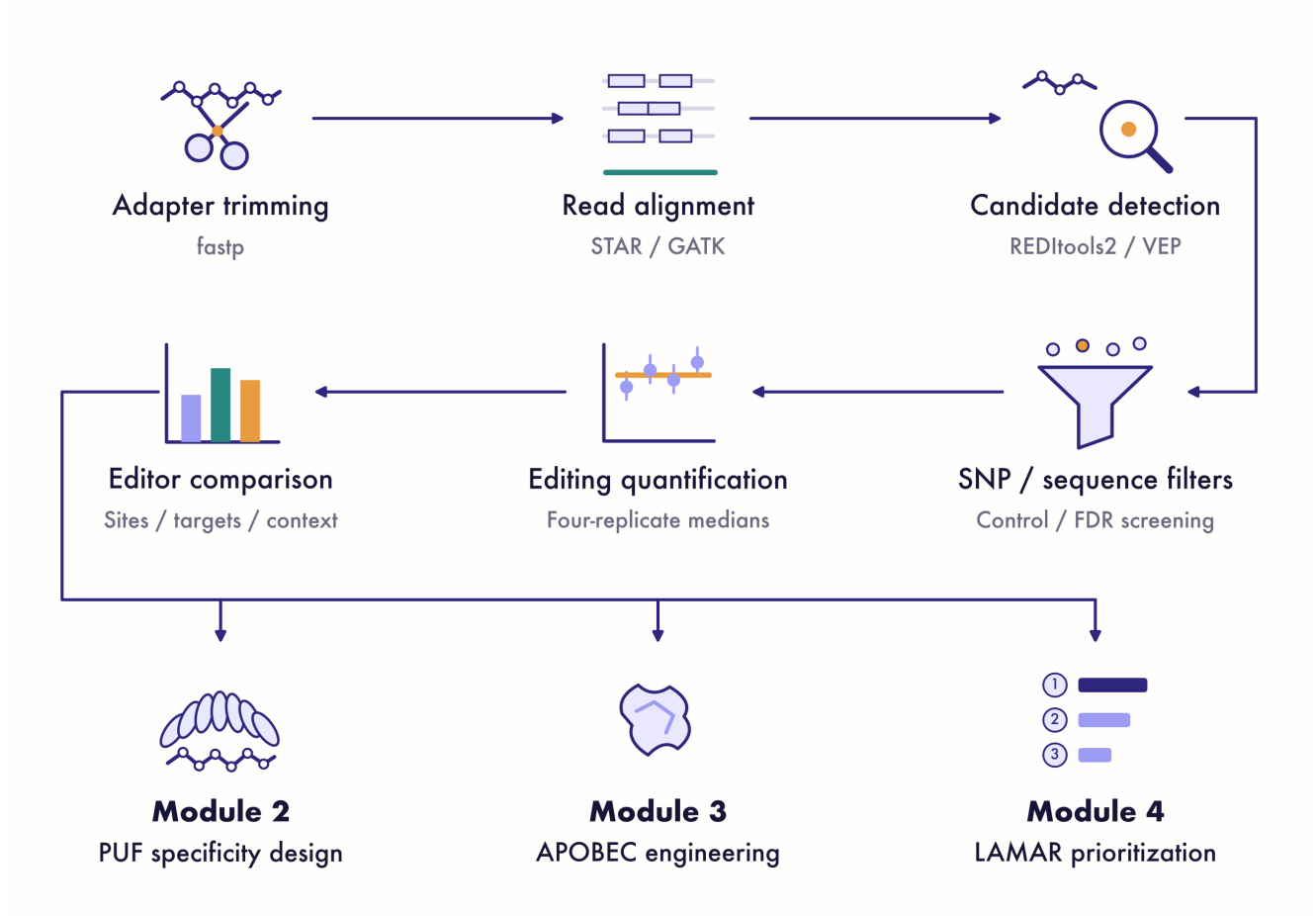


Figure 2. RNA-seq candidate-detection workflow for 36 libraries across nine groups, with four biological replicates per group. Read from sequencing to the retained-site set: alignment establishes read support, variant and strand filters remove alternative explanations, and agreement across replicates identifies reproducible candidates. The final branches connect this common evidence base to PUF recognition, APOBEC design and RNA-context modeling.

Model Figure 3

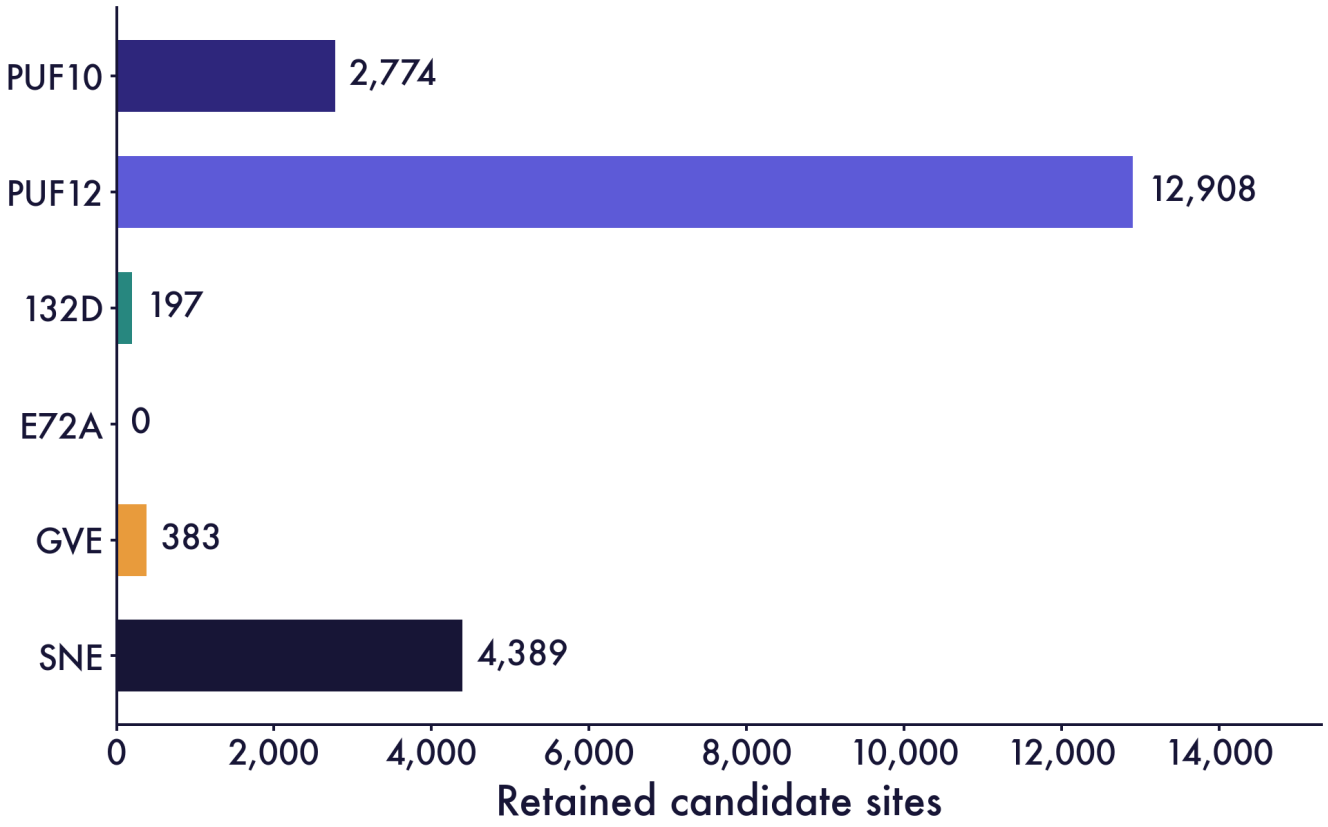


Figure 3. Retained candidate-site counts across six editors. C388 is excluded; E72A has no retained sites. PUF10/12 denote 10/12-repeat recognition domains; E72A denotes APOBEC Glu72→Ala, and 132D denotes Met132→Asp. Each bar counts sites passing the same filtering workflow, so a shorter bar indicates a narrower detected editing landscape. PUF12 retains more sites than PUF10, whereas the engineered variants narrow that landscape. Site counts complement the target and bystander readouts in Figure 5.

Model Figure 4

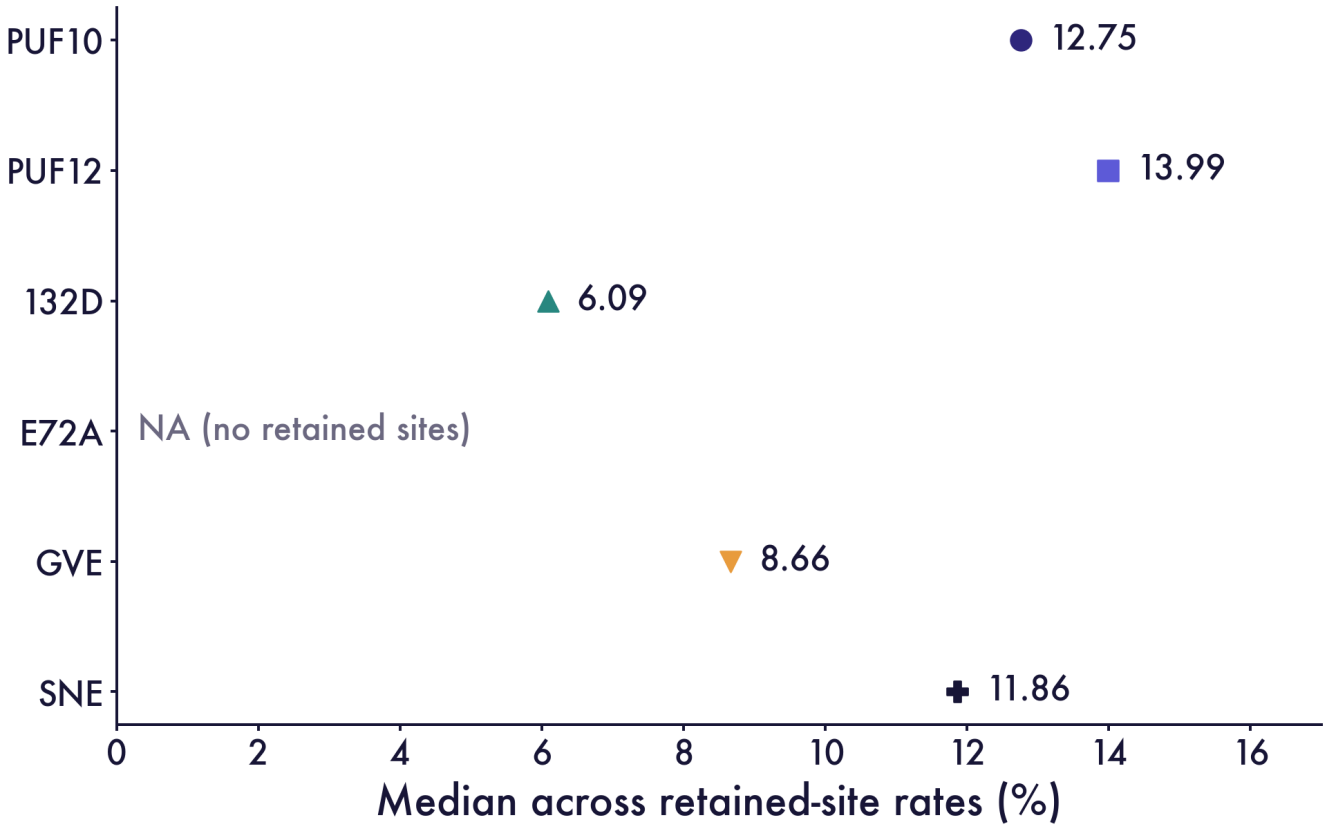


Figure 4. Median editing rates across retained sites, using each site’s median across four biological replicates. E72A has no retained sites. The horizontal axis measures editing strength at the sites retained for each editor. The 132D and GVE groups show lower typical rates than PUF10/12. Because each group retains a different site set, Figure 8 separately compares editing at identical sites.

Module 1 Table 2

PUF12 combined broader editing with a distinct target profile

Editor	Retained sites	Median across sites (%)	Apparent C388 rate (%)
PUF10	2,774	12.75	73.23
PUF12	12,908	13.99	81.01
132D	197	6.09	36.44
E72A	0	NA	0.32
GVE	383	8.66	64.98
SNE	4,389	11.86	76.91

Model Figure 5

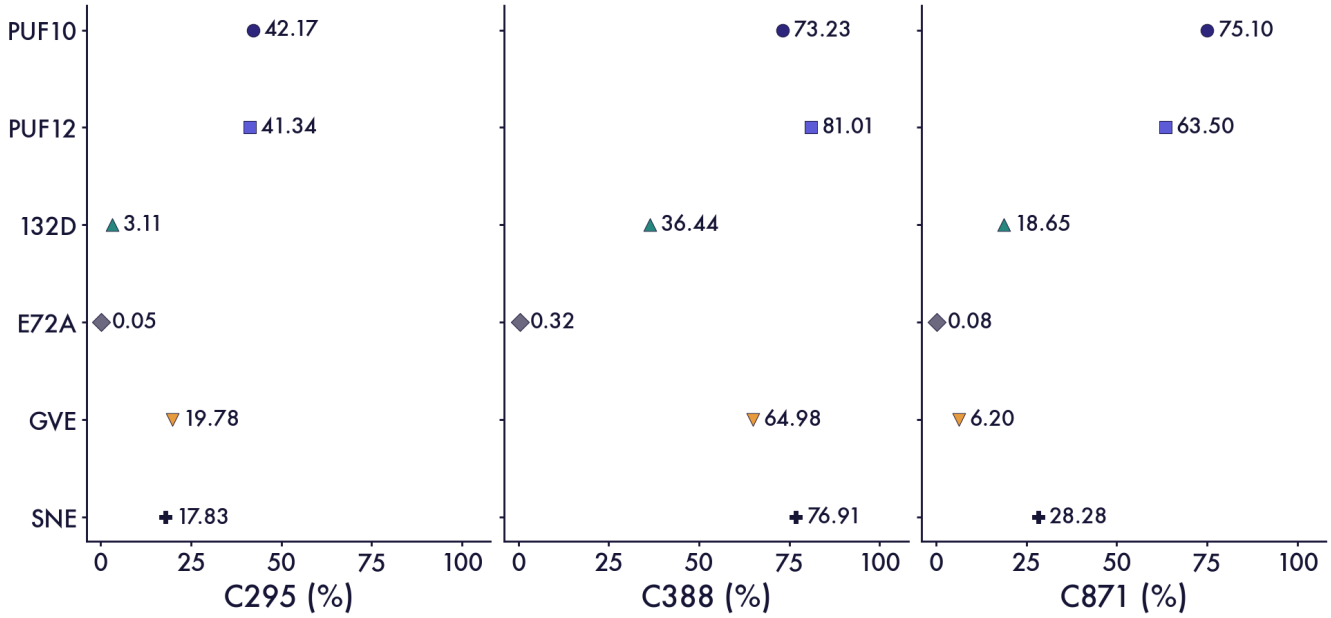


Figure 5. Apparent rates at C295, C388 and C871, shown as medians across four biological replicates. C388 includes endogenous reference T. C labels mark RNA nucleotide positions. P8-GVE denotes repeat-8 S288G (Ser→Gly) and Y289V (Tyr→Val); GVE is Gly–Val–Glu. P4+P7-SNE denotes Tyr→Asn at residues 145 and 253, producing Ser–Asn–Glu motifs in repeats 4 and 7. Read C388 as the target-associated signal and C295/C871 as monitored bystanders. GVE and SNE preserve a target-associated signal while reducing bystander signals relative to the parental editors.

Model Figure 6

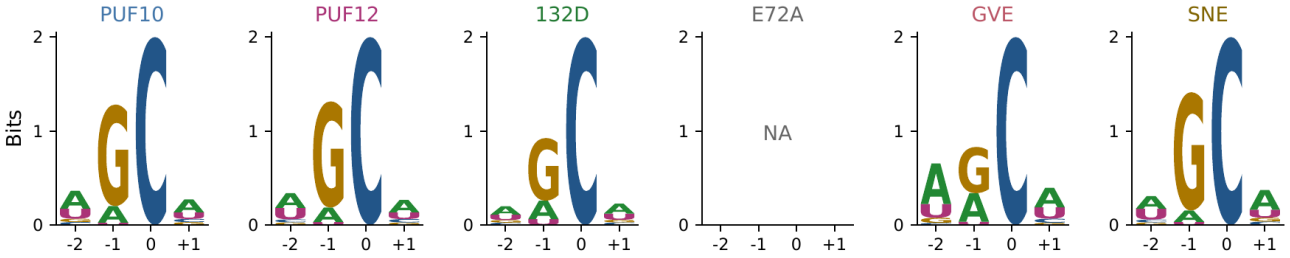


Figure 6. Sequence logos around retained cytosines, from −2 to +1. E72A has no retained sites. Position 0 is the edited C; −1 and +1 are its upstream and downstream neighbors. Letter heights summarize base enrichment. The shared upstream G preference across PUF variants supports a common APOBEC-associated local sequence bias. GVE shows weaker −1 G and greater −2 A enrichment, indicating an additional recruitment-associated context shift. The central C is fixed by the site-selection rule.

Model Figure 7

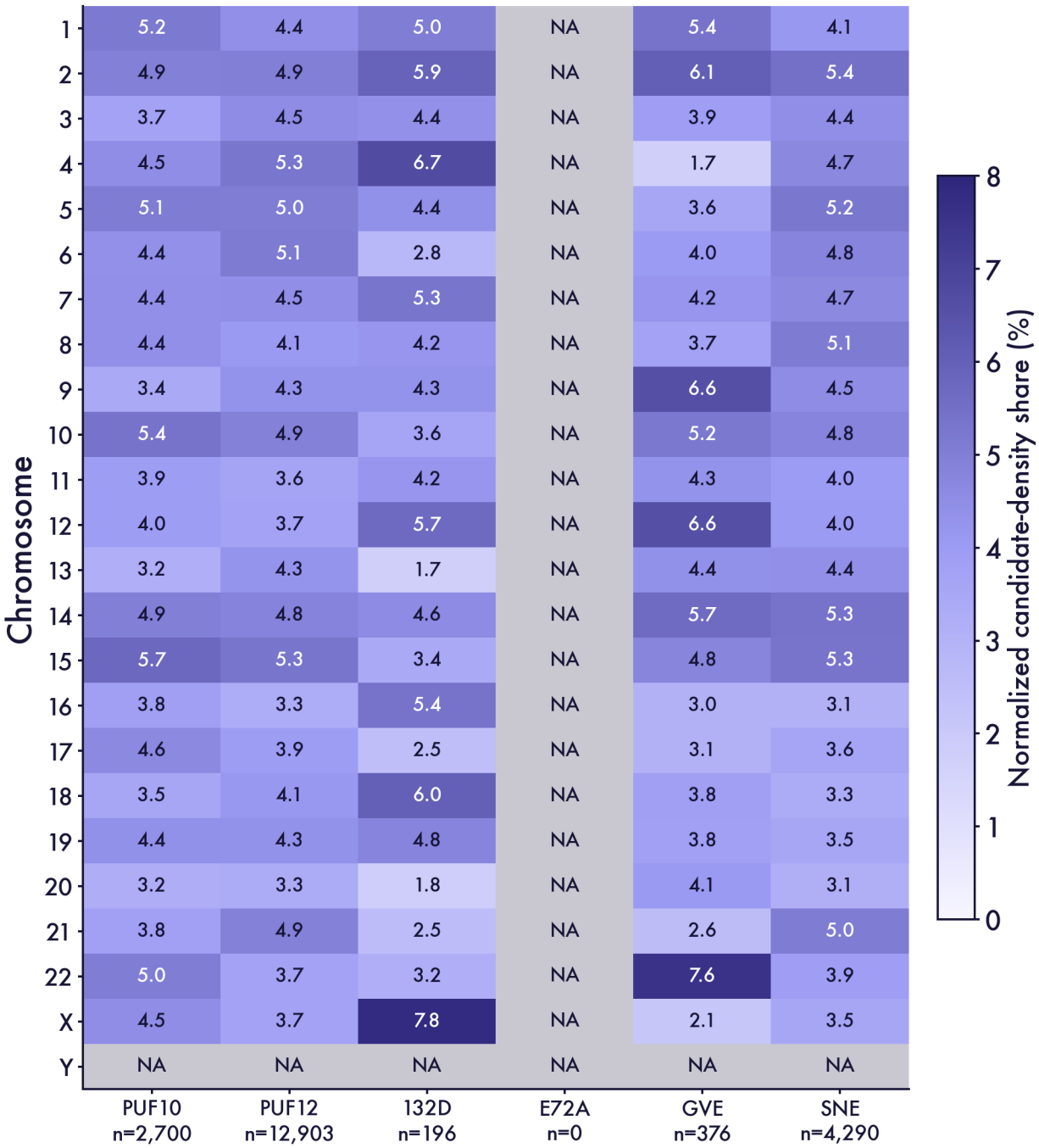


Figure 7. Chromosome-wise candidate densities within a shared callable-C reference, scaled to 100% per editor. n denotes reference-matched candidates; E72A and chromosome Y are NA. Each editor column describes relative chromosome distribution, rather than total editing burden. A higher cell value means greater representation after callable-C normalization. The broadly distributed signal shows that the retained landscape extends across chromosomes; overall site numbers are compared separately in Figure 3.

Model Figure 8

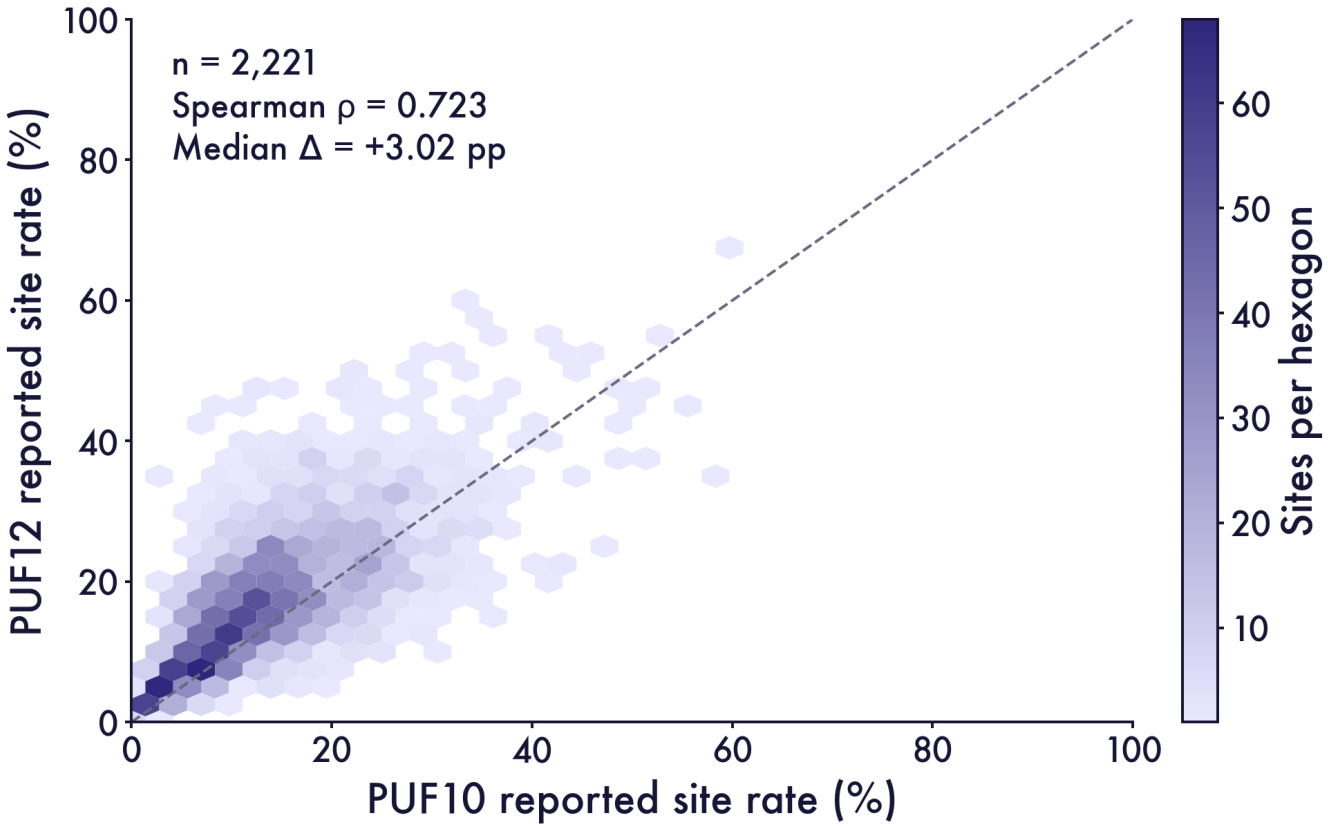


Figure 8. Editing rates at 2,221 sites retained by both PUF10 and PUF12. Each rate is a four-replicate median; hexagons show site counts and the dashed line marks equality. Δ is PUF12 minus PUF10 in percentage points. Sites above the equality line have stronger editing with PUF12. This occurs at 73.1% of shared sites, showing that PUF12 both reaches more retained sites and increases editing at many sites already reached by PUF10.

Model Figure 9

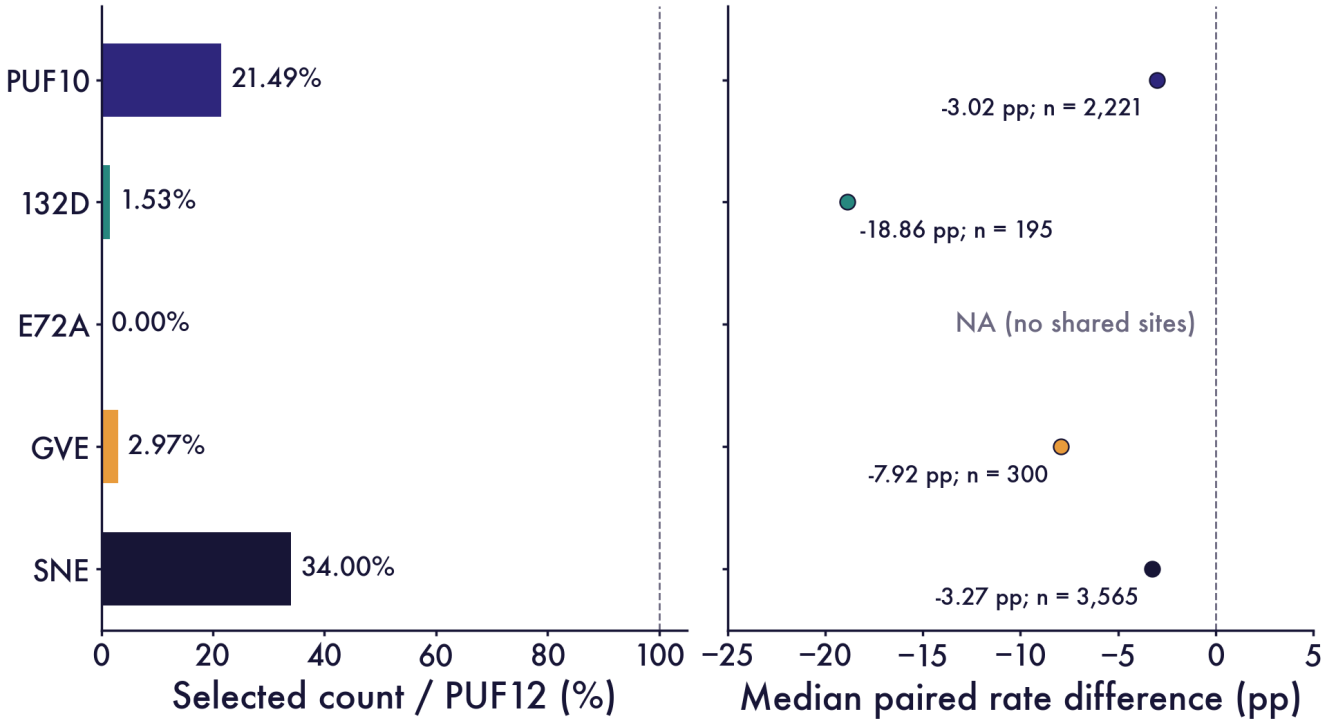


Figure 9. Retained-site counts as a percentage of PUF12 (left) and median group-minus-PUF12 rate differences among pairwise shared sites, in percentage points (right). n denotes shared sites; E72A’s rate comparison is NA. Read the two panels together: a smaller left value means fewer detected sites, while a negative right value means weaker editing at shared sites. The 132D group combines both reductions; NA for E72A means there is no shared-site rate to calculate.

Module 2: Recognition Code

Model Figure 10

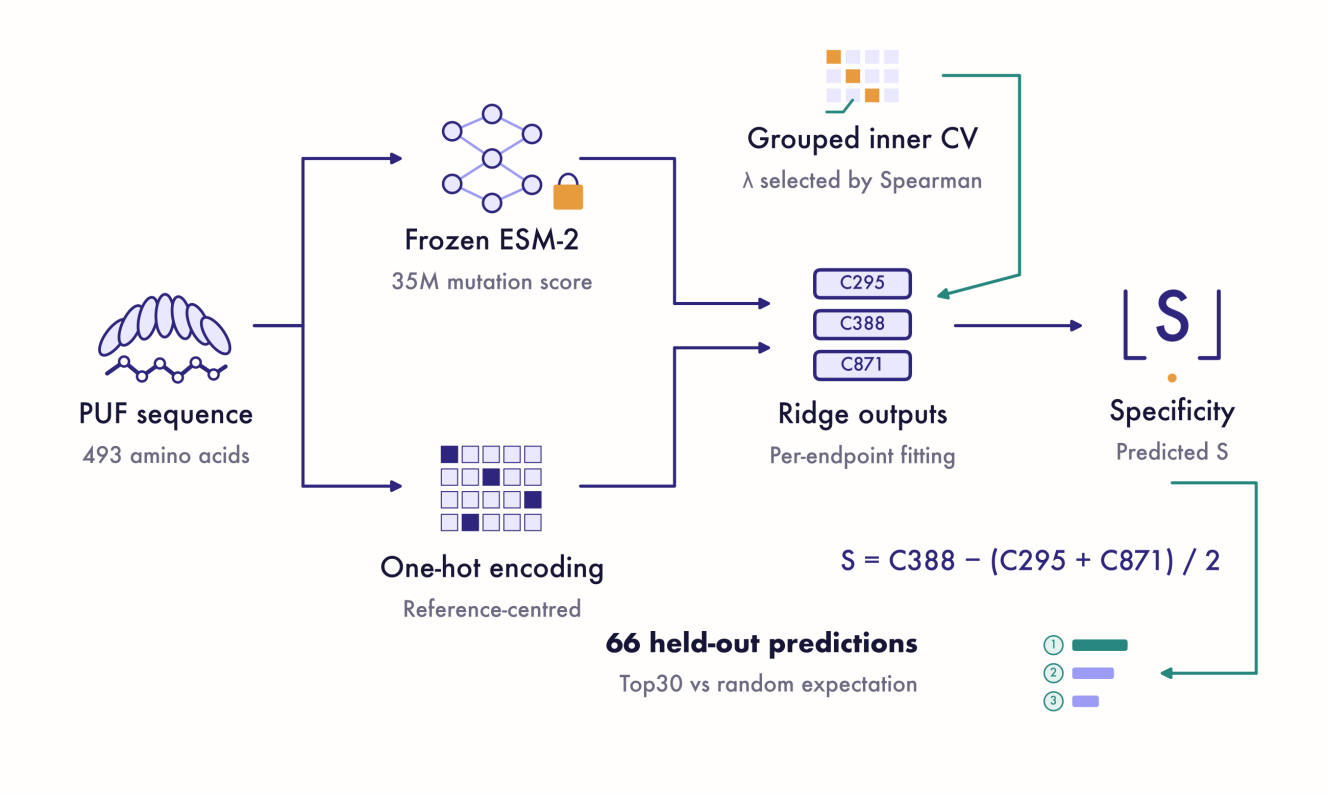


Figure 10. PUF-ESM combines ESM-2 mutation scores and one-hot features in Ridge regression to predict C295, C388 and C871 rates and derive specificity score S. ESM-2 is a protein language model; one-hot features record residue identity. C295/C388/C871 are RNA assay positions, with C388 the target. The score subtracts the mean bystander readout from the target readout: S = C388 − (C295 + C871)/2. Higher S therefore rewards target activity together with lower bystander editing. Assay measurements teach the regressor how sequence features relate to this balance.

Model Figure 11

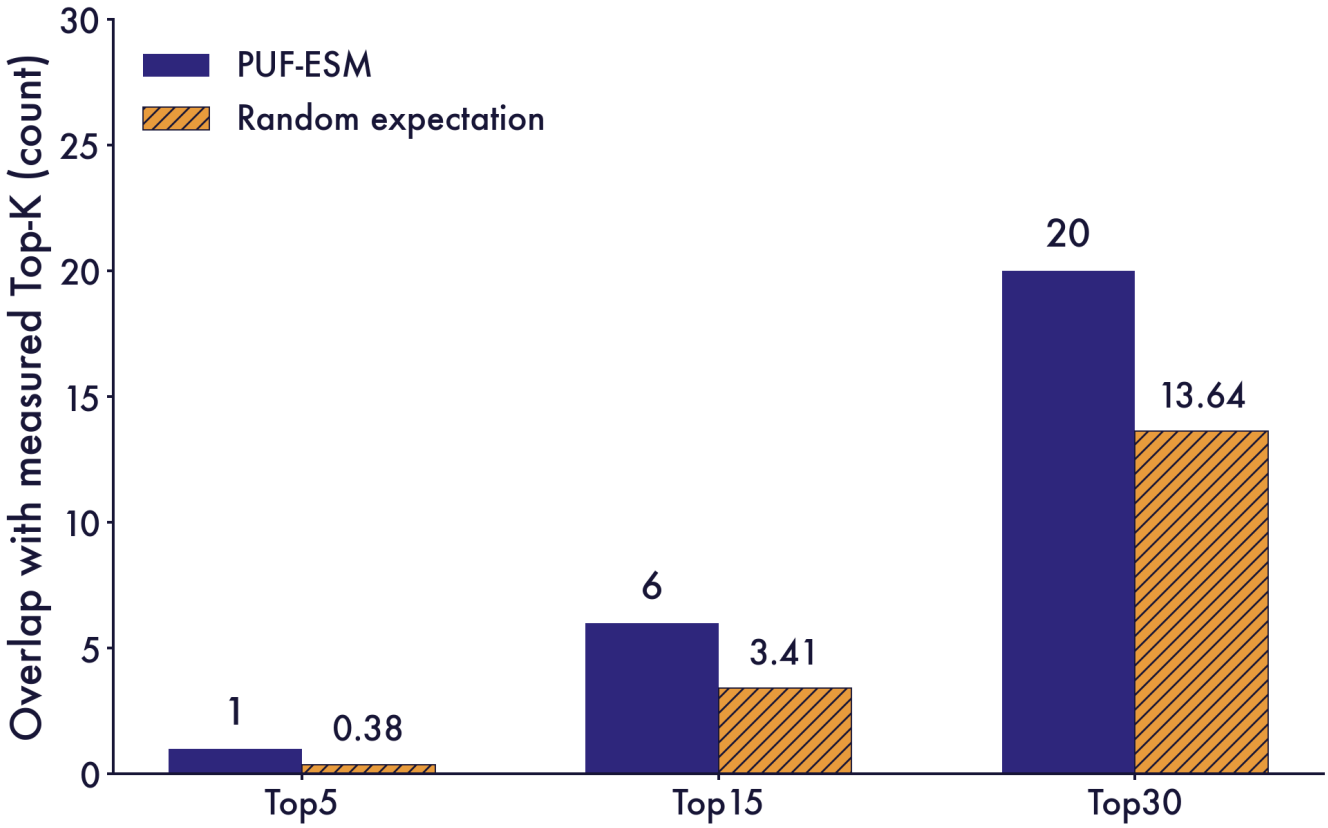


Figure 11. Top-K overlap between predicted and measured rankings among 66 constructs. Random expectation is $K^2/66$. K is the number of candidates selected from each ranking; overlap counts candidates present in both lists. At K = 30, prediction recovers 20 experimentally top-ranked constructs, above the random expectation of 13.64. This measures enrichment of a screening panel, rather than exact agreement on every rank.

Model Figure 12

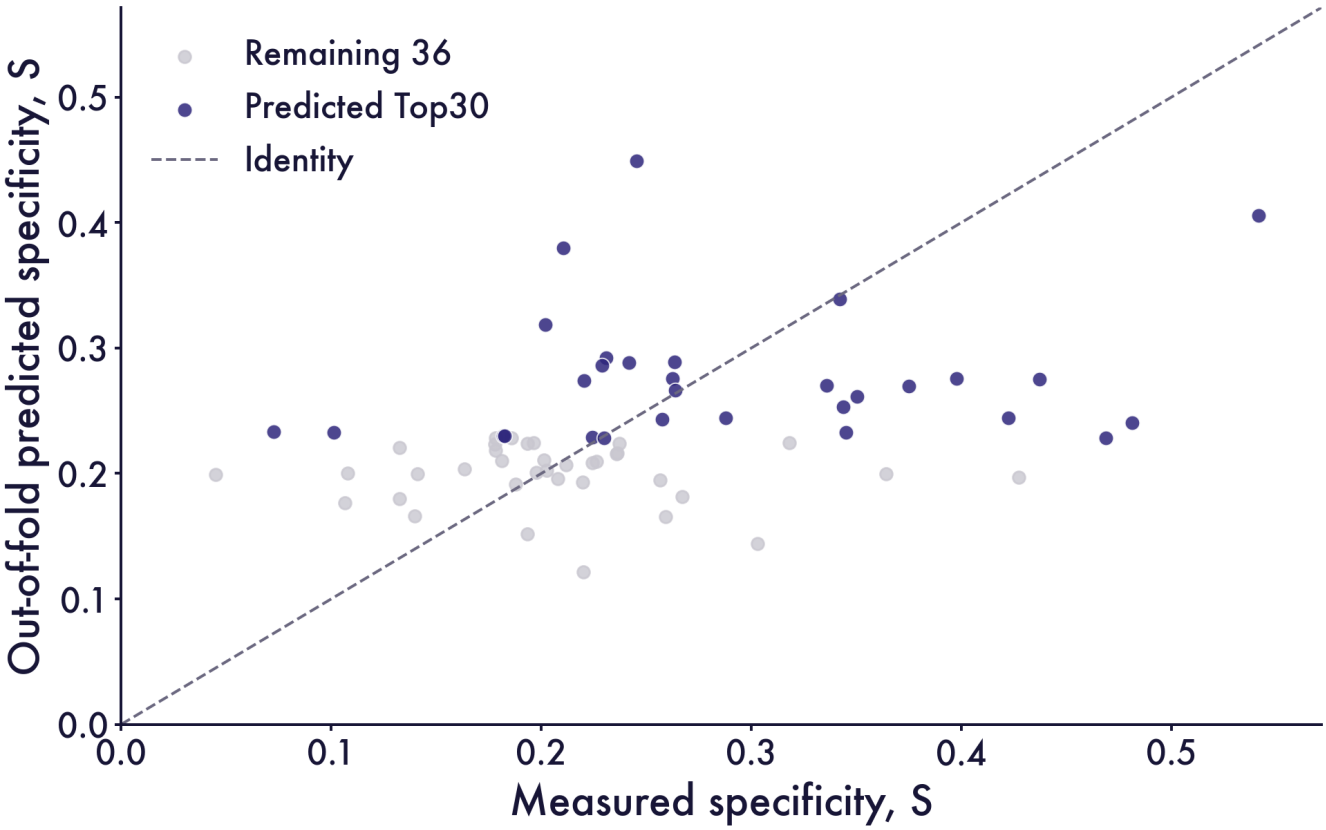


Figure 12. Measured versus out-of-fold predicted specificity score S for 66 constructs, highlighting the predicted Top30. The dashed line marks equality. Out-of-fold predictions are made for constructs held out of the corresponding model fit. Each point is one construct; distance from the diagonal measures score error. S rewards target C388 editing while subtracting the mean of C295/C871 bystander editing. The highlighted candidates illustrate selection of a promising panel even when predicted and measured scores differ.

Model Figure 13

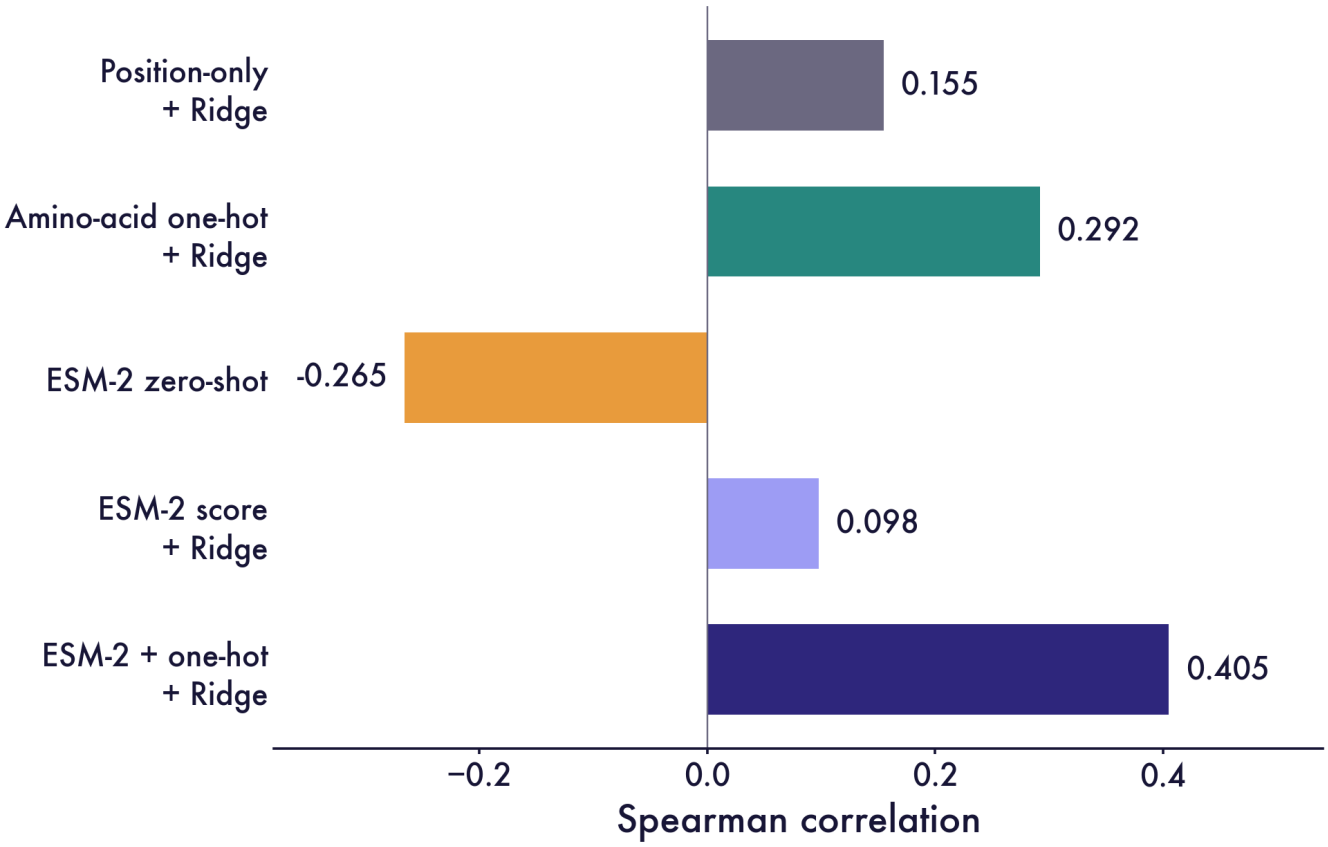


Figure 13. Spearman correlations for five feature ablations, using grouped out-of-fold predictions for the same 66 constructs. An ablation removes or replaces a feature source to test its contribution. Higher Spearman correlation means better agreement in rank order. Combining mutation identity with the ESM-2 score outperforms mutation-position-only encoding, linking the ranking gain to amino-acid information and assay calibration.

Model Figure 14

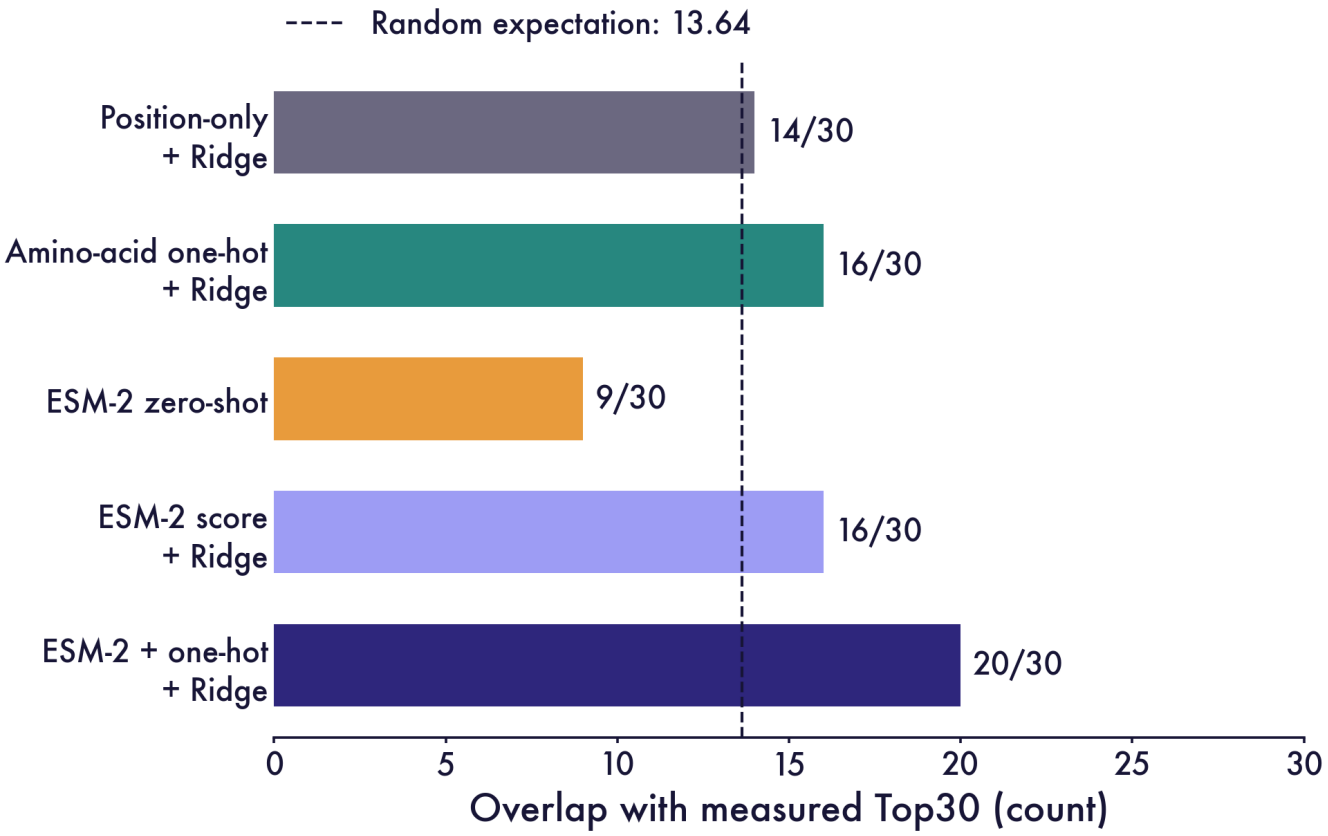


Figure 14. Measured Top30 recovery across five feature ablations. The dashed line marks random expectation: 13.64 of 30 constructs. Each bar counts experimentally top-ranked constructs recovered by a predicted 30-candidate panel. The combined model recovers 20, compared with 14 for position-only features. This translates the feature comparison into how many strong candidates a fixed-size screen captures.

Model Figure 15

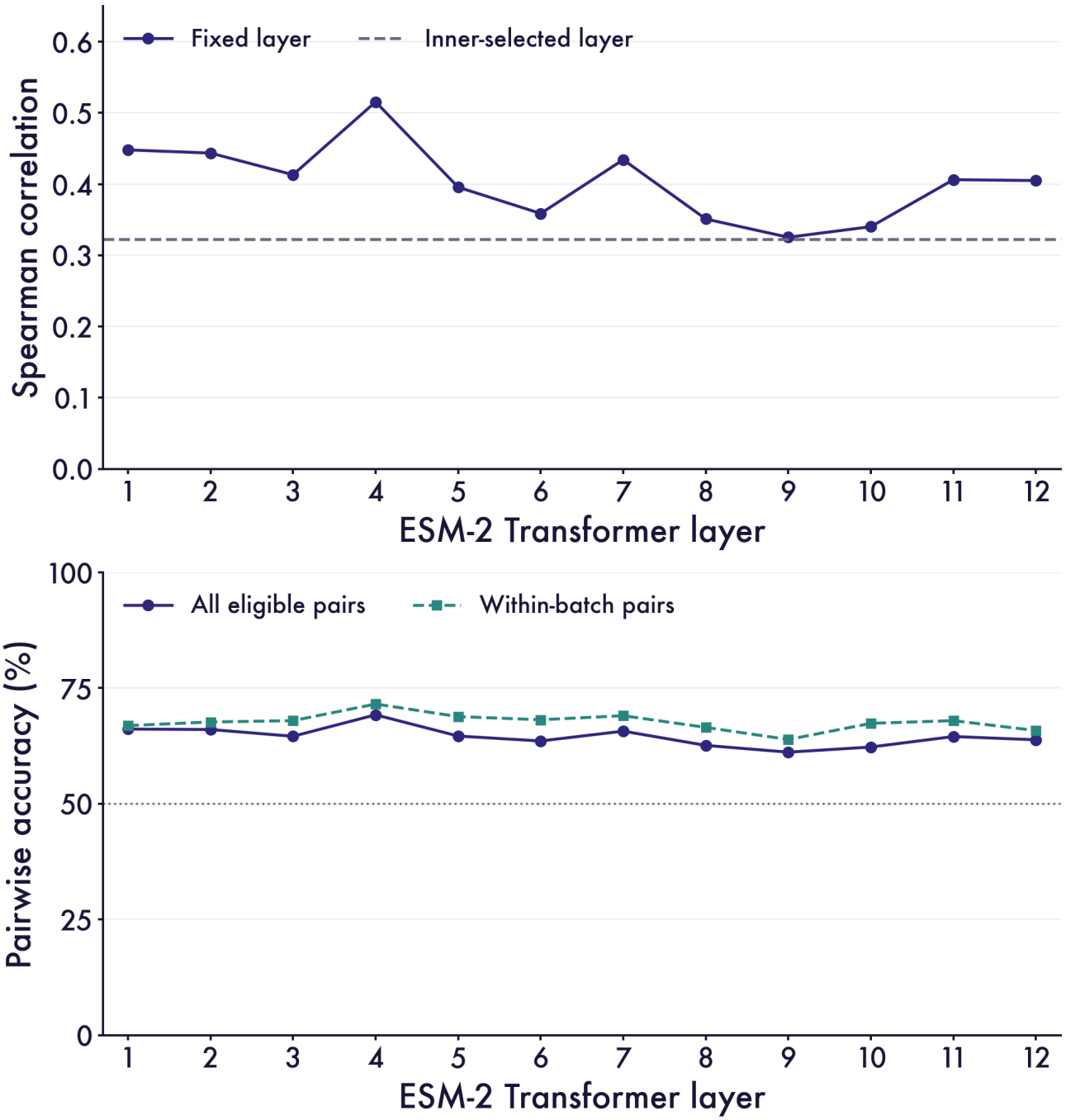


Figure 15. ESM-2 readout-layer sensitivity: grouped outer-OOF Spearman correlation above and all-pair/within-batch accuracy below. Pairs with $|\Delta S| \leqslant 0.01$ are excluded; ties score 0.5. The dashed reference selects layers within inner validation. OOF means out-of-fold: predictions are evaluated on held-out constructs. Spearman measures ordering across the library; pairwise accuracy asks whether the higher-scoring member of a pair is ranked correctly. The within-batch comparison controls the comparison context. Layer choice affects both views of ranking performance.

Model Figure 16

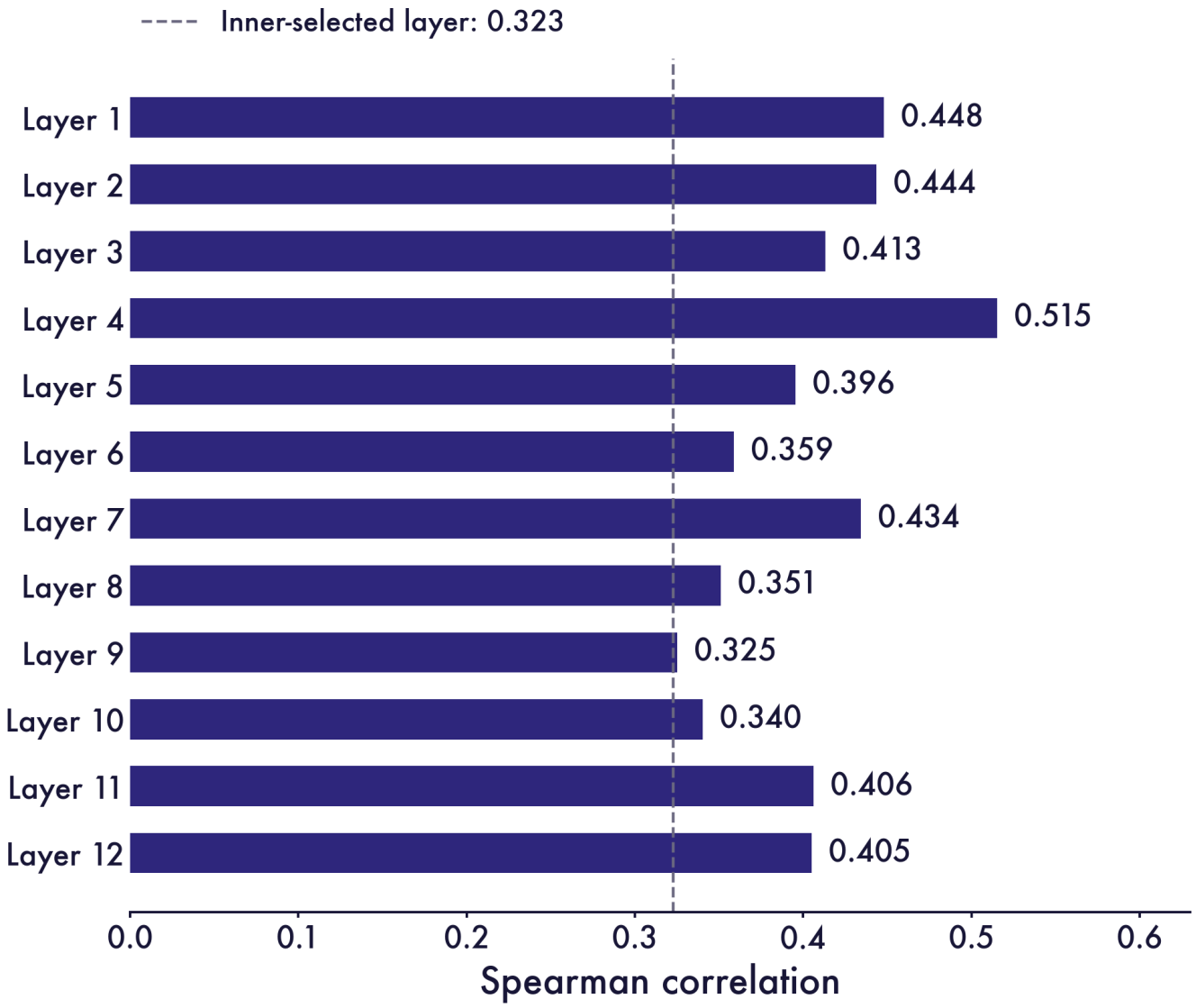


Figure 16. Grouped outer-OOF Spearman correlations across twelve ESM-2 readout layers for 66 constructs. The dashed line marks the inner-selected-layer pipeline (ρ =0.323). The horizontal axis selects which internal ESM-2 layer supplies the sequence score; the vertical axis measures agreement with experimental rank order. Layer 4 has the highest observed correlation. The strongest observed result comes from an early layer, making representation choice a useful part of task-specific ranking.

Model Figure 17

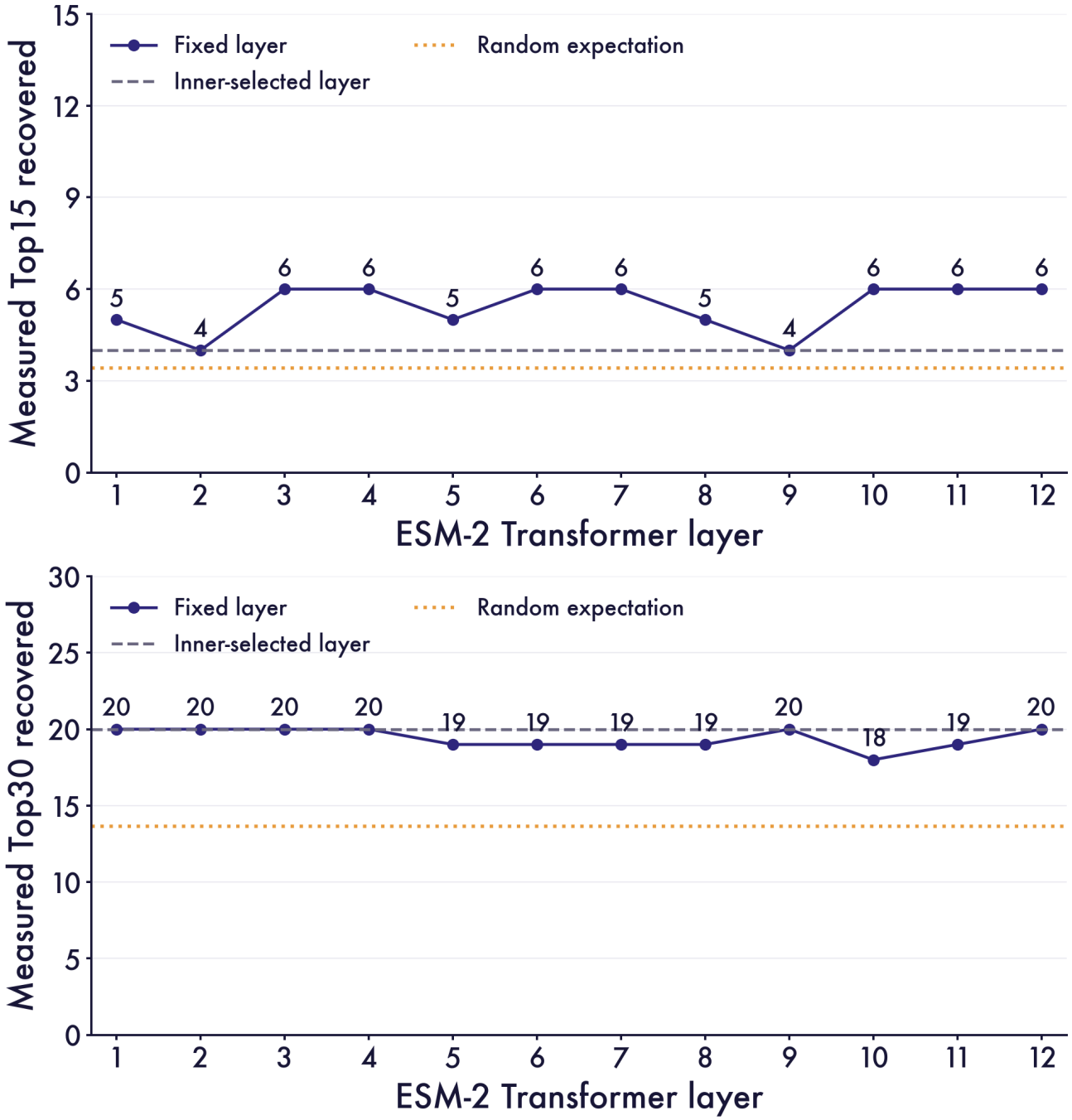


Figure 17. Top15 and Top30 recovery across ESM-2 readout layers for 66 constructs. Random expectations are 3.41 and 13.64; dashed lines show the inner-selected-layer pipeline. Recovery counts overlap between predicted and experimentally measured top-ranked sets. Several layers reach the same best Top30 recovery of 20 constructs despite different rank correlations. Overall rank agreement and candidate-panel recovery therefore provide complementary criteria for selecting the representation.

Model Figure 18

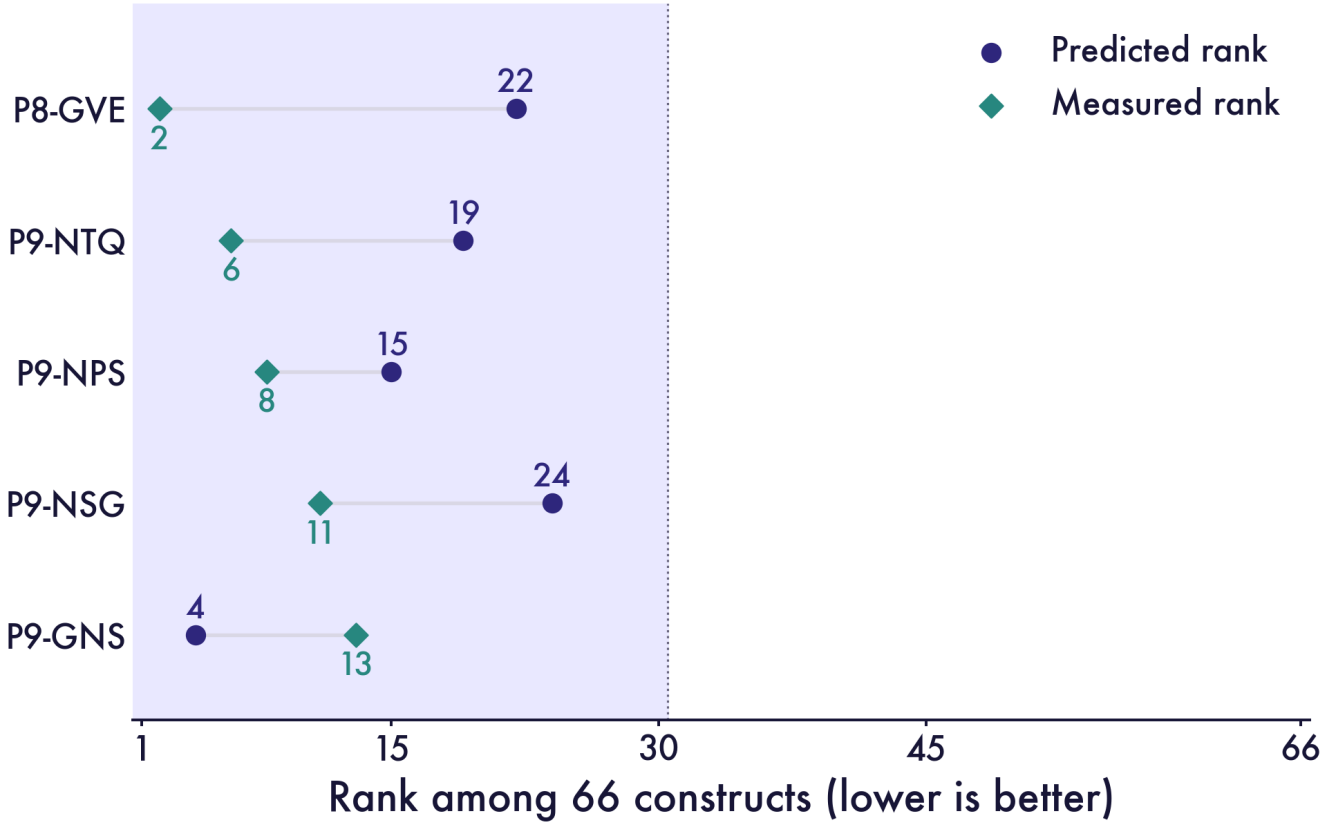


Figure 18. Current out-of-fold and measured ranks of five highlighted constructs among 66 candidates. All five enter the predicted Top30. P4-SNE + P7-SNE specifies Ser–Asn–Glu recognition motifs in repeats 4 and 7 (Y145N and Y253N: Tyr→Asn). Rank 1 is best; the separation between each construct’s predicted and measured ranks shows ordering error. All five also belong to the measured Top30, illustrating successful panel inclusion alongside variation in exact rank.

Model Figure 19

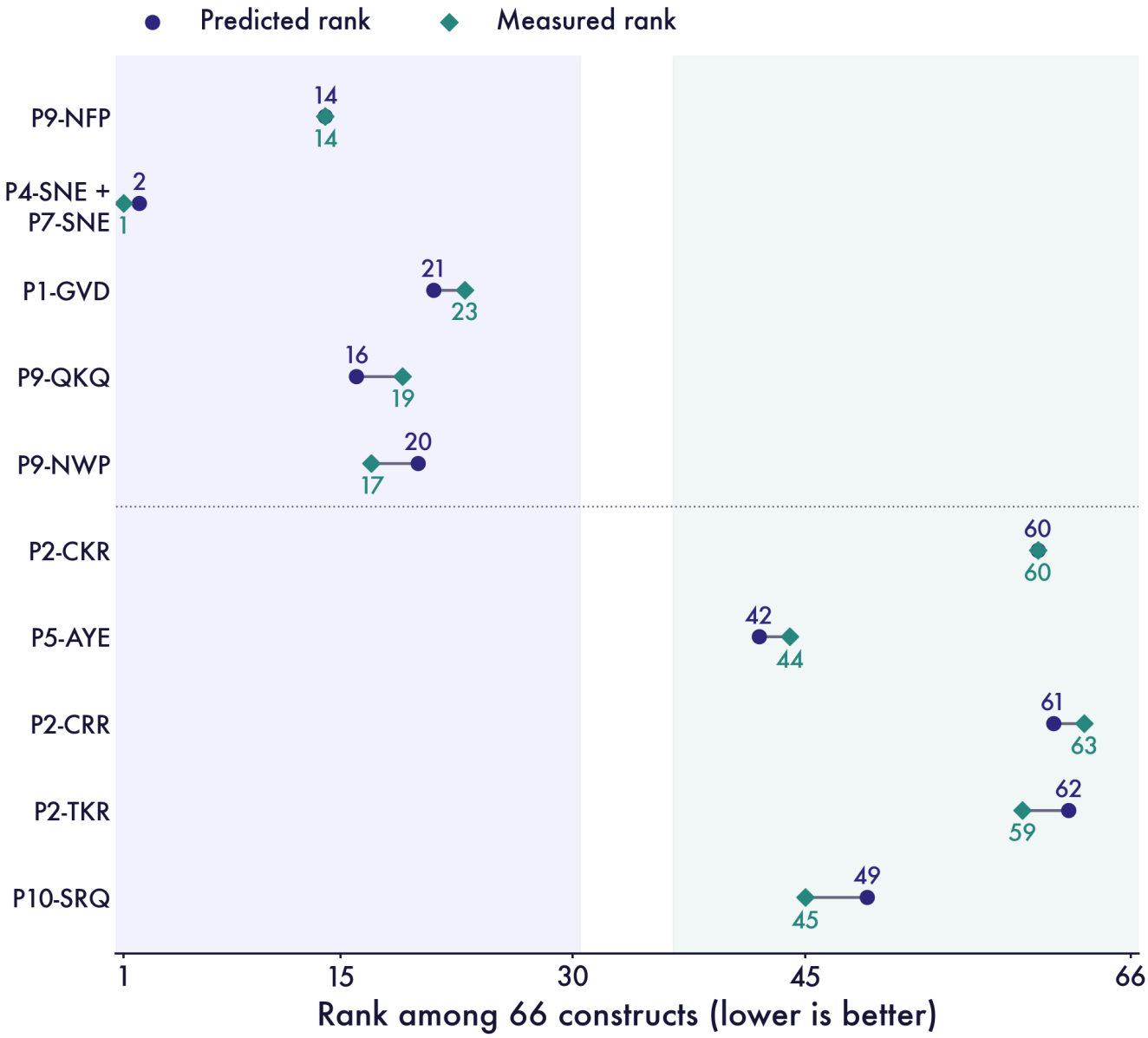


Figure 19. Five higher-ranked and five lower-ranked examples selected retrospectively for close predicted and measured ranks. P numbers identify PUF repeats; the three-letter suffix gives the recognition motif in one-letter amino-acid codes. Closer predicted and measured ranks indicate stronger agreement for an example. SNE means Ser–Asn–Glu; P4-SNE + P7-SNE carries Y145N/Y253N (Tyr→Asn) in repeats 4 and 7. These deliberately concordant cases illustrate ranking behavior; Figures 11–14 evaluate the full library.

Model Figure 20

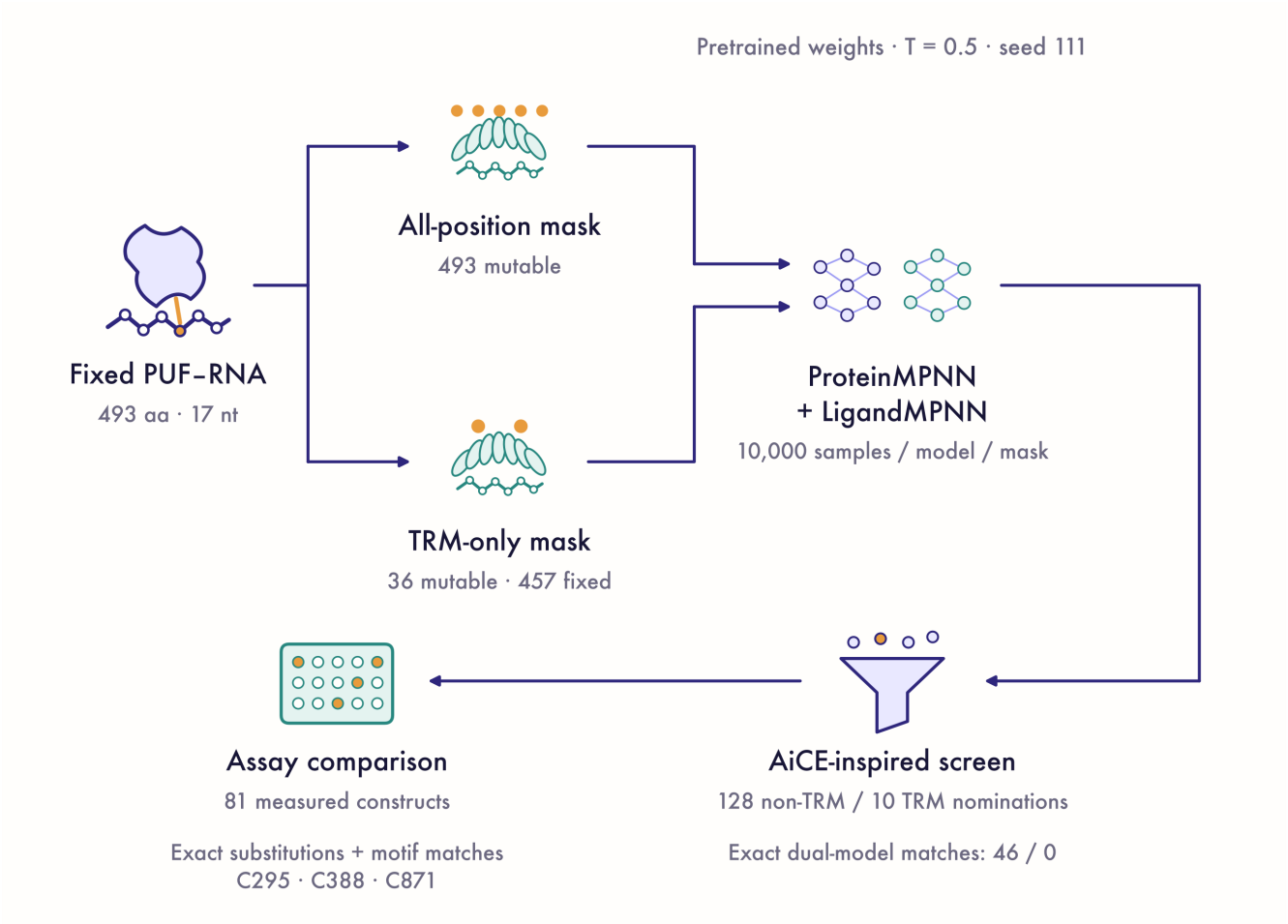
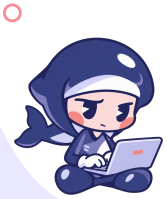


Figure 20. All-position and TRM-only design with ProteinMPNN and LigandMPNN, using 10,000 sequences per model and design space, followed by structural screening and assay comparison. TRM means tripartite recognition motif: three residues that recognize one RNA base. Both models generate sequences on a fixed protein backbone. ProteinMPNN uses the protein structure; LigandMPNN also incorporates RNA atoms. Comparing their nominations separates scaffold-sequence preferences from RNA-conditioned preferences and produces candidate sequences for subsequent experimental reference.

Module 2 Table 1

Models and two design spaces

Analysis	Mutable protein positions	Fixed protein positions	Sampling source	Reported screen
All-position sampling	493	0	Archived 10,000 sequences per model	Non-TRM substitutions after excluding 36 recognition positions
TRM-only sampling	36	457	Newly generated 10,000 sequences per model	All 36 recognition positions; all amino-acid substitutions permitted



Model Figure 21

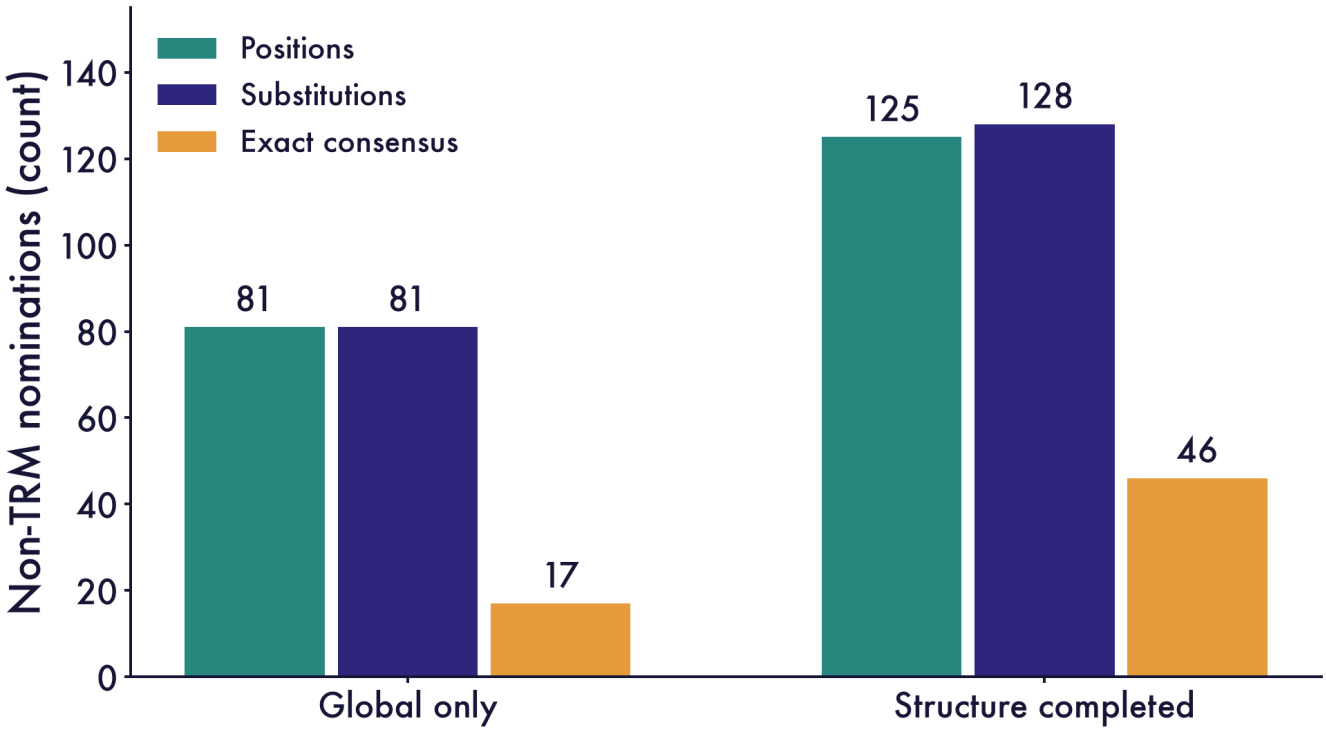


Figure 21. Non-TRM nominations before and after structural screening. Exact consensus requires both models to support the same substitution. Non-TRM positions lie outside the three-residue recognition motifs; N12G, for example, replaces Asn12 with Gly. The structurally screened set contains 128 substitutions at 125 positions, including 46 exact two-model agreements. Counts distinguish substitutions, affected positions and shared nominations. This organizes a scaffold-sequence candidate set beyond the RNA-contacting motifs.

Model Figure 22

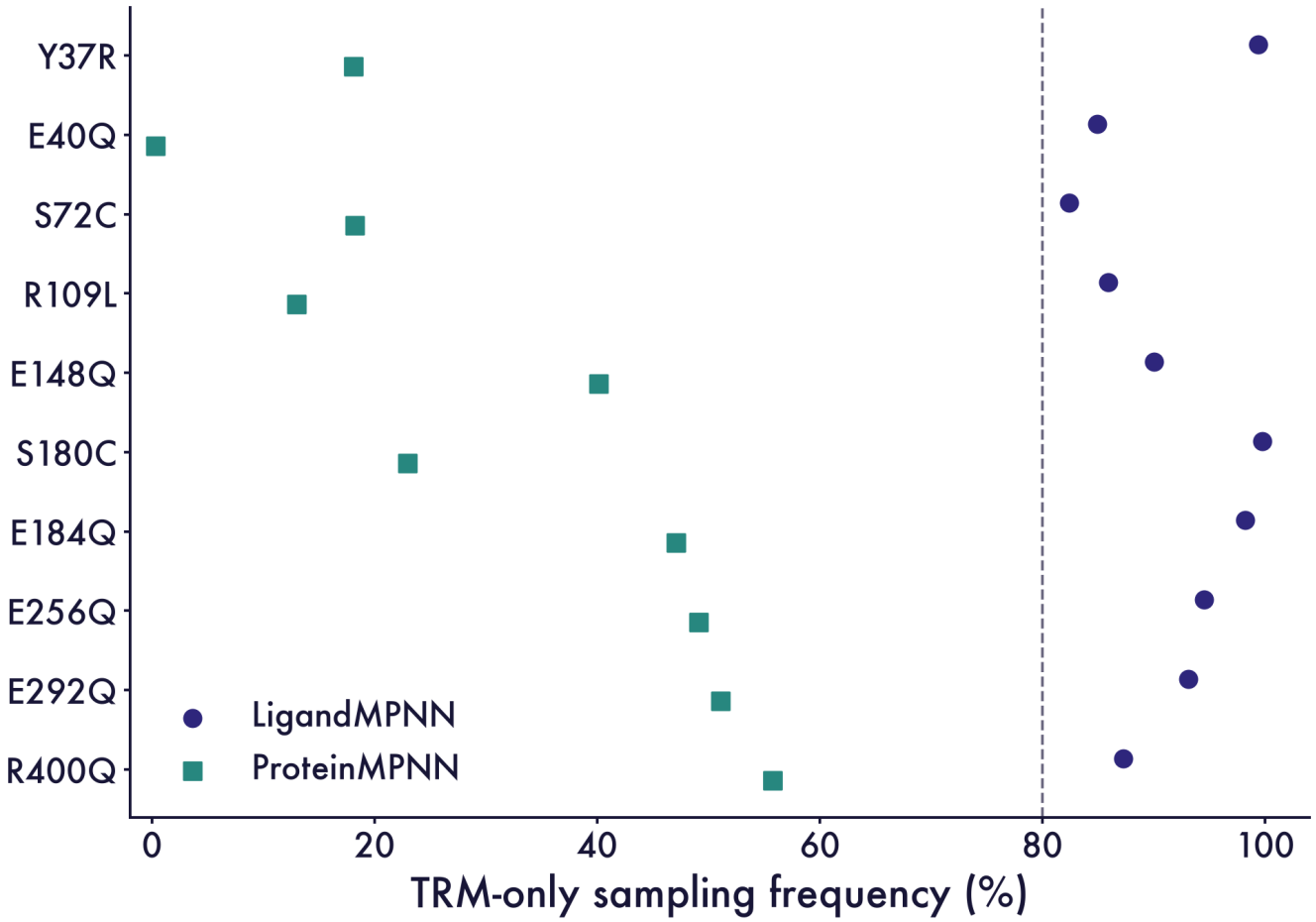


Figure 22. Frequencies of ten nominated substitutions in 10,000 TRM-only samples per model. The dashed line marks the 80% threshold. Substitution labels give original residue, position and replacement; Y37R means Tyr37→Arg. Frequency is the fraction of generated sequences carrying a substitution, rather than an experimental editing rate. LigandMPNN supplies all ten above-threshold nominations, whereas ProteinMPNN supplies none, showing that the two structural contexts lead to different recognition-residue preferences.

Model Figure 23

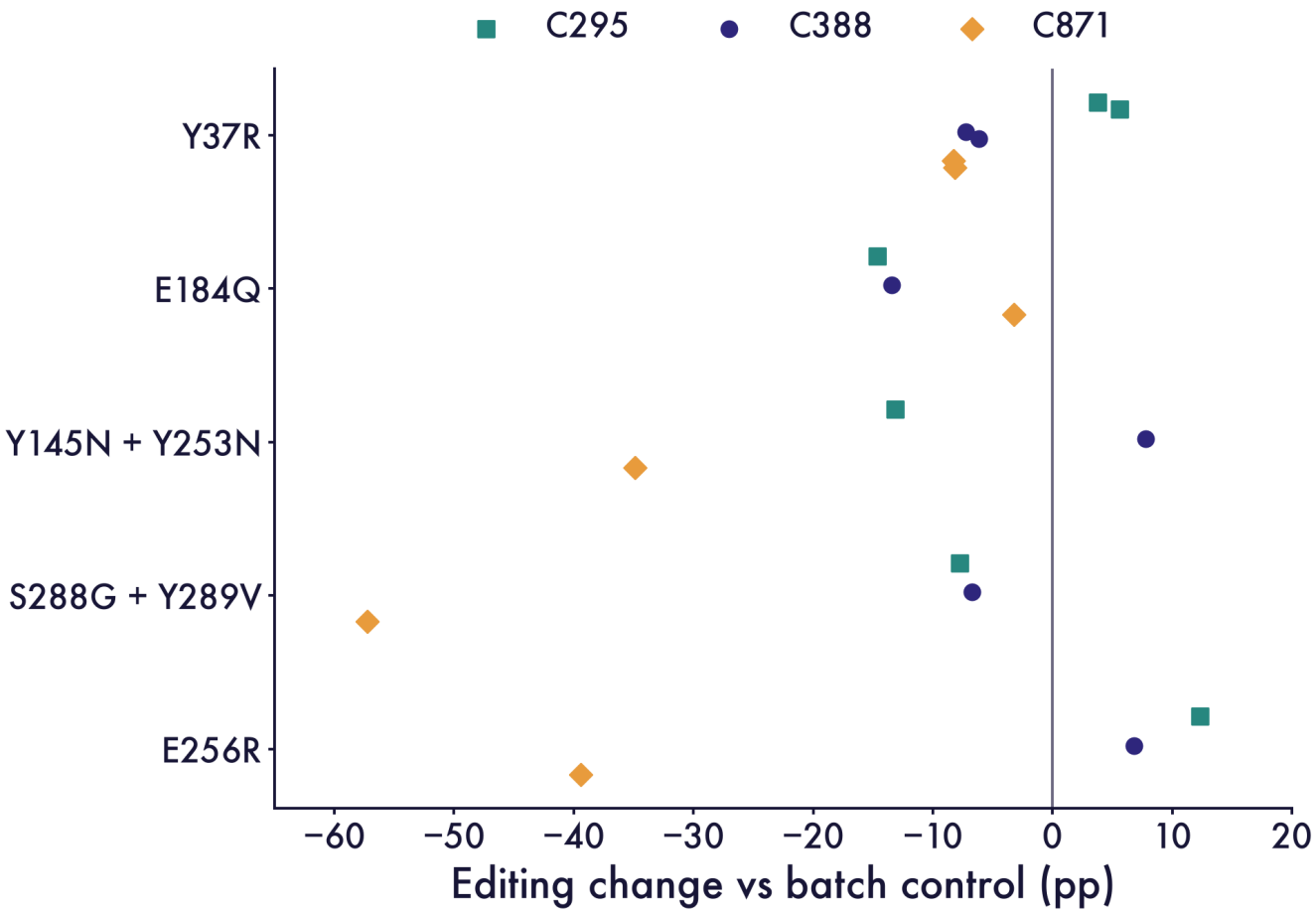
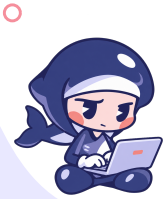


Figure 23. Editing changes relative to same-batch controls, with two records per mean. Y37R is compared separately with both batch-2 controls. P8-GVE changes repeat 8 via S288G and Y289V (Ser→Gly; Tyr→Val); Y145N + Y253N (Tyr→Asn) introduces SNE (Ser–Asn–Glu) motifs in repeats 4 and 7. Negative horizontal values mean less editing than the control; pp means percentage points. Purple circles mark target C388, while teal squares and orange diamonds mark C295/C871 bystanders. The SNE double mutant increases target editing while lowering both bystander readouts.

Module 2 Table 2

Experimental overlap, similarity and disagreement

Match definition	LigandMPNN construct coverage	ProteinMPNN construct coverage
All specified substitutions present; other residues unconstrained	23/81 (28.4%)	35/81 (43.2%)
All affected TRM triplets match; other triplets unconstrained	13/81 (16.0%)	20/81 (24.7%)
Entire 493-residue experimental sequence matches	0/81	0/81



Model Figure 24

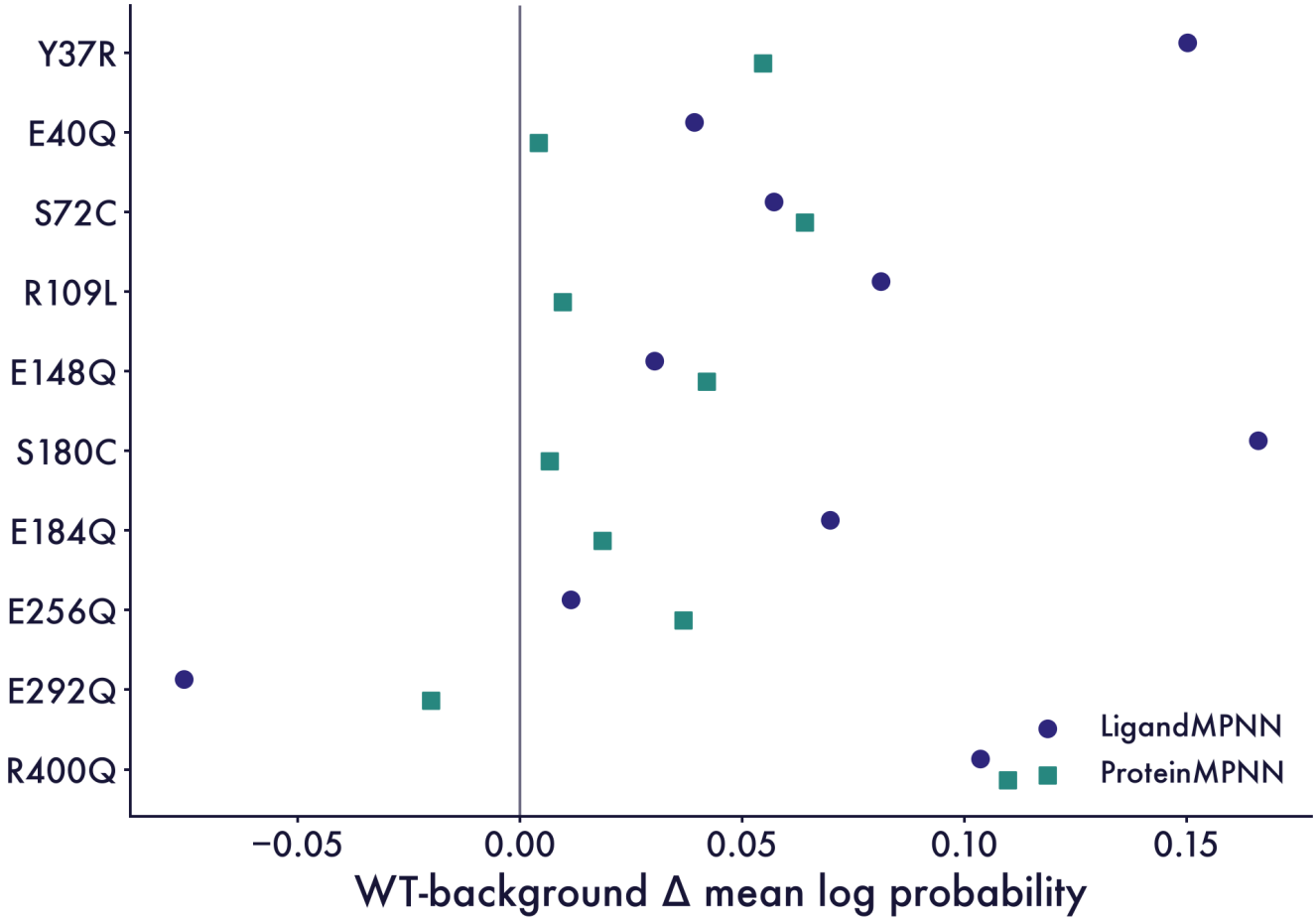


Figure 24. Mean TRM log-probability changes versus WT, averaged over eight decoding orders. Positive values indicate model preference. TRM denotes the three-residue recognition motif, and WT is the parental sequence. Values compare how strongly each model favors the mutant motif over WT. Most nominations score positively in both models, whereas E292Q (Glu292→Gln) scores negatively; the assay comparison in Figure 23 separately evaluates editing.

Module 3: Catalytic Design

Model Figure 25

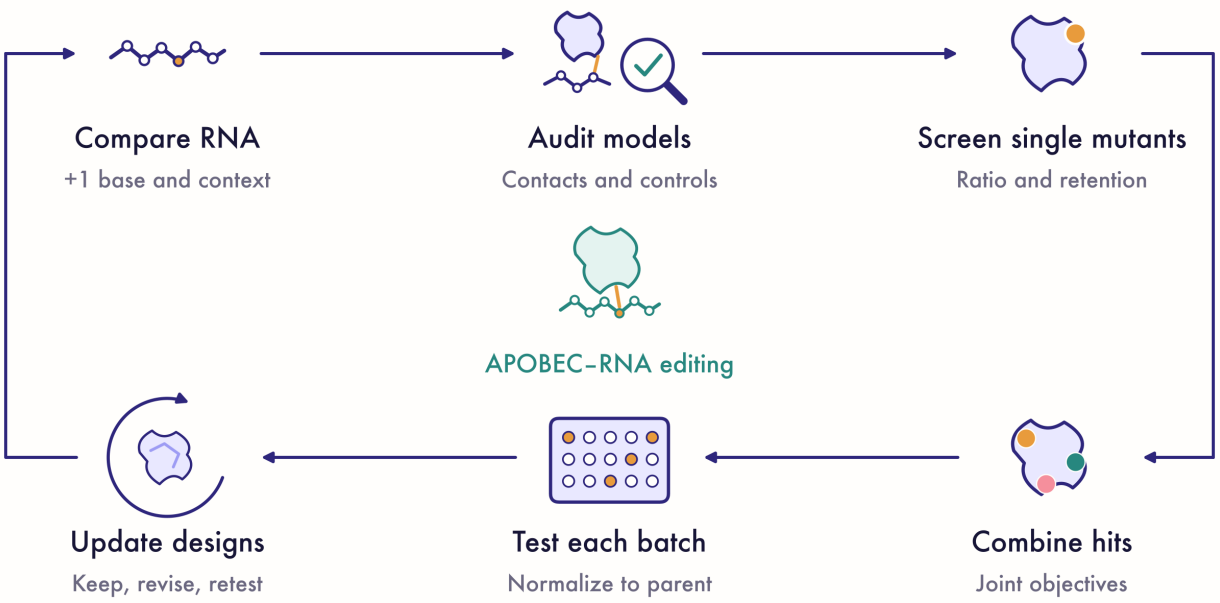
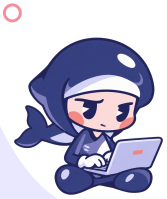


Figure 25. Structural modelling and experimental screening workflow. The central protein–RNA image is conceptual. The workflow compares GCG and GCA RNA contexts, which differ at the base after the edited C, then connects predicted contacts to measured target and bystander editing. Single-mutant screens inform combinations, and same-batch comparisons identify designs that improve preference while retaining target activity.

Module 3 Table 1

Experimental readouts

Metric	Definition	Purpose
+1 ratio	Editing at c.388 / editing at c.871	Compare the two GCG/GCA-context sites
Activity retention	Mutant c.388 editing / same-batch parental c.388 editing	Quantify retained on-target activity
ROF	Mutant +1 ratio / same-batch parental +1 ratio	Normalize preference within a batch
Amplicon targeting ratio	Summed GCG-site editing / summed non-GCG-site editing	Assess editing distribution beyond two sites



Model Figure 26

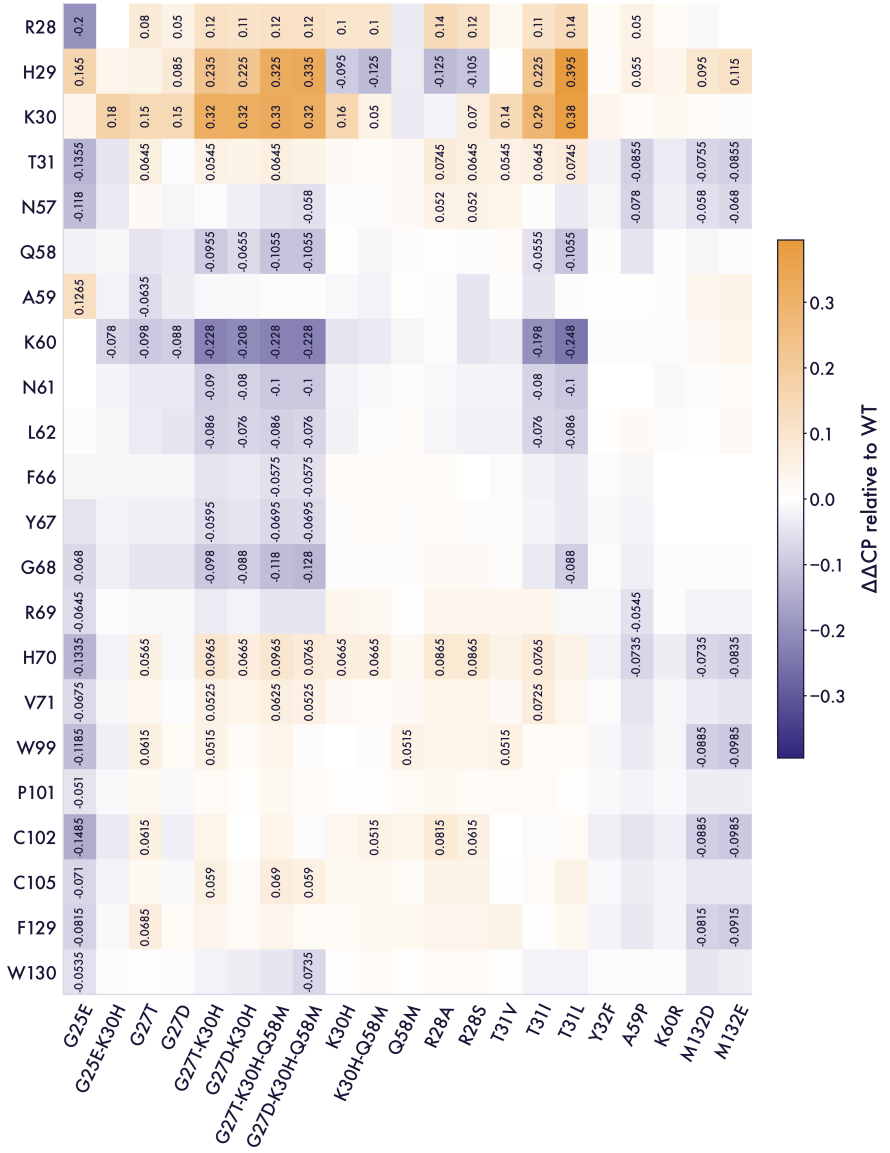


Figure 26. Δ Δ CP across 21 variants for the 22 residues with the largest maximum absolute changes. Colours span −0.395 to +0.395. CP is predicted contact probability; Δ Δ CP compares each mutant’ s GCA-versus-GCG contact preference with its parent. T31A means Thr31→Ala. Positive values indicate a shift toward A-associated contacts relative to G-associated contacts; negative values indicate the reverse. The heatmap identifies where a mutation redistributes predicted contacts across the interface. It guides structural interpretation alongside the activity measurements, rather than assigning an editing rate.

Model Figure 27

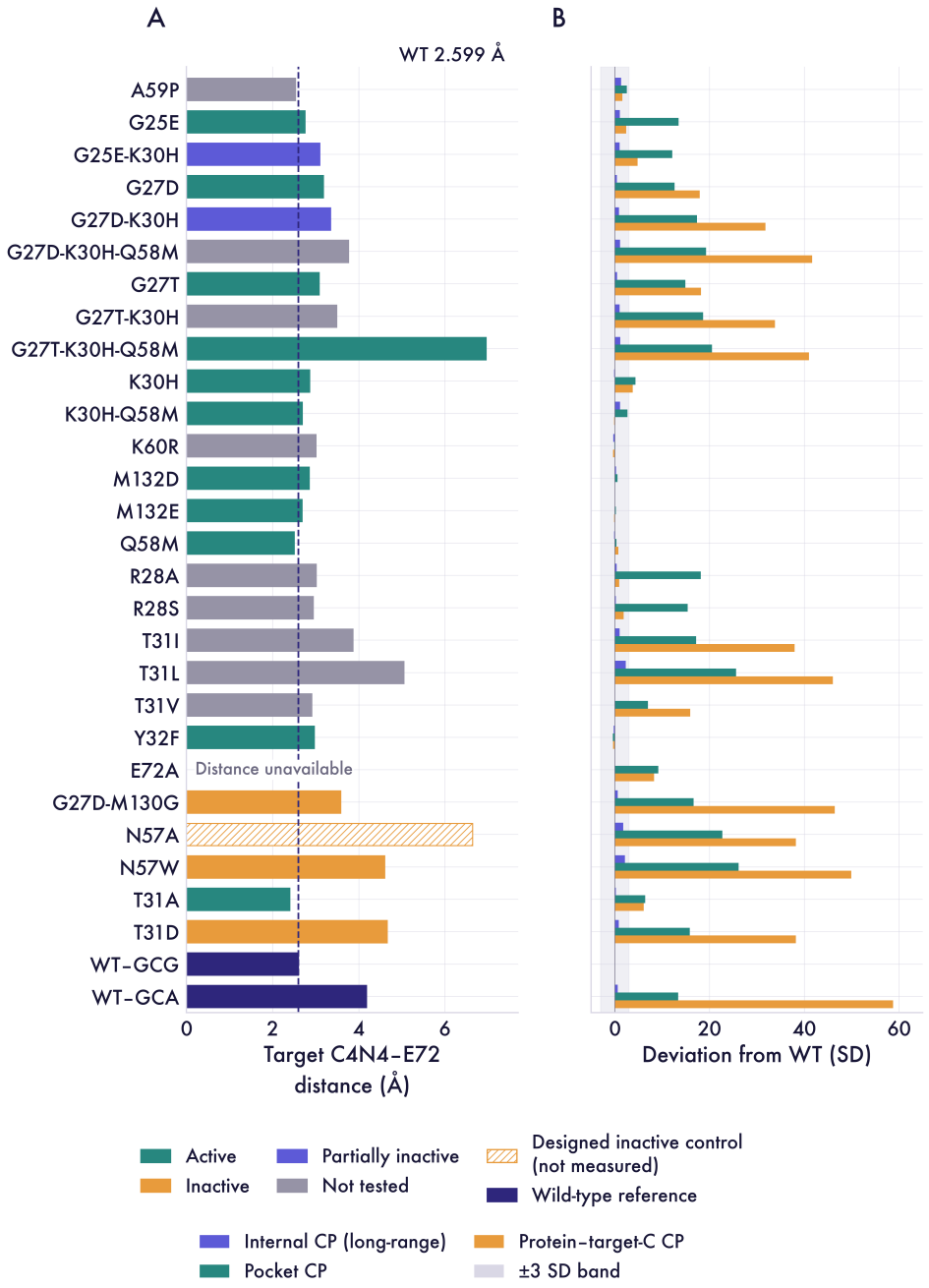


Figure 27. Catalytic-distance and contact-deviation summaries for 29 construct records. E72-based distances are shown with E72A marked NA. E72A replaces catalytic Glu72 with Ala; NA marks an unavailable E72-based distance. Distances describe catalytic geometry, while contact deviations describe changes in the predicted interface. Similar structural summaries accompany very different measured retention for T31A and T31D (Thr31→Ala/Asp), linking structural inspection to the paired functional screen rather than treating geometry alone as activity.

Model Figure 28

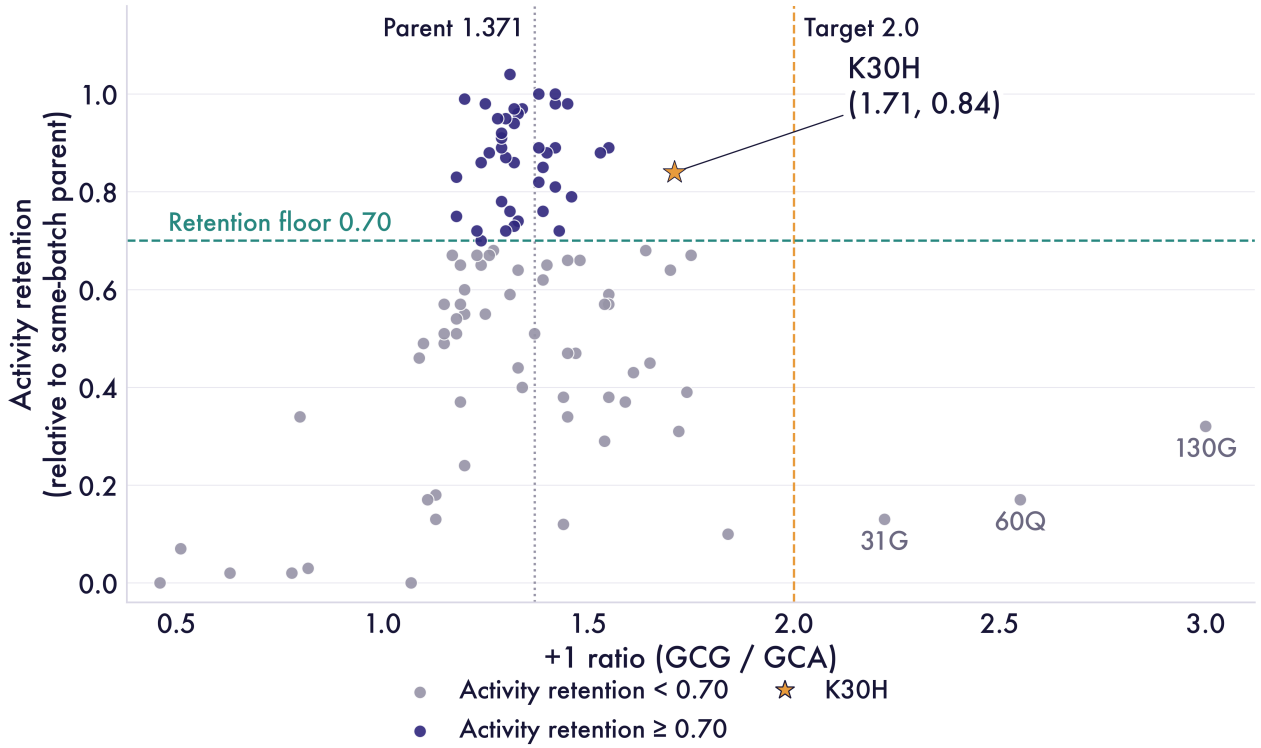


Figure 28. Single-mutant preference ratio and activity retention. No tested mutant meets both ratio ⩾2.0 and retention ⩾0.70. Preference is C388/C871 editing; retention compares target C388 with the same-batch parent. K30H means Lys30→His. Moving right indicates greater target retention; moving upward indicates a higher target-to-bystander ratio. The upper-right region meets both criteria. K30H retains activity but misses the ratio criterion, while higher-ratio alternatives lose activity, motivating the combination screen.

Model Figure 29

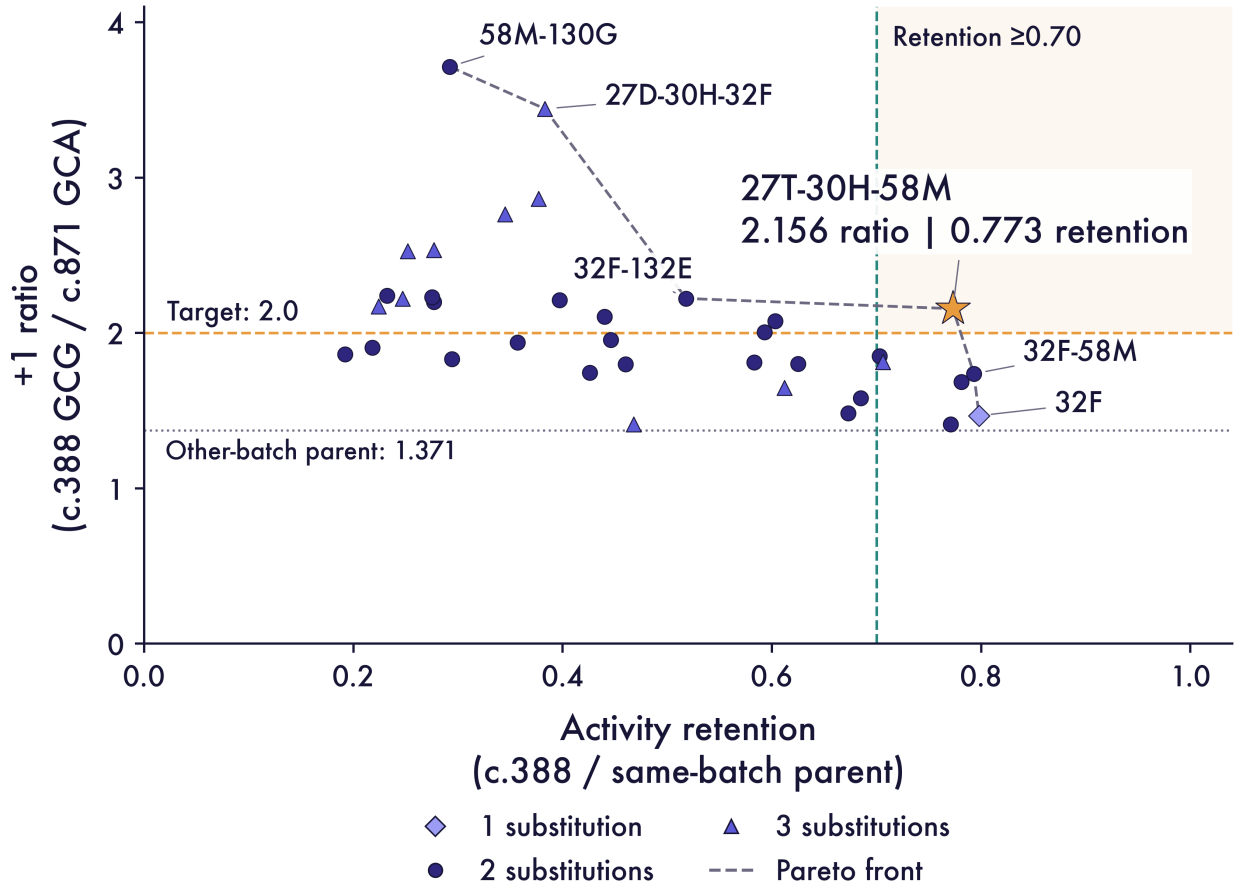


Figure 29. Combination-batch preference ratio and activity retention, with the empirical Pareto frontier and highlighted G27T−K30H−Q58M lead. The lead combines Gly27→Thr, Lys30→His and Gln58→Met. The frontier marks variants for which neither plotted readout improves without worsening the other. Moving upward improves C388/C871 preference and moving right retains more target activity relative to the same-batch parent. The highlighted lead is the only tested combination meeting both plotted criteria, illustrating the gain from combining substitutions rather than selecting on ratio alone.

Model Figure 30

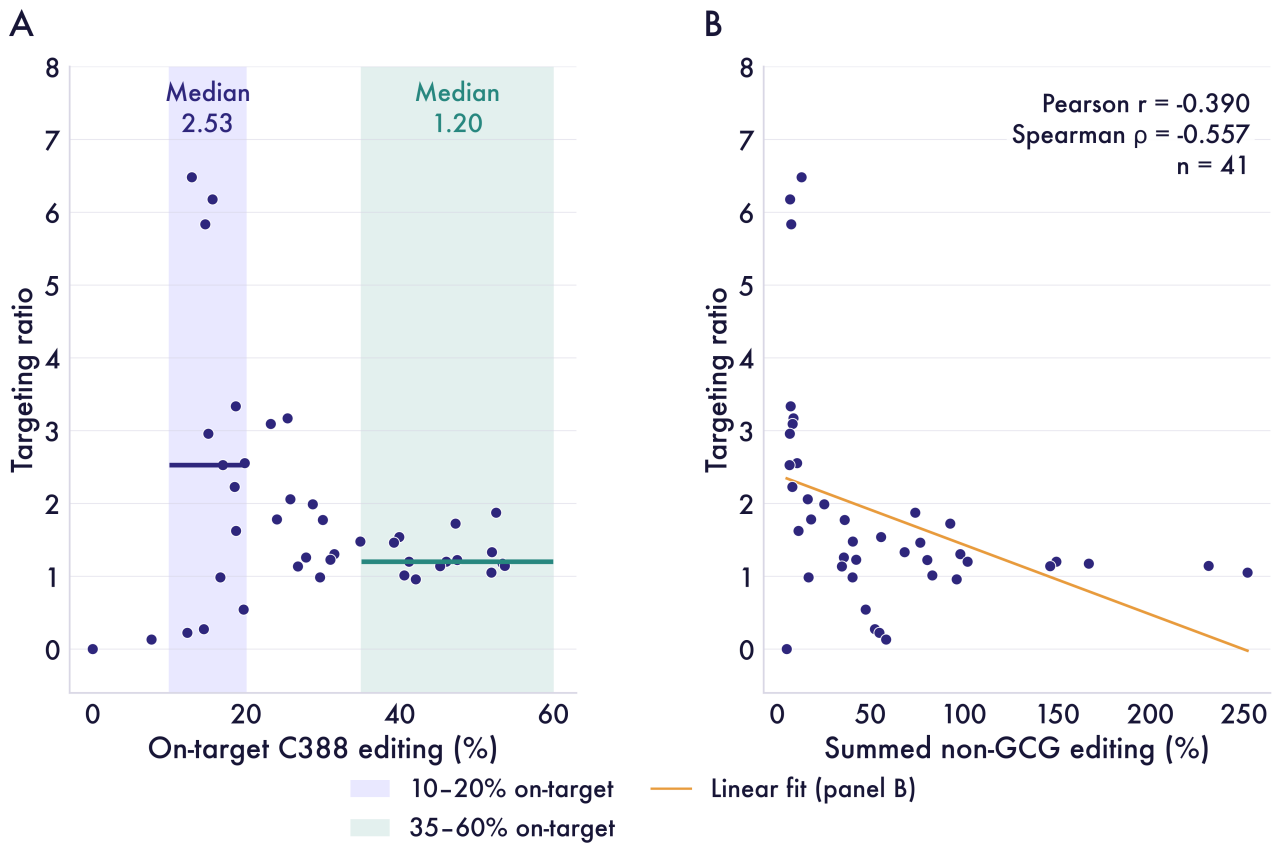


Figure 30. Amplicon targeting ratio versus editing output. Non-GCG editing also contributes to the ratio’s denominator. The targeting ratio sums editing at GCG-context sites relative to non-GCG sites across the assayed RNA segment. Read target output and non-GCG output separately: a higher ratio can arise from suppressing unwanted edits. The negative association with non-GCG editing is consistent with that pattern, but the shared denominator contributes to the association. Absolute target output remains necessary to assess retained activity.

Model Figure 31

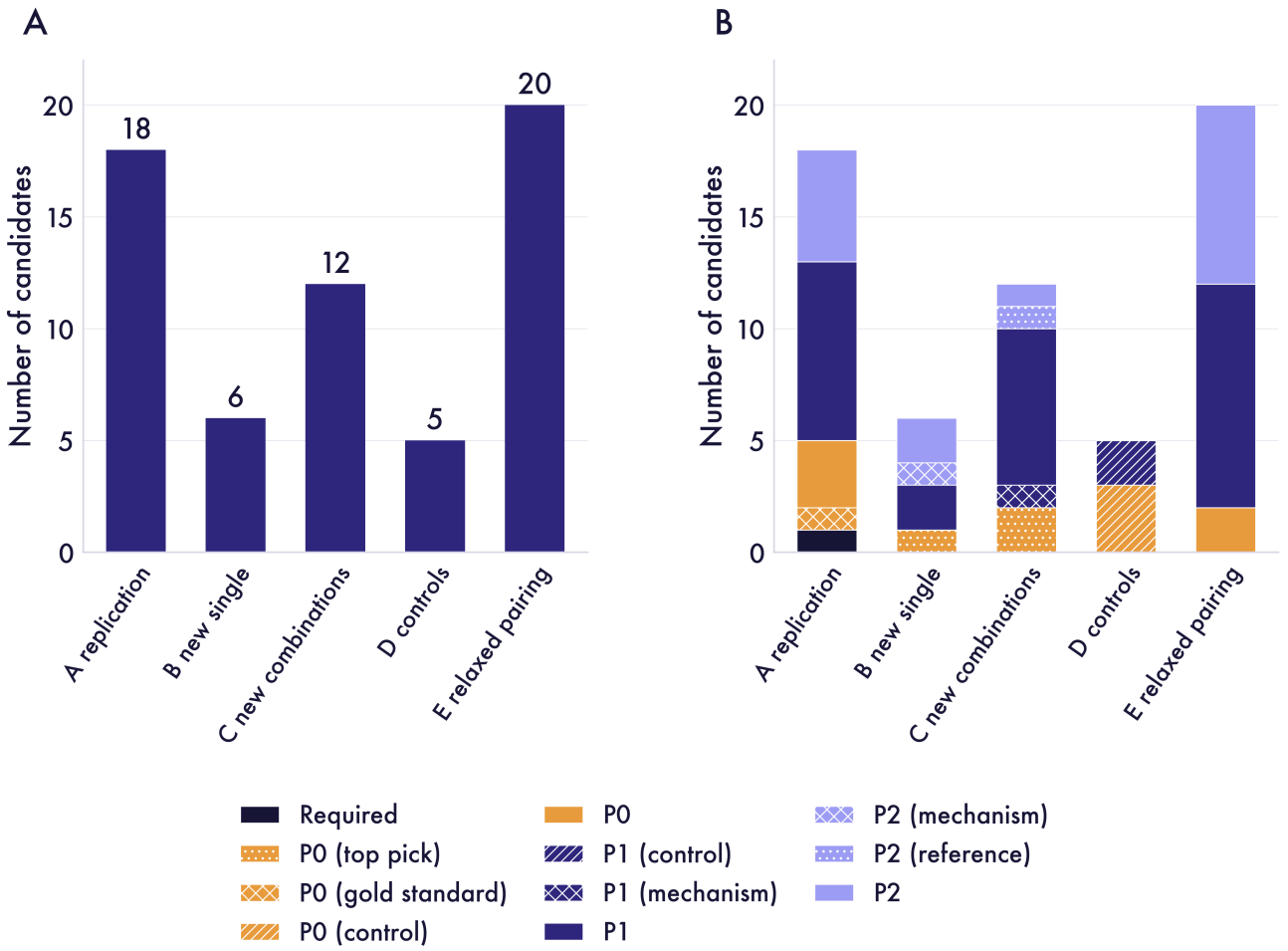


Figure 31. Five design groups organize 61 candidate entries and their priorities. The groups separate replication, new single substitutions, new combinations, controls and relaxed or anomalous cases. Their sizes describe proposed candidate entries, not completed experimental hits. This converts the preceding structural and assay comparisons into an organized follow-up panel.

Model Figure 32

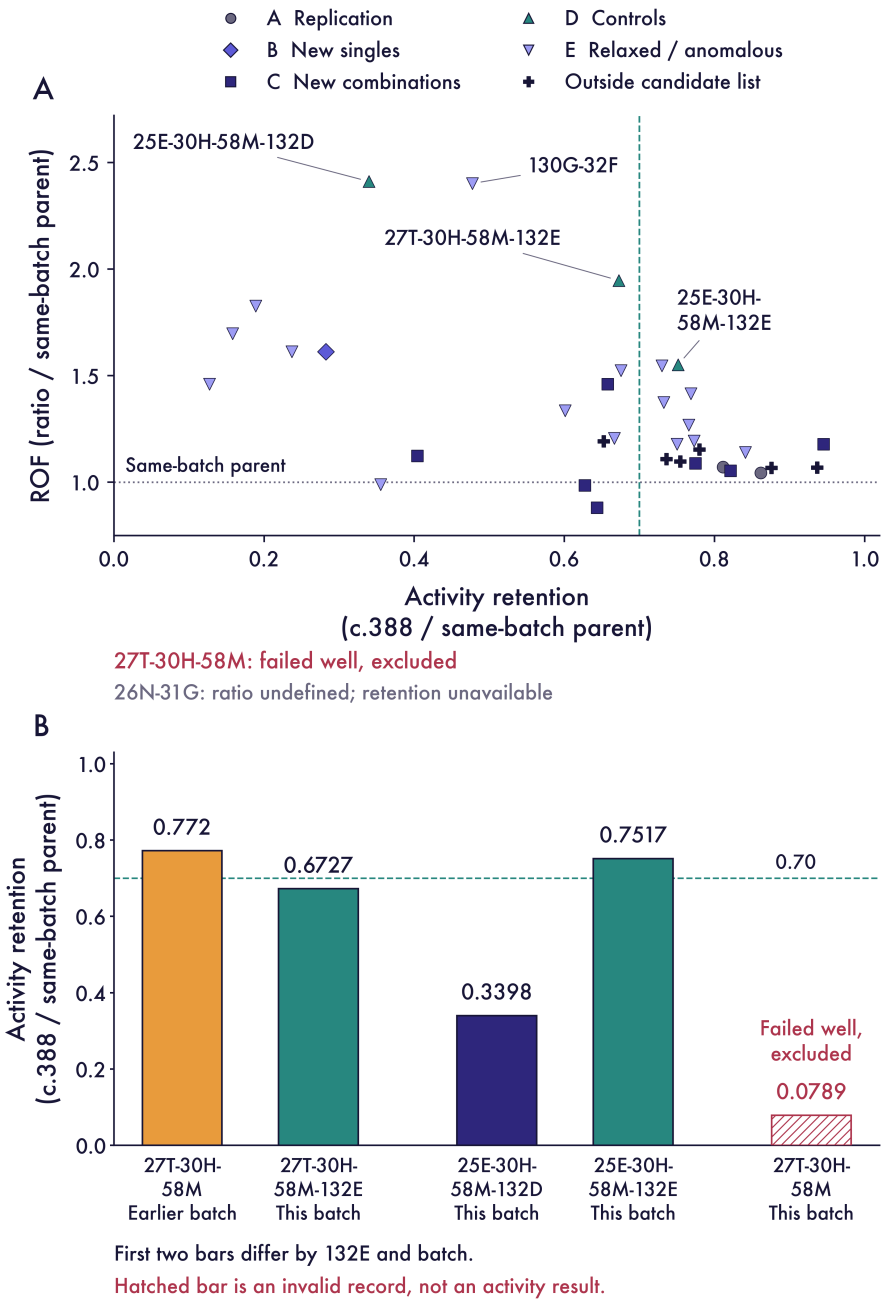


Figure 32. Follow-up preference ratios and activity retention, including position-132 variants, with batch-specific controls. M132D and M132E replace Met132 with Asp and Glu, respectively; each is assessed against its same-batch parent. Panel A shows activity retention against ROF, the mutant preference ratio divided by its same-batch parent's ratio; ROF above 1 indicates improved preference. Panel B contrasts selected retention values: replacing 132D with 132E on the matched G25E–K30H–Q58M background increases retention. The hatched failed-well record is excluded.

Module 4: LAMAR

Model Figure 33

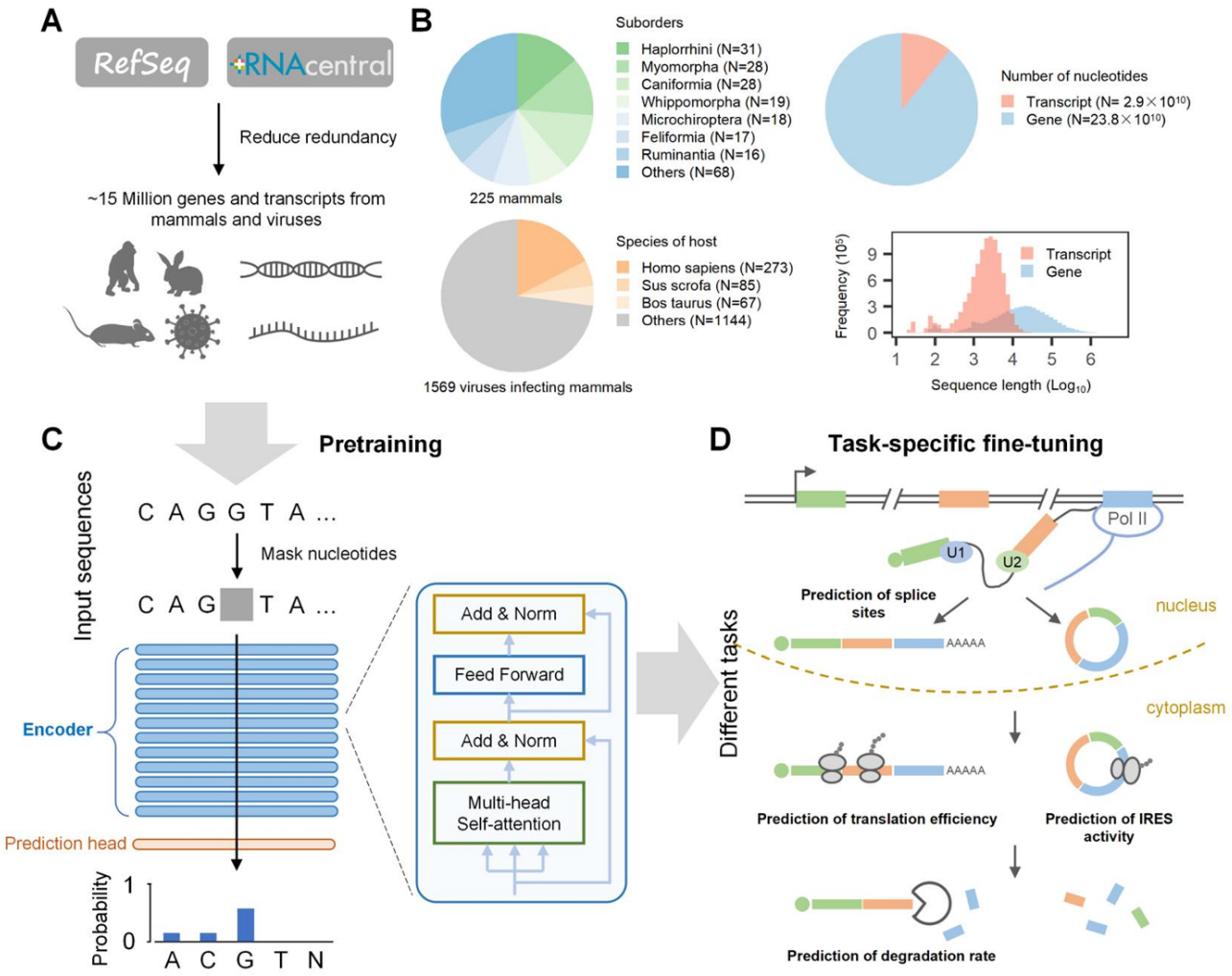
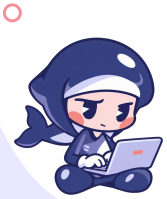


Figure 33. LAMAR pretraining and masked nucleotide prediction. Adapted from Zhou et al. (2025). Masked prediction hides selected nucleotides and trains the model to recover them from surrounding sequence. This published framework learns RNA sequence representations from context. Our downstream adaptation uses those representations first to identify editing-associated regions and then to rank cytidines within them, connecting transcriptome observations to a shorter list of RNA sites.

Module 4 Table 1

Rationale

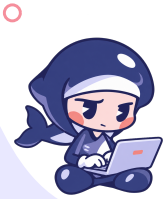
Input	Source
Gene annotation	gencode.v50.chr_patch_hapl_scaff.annotation.gtf
Genome sequence	GRCh38.primary_assembly.genome.fa
Generated transcriptome	gencode.v50.transcripts.fa



Module 4 Table 2

Rationale

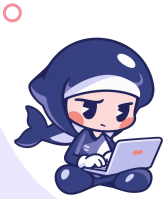
Property	Value
Transcript sequences	644,292
Total nucleotides	1,443,754,799



Module 4 Table 3

Continued pretraining result

Model	Masked prediction accuracy on human transcript sequences
Original LAMAR	0.505
Human-adapted LAMAR	0.575



Model Figure 34

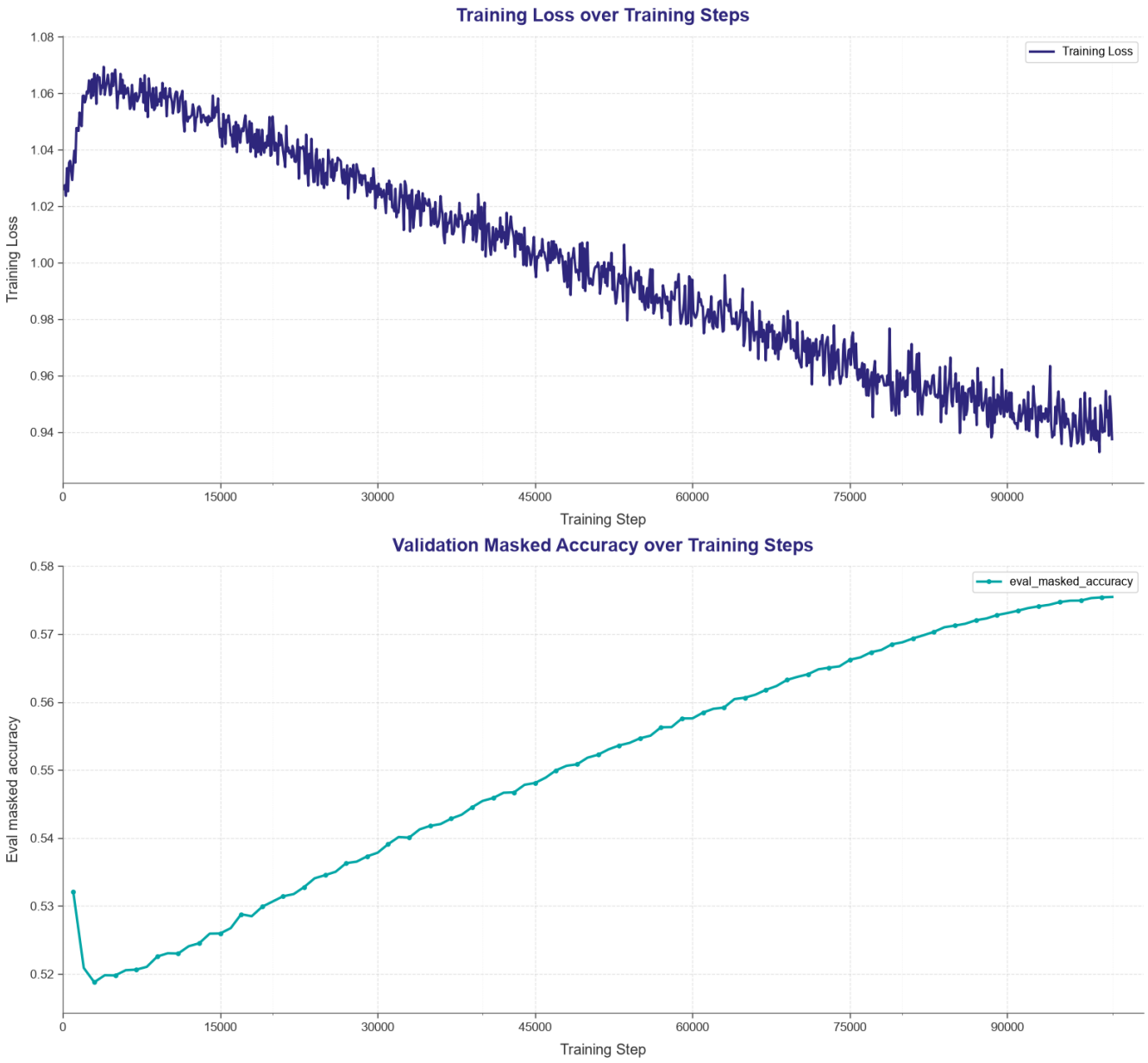
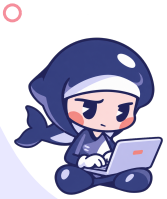


Figure 34. Training loss and masked nucleotide accuracy during continued LAMAR pretraining on human transcripts. The upper trace measures training error; the lower trace measures recovery of hidden nucleotides in validation sequences. After the initial adjustment, loss falls while validation accuracy rises. This shows improved sequence reconstruction during adaptation; Figures 35–36 evaluate the downstream editing tasks separately.

Module 4 Table 4

Stage I: 1024-nt candidate-region classifier

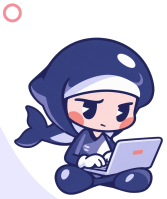
Split	Total windows	Positive windows	Negative windows
Training	3,835	2,643	1,192
Validation	213	150	63
Test	212	140	72



Module 4 Table 5

Model training

Parameter	Value
Hidden size	768
Intermediate size	3,072
Attention heads	12
Transformer layers	12
Positional embedding	Rotary positional embedding
Output labels	2
Classification head	Linear
Maximum sequence length, Stage I	1,026 tokens including special tokens
Random seed	42



Module 4 Table 6

Stage I optimization

Hyperparameter	Selected value
Batch size per device	4
Gradient accumulation steps	2
Effective batch configuration	batch4_accum2
Learning rate	5e-4
Weight decay	7.9194e-6
Token dropout	0.0
Warmup ratio	0.05338
Maximum epochs	15

Module 4 Table 7

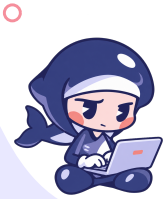
Stage II optimization

Hyperparameter	Selected value
Optuna trial	10
Batch configuration	batch4_accum2
Learning rate	1.0139e-5
Weight decay	6.6122e-5
Token dropout	8.9809e-4
Warmup ratio	0.09762

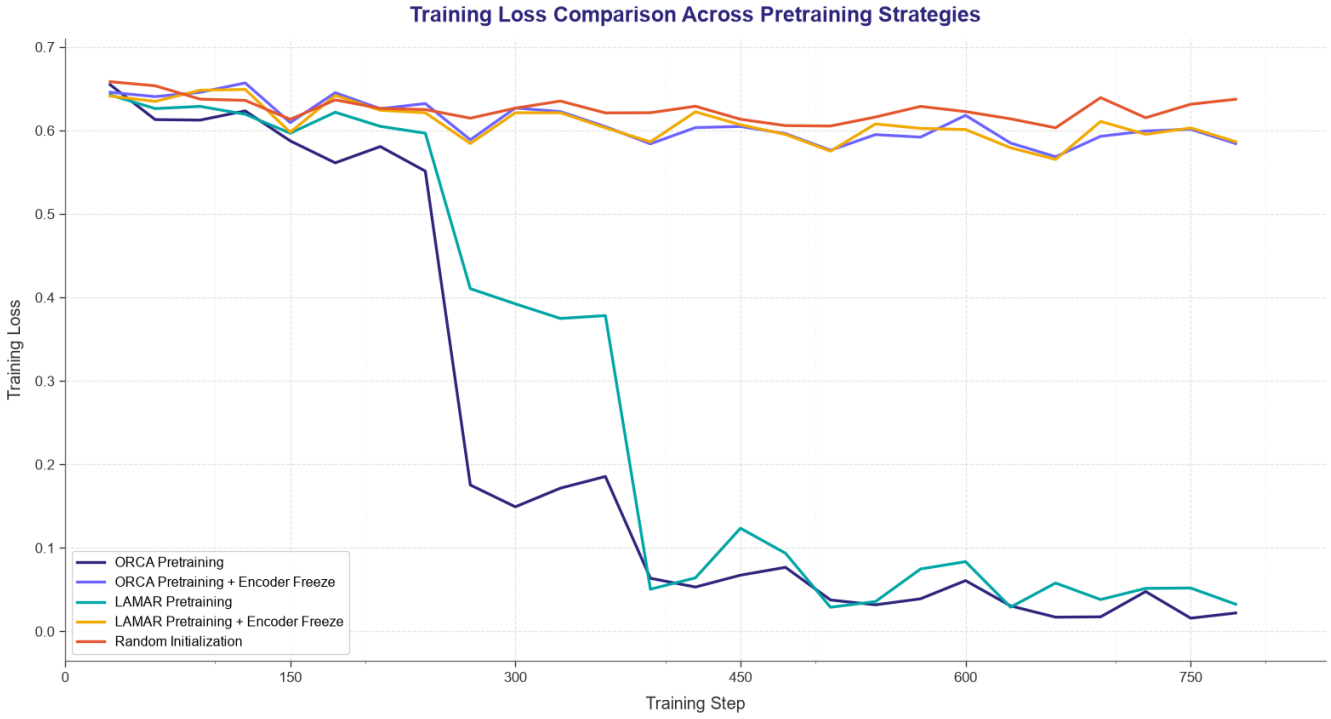
Module 4 Table 8

Stage I results: candidate-region enrichment

Dataset	Best PR-AUC
Validation	0.813
Test	0.722



Model Figure 35



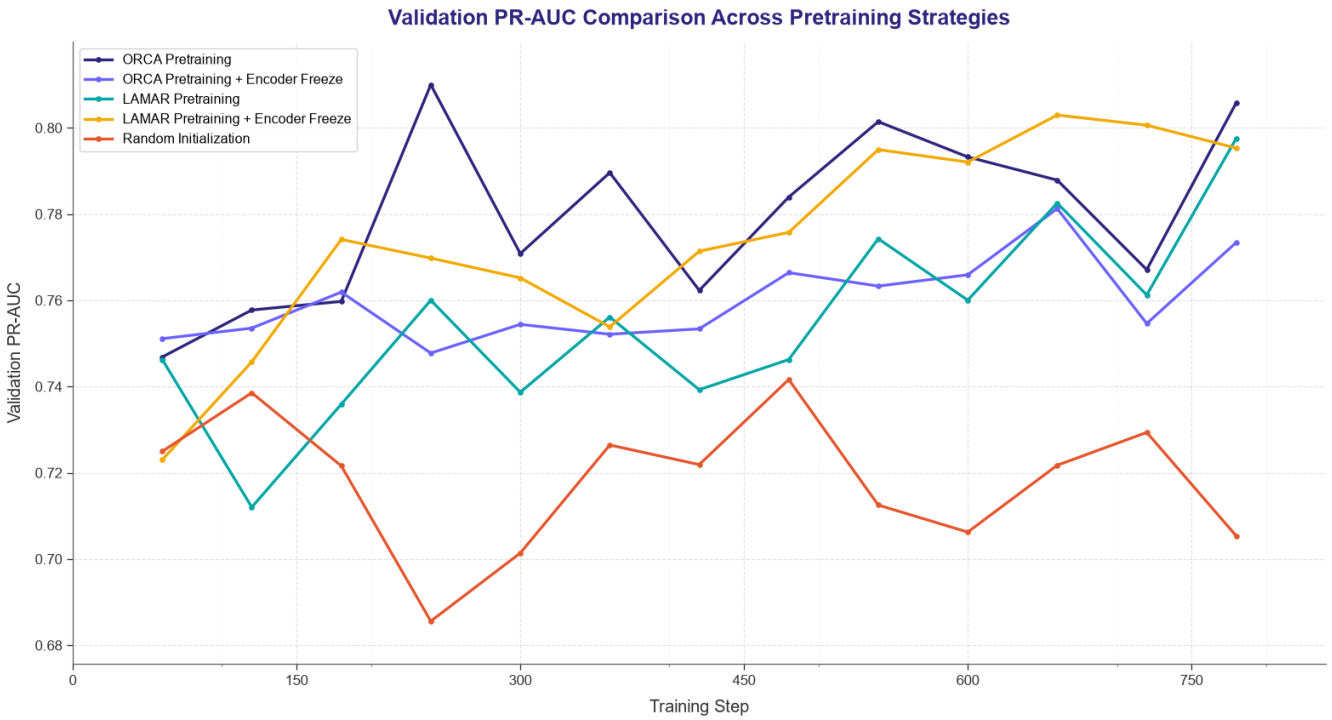


Figure 35. Stage I training loss and validation PR-AUC across continued-pretrained, original, encoder-frozen and randomly initialized LAMAR configurations. PR-AUC is the area under the precision–recall curve, summarizing identification of positive regions. A frozen encoder keeps the pretrained RNA representation fixed. Read the upper plot for fit to training data and the lower plot for ranking of positive validation regions. Although fine-tuning lowers training loss, the original frozen encoder gives the strongest validation performance, supporting retention of the pretrained representation for regional screening.

Module 4 Table 9

Validation results

Metric	Validation result
Hit@1	0.247
Hit@5	0.607
Hit@10	0.847
Hit@20	0.953
Recall@5	0.381
Recall@10	0.635
Recall@20	0.841
MRR	0.409
Mean best rank	6.527
Median best rank	4
Evaluated positive windows	150

Module 4 Table 10

Test results

Metric	Test result
Hit@1	0.243
Hit@5	0.607
Hit@10	0.779
Hit@20	0.921
Recall@5	0.373
Recall@10	0.573
Recall@20	0.792
MRR	0.406
Mean best rank	7.429
Median best rank	4
Evaluated positive windows	140

Model Figure 36

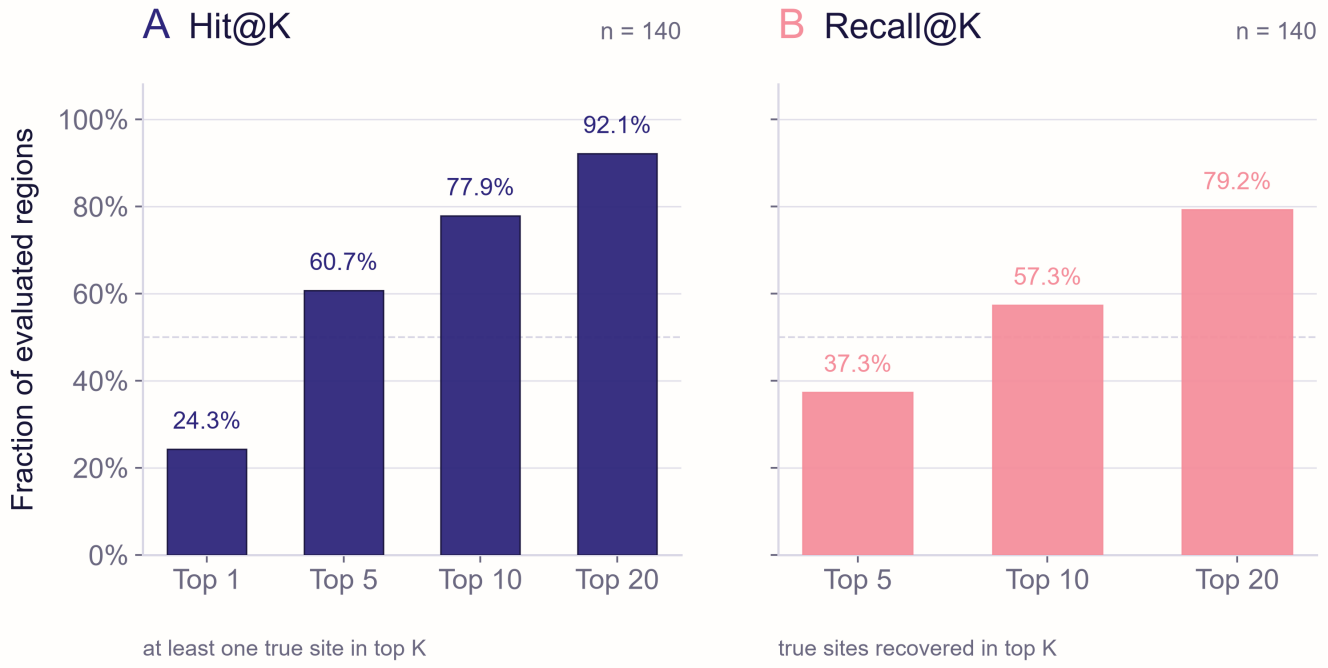


Figure 36. Reported Hit@K and Recall@K for 140 positive test windows. Hit@K indicates any supported site within the top K; Recall@K is the fraction of supported sites recovered within the top K. K is the number of highest-ranked cytidines examined per window. Hit@10 = 0.779 means the top ten include at least one supported site in 77.9% of positive windows. Increasing K improves recovery while requiring more sites to be examined, making the screening-budget trade-off explicit.

Model Figure 37

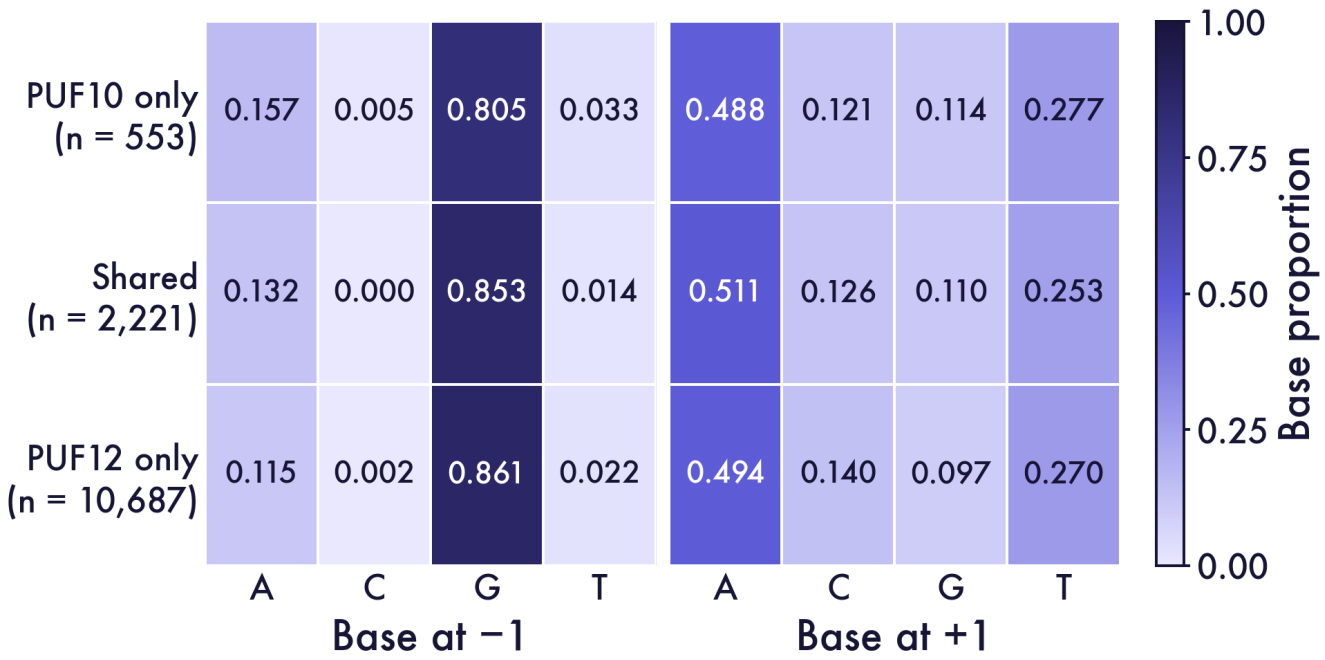


Figure 37. Base proportions at −1 and +1 around edited C in PUF10-only (553), shared (2,221) and PUF12-only (10,687) retained-site sets. Position −1 is immediately upstream of the edited C; +1 is immediately downstream. All three sets favor GCA, supporting a shared APOBEC-associated local preference across the PUF comparison. Their distinct membership and flanking profiles connect PUF-mediated recruitment with the longer RNA context used for site ranking.

Model Figure 38

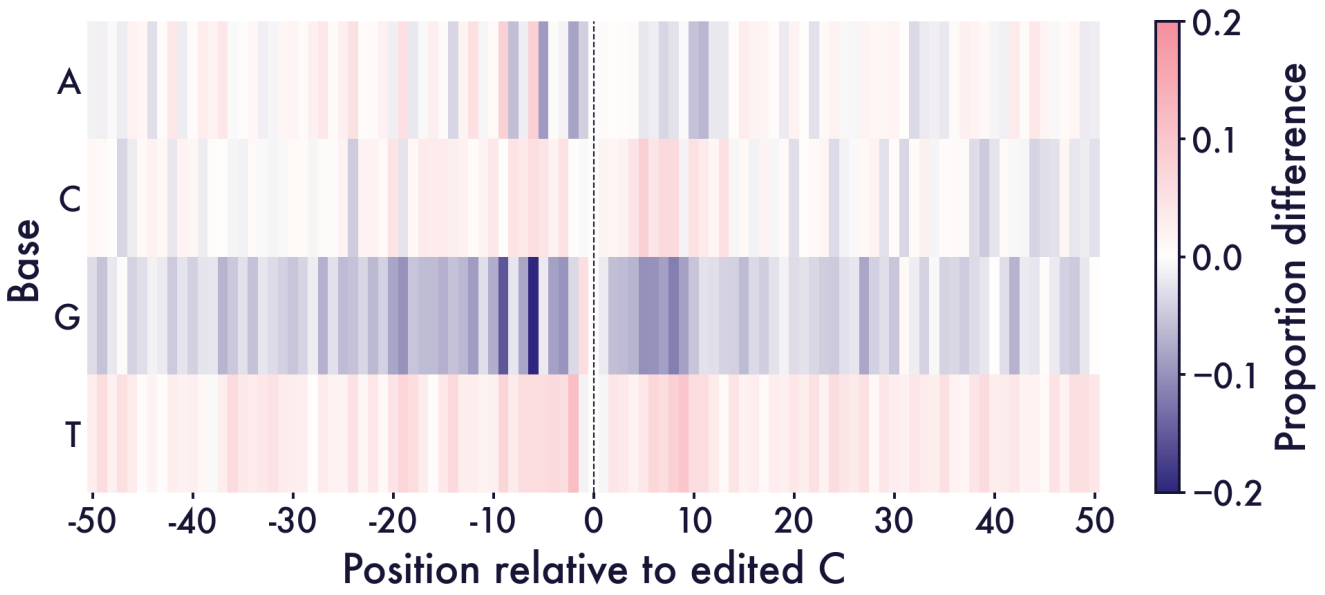


Figure 38. Base-proportion differences, PUF12-only minus PUF10-only, across 101 positions centred on edited C. Indigo is negative; coral is positive. Zero means equal base proportions in the two exclusive sets. A difference of −0.198 corresponds to 19.8 percentage points less G at −6 in PUF12-only sites, the largest observed shift. Differences beyond the adjacent bases motivate the longer RNA-context representation used for site ranking.

Delivery Dynamics

Delivery Figure 1

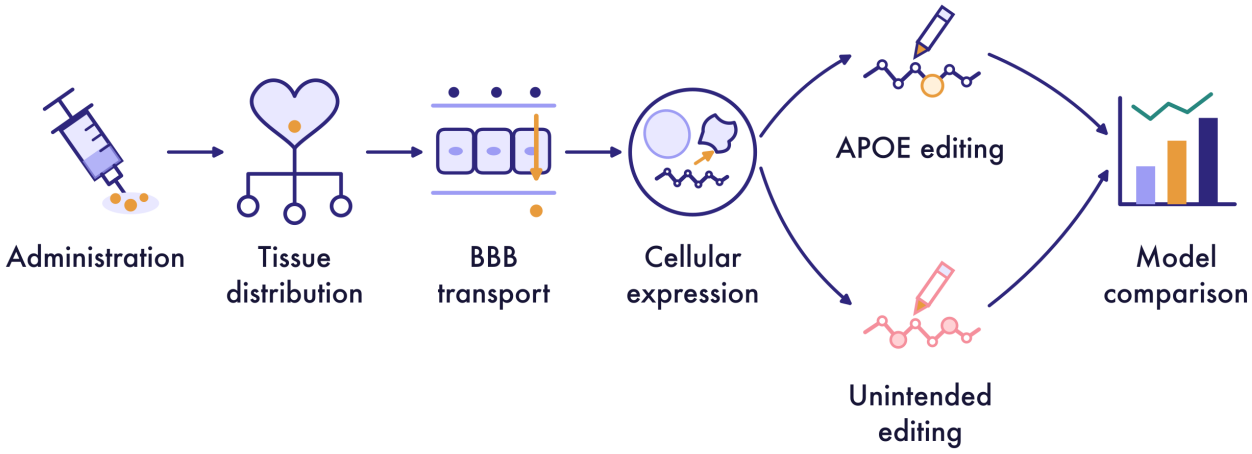


Figure 1. Delivery-to-editing model. Transport and expression connect the administered input to APOE editing and three unwanted-editing pathways. AAV means adeno-associated virus. BBB means blood–brain barrier; ODEs are ordinary differential equations describing changes over time. Read left to right: delivery sets editor exposure, expression supplies active protein, and competing reactions determine the RNA product mixture. The same chain lets us trace a change in dose to both intended editing and unwanted products.

Delivery Figure 4

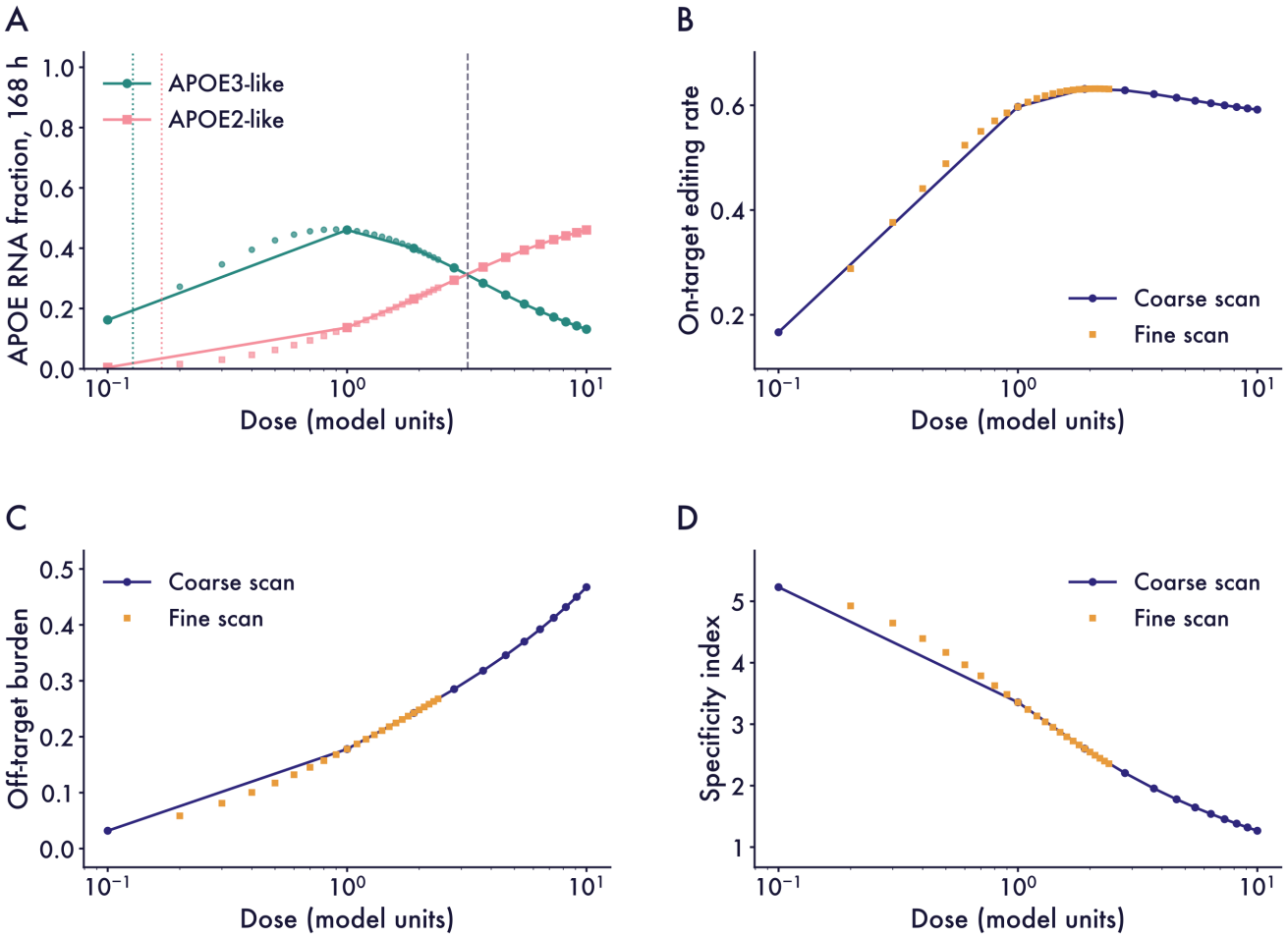


Figure 4. Dose response: editing and specificity. A: APOE product fractions at 168 h. B–D: on-target editing, off-target burden and specificity. Coarse and fine scan values are retained; thresholds are model criteria. The horizontal dose axis is logarithmic. Panel B counts both APOE3-like and APOE2-like products, whereas panel A separates them. APOE3-like RNA peaks at an intermediate dose and then falls as APOE2-like RNA accumulates: product composition identifies the intermediate-dose advantage.

Delivery Figure 5

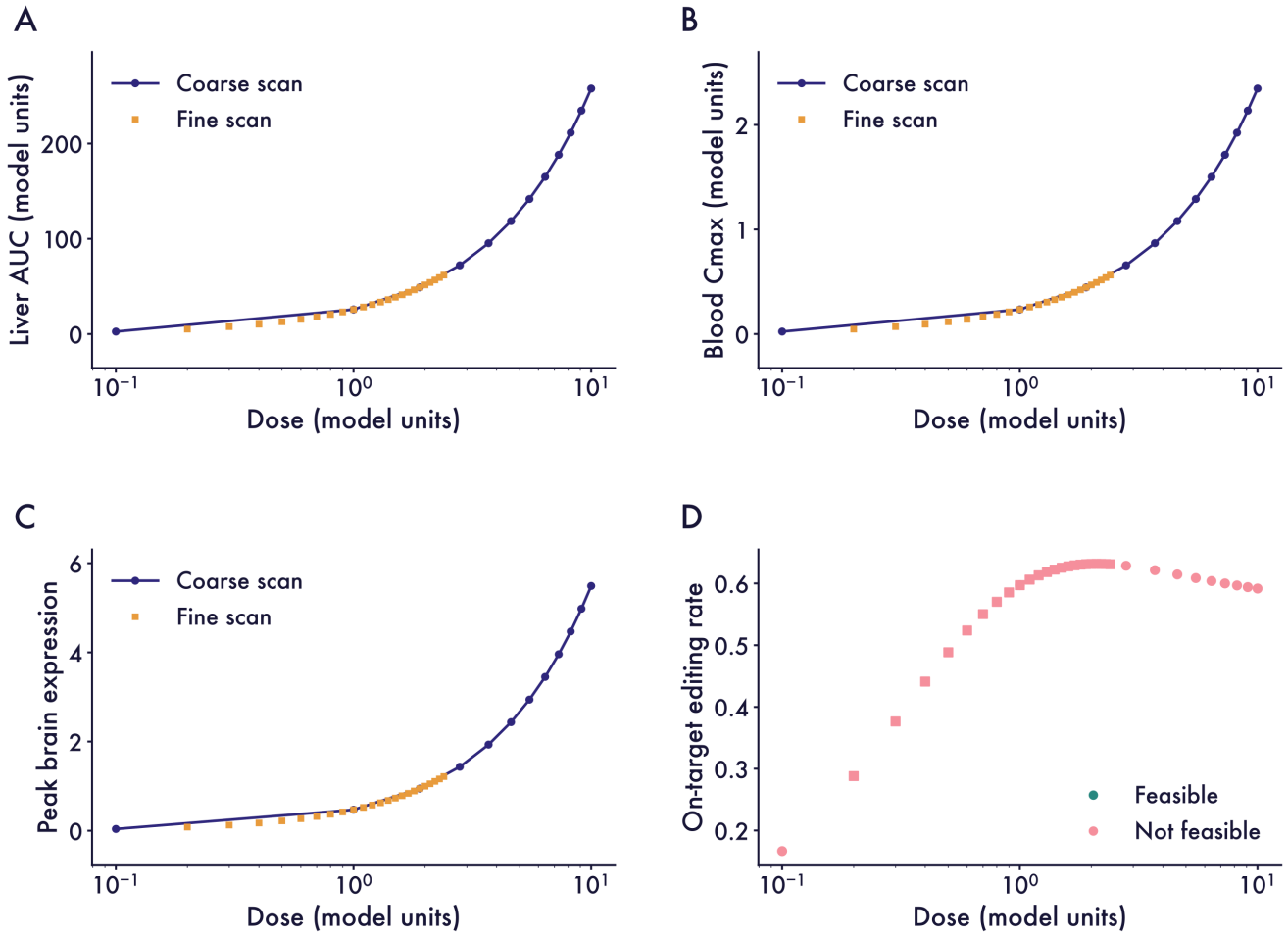


Figure 5. Dose response: exposure and feasibility. A–C: liver exposure, peak blood concentration and peak brain expression. AUC is area under the concentration–time curve; Cmax is peak concentration. D: the original feasibility classification, using source model units and criteria. Read exposure alongside editing, rather than selecting the highest-editing dose alone. The dose scan exposes a trade-off across the joint criteria, prioritizing improved delivery and selectivity to balance intended yield with exposure and unwanted editing.

Delivery Figure 6

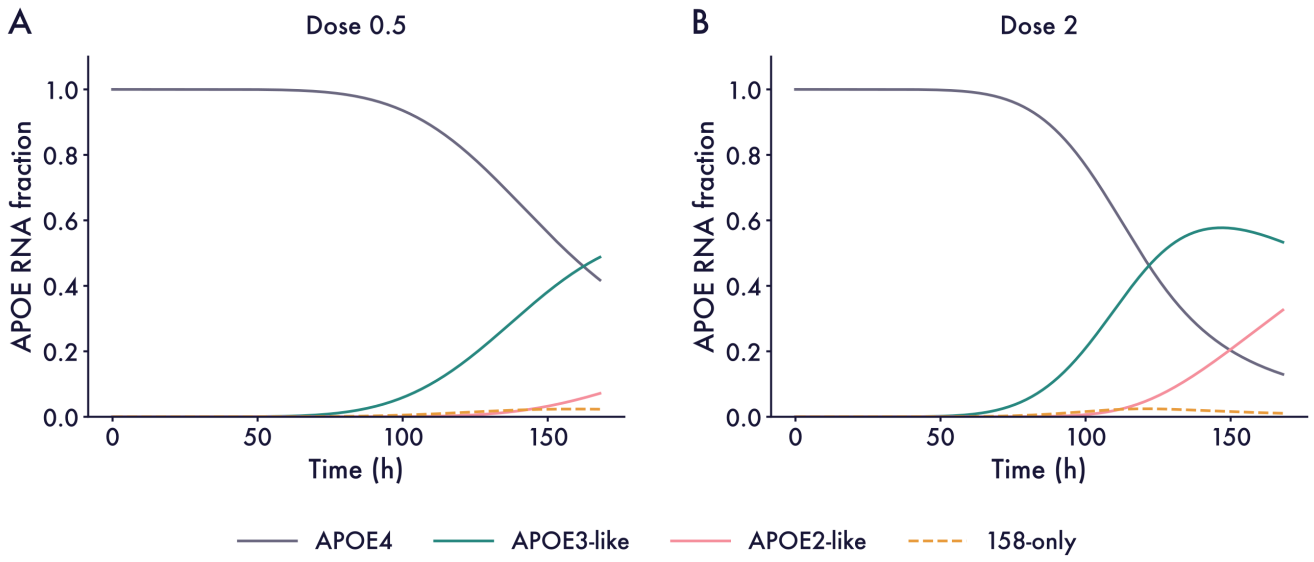


Figure 6. APOE product trajectories. A, dose 0.5; B, dose 2.0. Both trajectories span 168 h. Fractions use the total APOE RNA pool. Follow each RNA state over time: APOE4 is the starting pool, APOE3-like has the residue-112 edit, APOE2-like has both 112/158 edits, and 158-only carries the other edit alone. Comparing the panels shows how higher exposure shifts product composition through sequential editing.

Delivery Figure 7

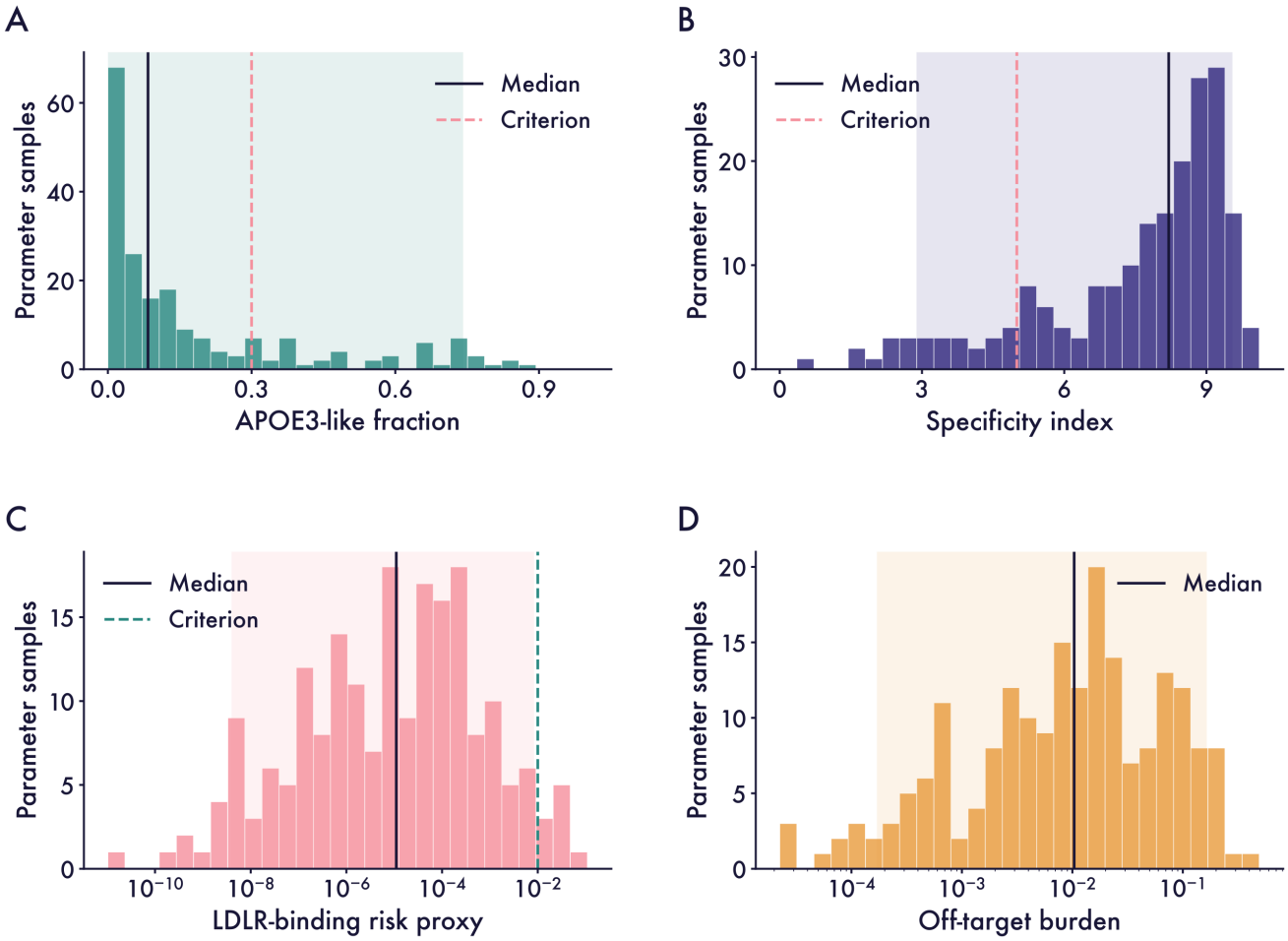


Figure 7. Parameter uncertainty. Two hundred parameter samples for 10R-proapobec at dose 0.9 and 168 h. Shading spans the 5th–95th percentiles; lines mark medians and model criteria. Each histogram counts parameter samples by their predicted output; the spread shows sensitivity to uncertain parameters. The APOE3-like fraction varies widely even at fixed dose, making its yield less stable across sampled assumptions than a single baseline trajectory suggests. LDLR is the low-density lipoprotein receptor; its binding-risk proxy here is the modeled APOE2-like fraction.

Delivery Figure 8

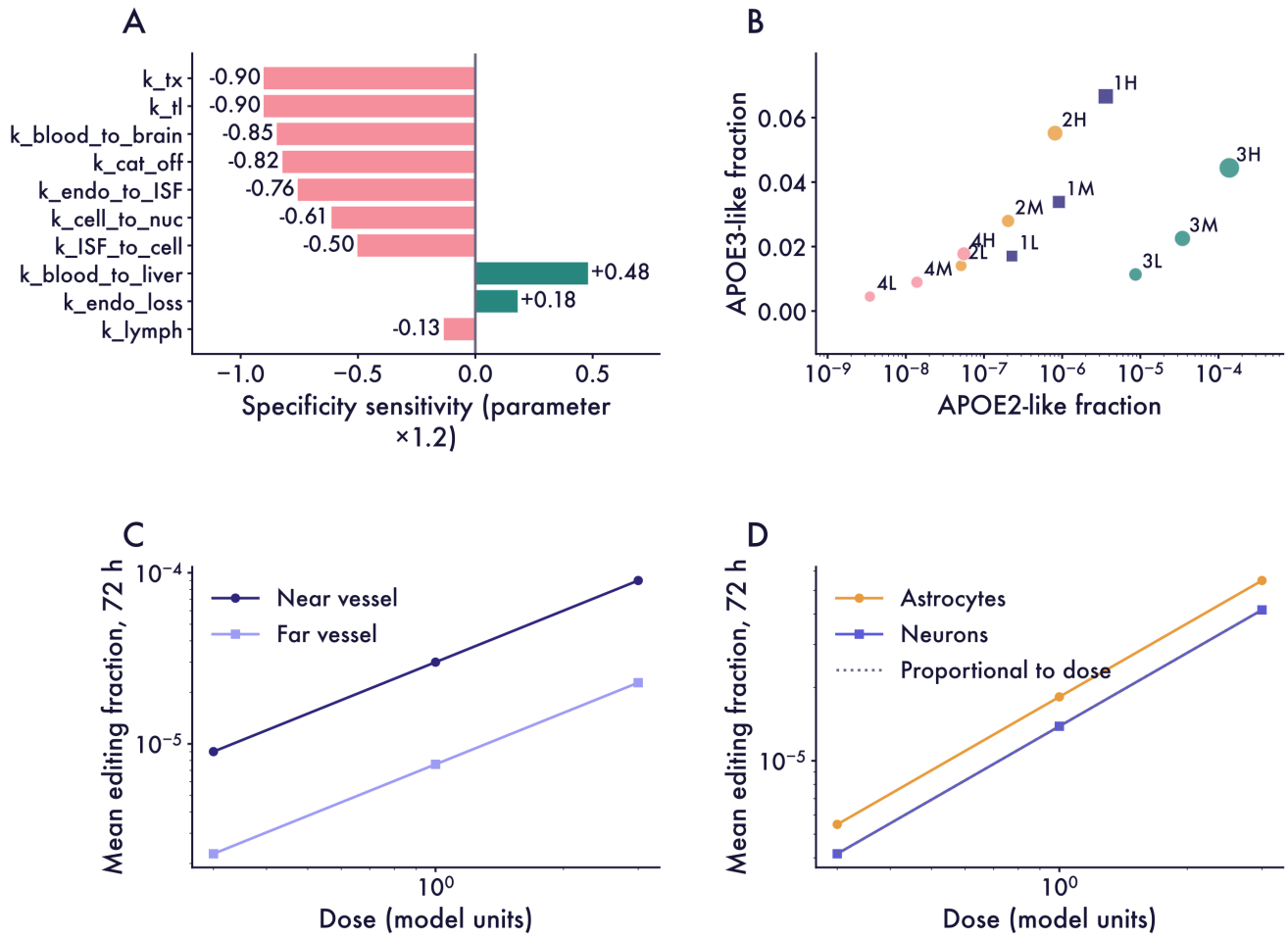


Figure 8. Sensitivity, design, and spatial comparisons. A: local sensitivity after a 20% parameter increase (scale 1.2). B: the 72-hour design comparison. C–D: spatial summary means by vessel distance and cell type. In A, negative sensitivity means reduced specificity. In B, upper-left positions combine more APOE3-like with less APOE2-like RNA; marker area represents unwanted-edit burden, and L/M/H mark dose categories. Designs 1–4 denote 10R-proapobec, 10R-design-4, 10R-dual-site and 8R-design-2; R counts PUF repeats. In C, near-vessel cells remain more strongly edited across doses, identifying tissue coverage as a distinct delivery objective.

Delivery Figure 9

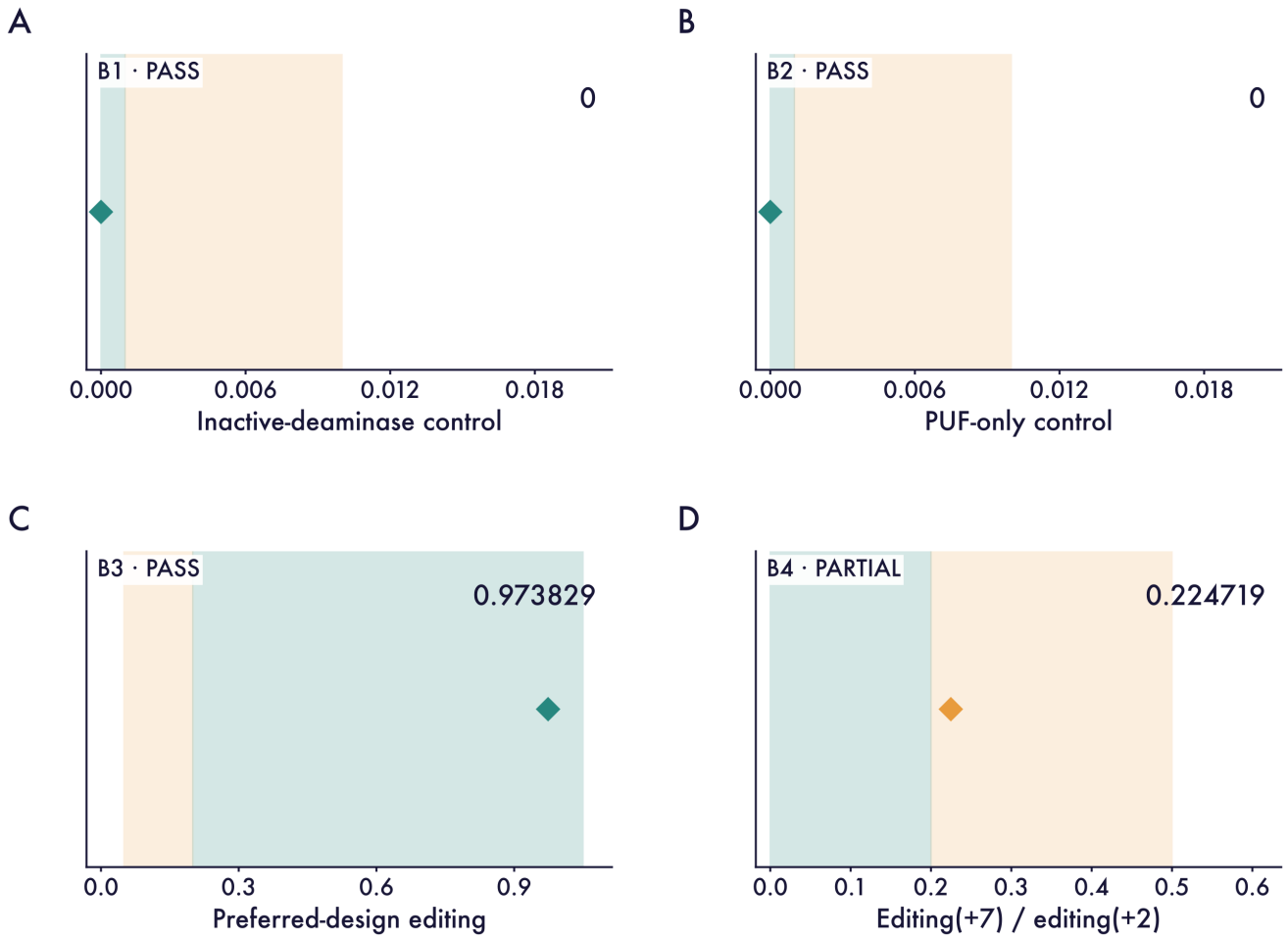


Figure 9. Benchmark checks B1–B4. Teal and orange indicate the supplied target bands. Each panel uses its own numeric scale. B1/B2 test inactive-deaminase and PUF-only controls; B3 checks preferred-design editing; B4 compares editing at offsets +7 and +2. Read each model value against its own band. The partial B4 result separates the captured control behavior from the incompletely reproduced positional effect.

Delivery Figure 10

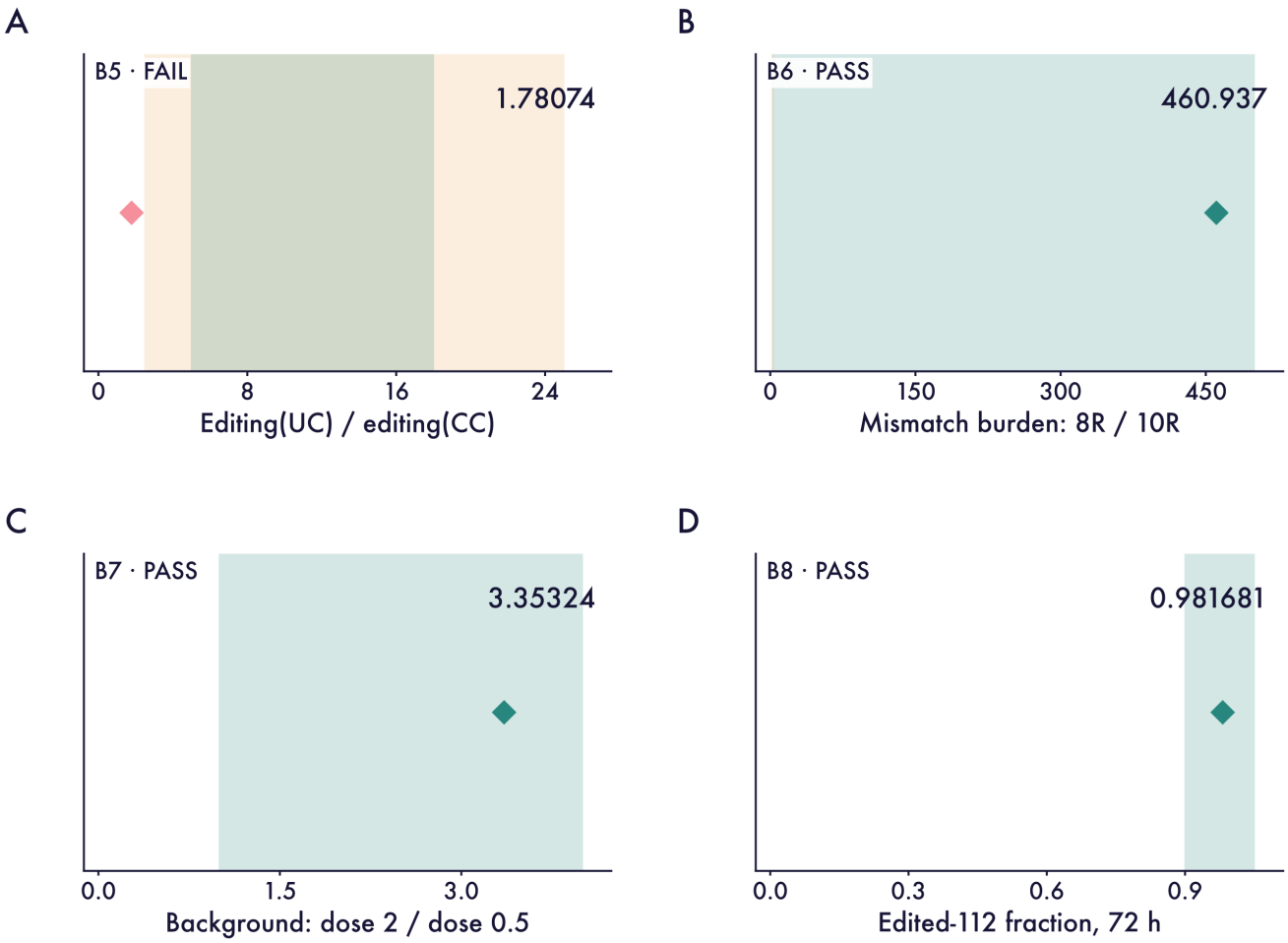


Figure 10. Benchmark checks B5–B8. The motif-ratio comparison fails its supplied criterion. B6 retains the full value, 460.937. B8 uses the original time-course criterion. B5 compares UC- and CC-context editing; B6 compares mismatched-PUF burden for 8- versus 10-repeat designs; B7 compares background at two doses; B8 checks editing at residue 112 over time. The scorecard exposes a specific mismatch in motif preference alongside the behaviors reproduced by the model.

Delivery Figure 11

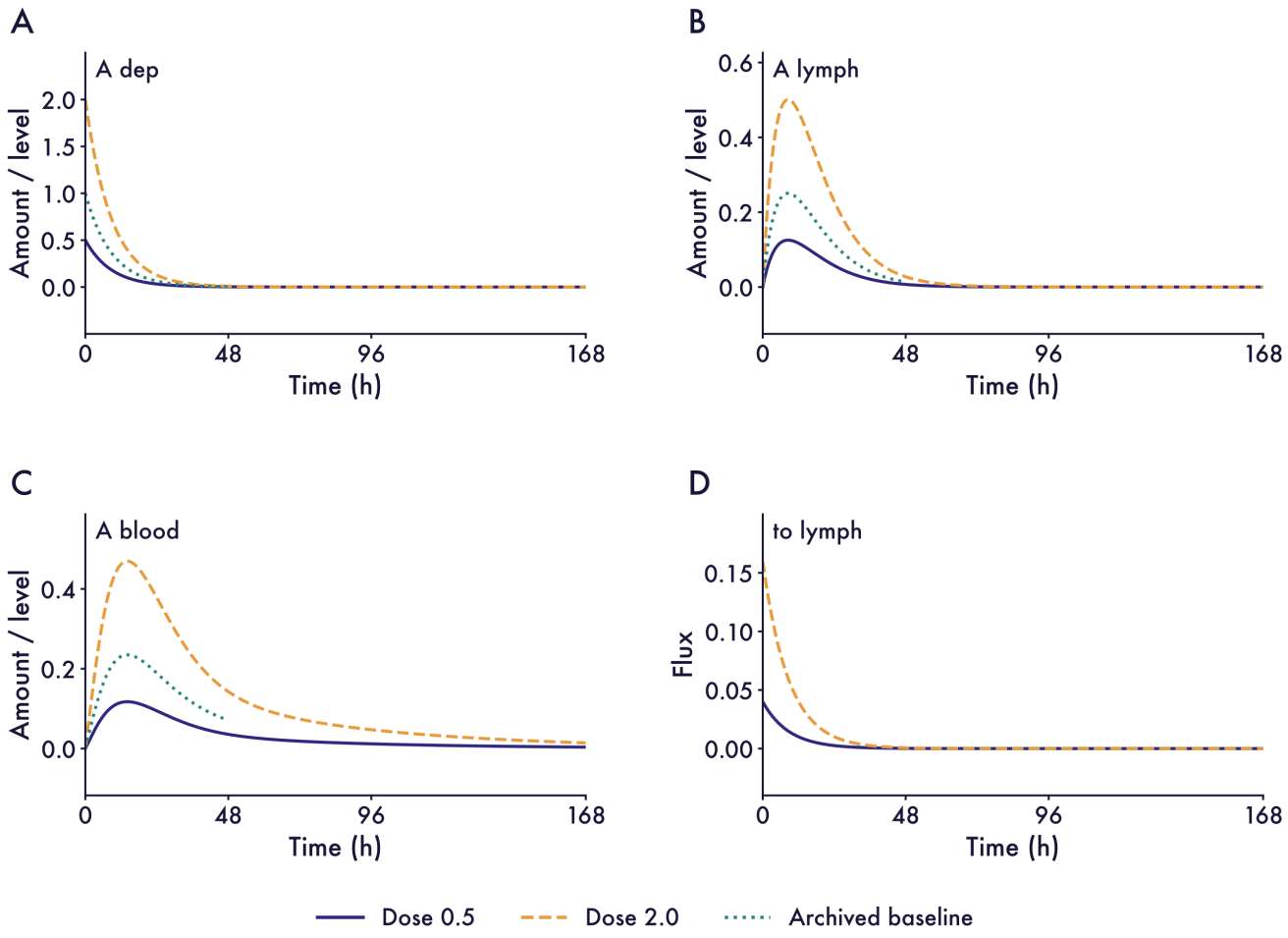


Figure 11. Absorption states and uptake. A–C: AAV amounts in the administration depot (A_dep), lymph (A_lymph) and blood (A_blood). D: depot-to-lymph transfer rate. Fluxes describe amounts transferred per unit time. Read A–C as stored material and D as flow: the depot supplies circulation through the lymphatic route, with timing and magnitude distinguished from the amount present.

Delivery Figure 12

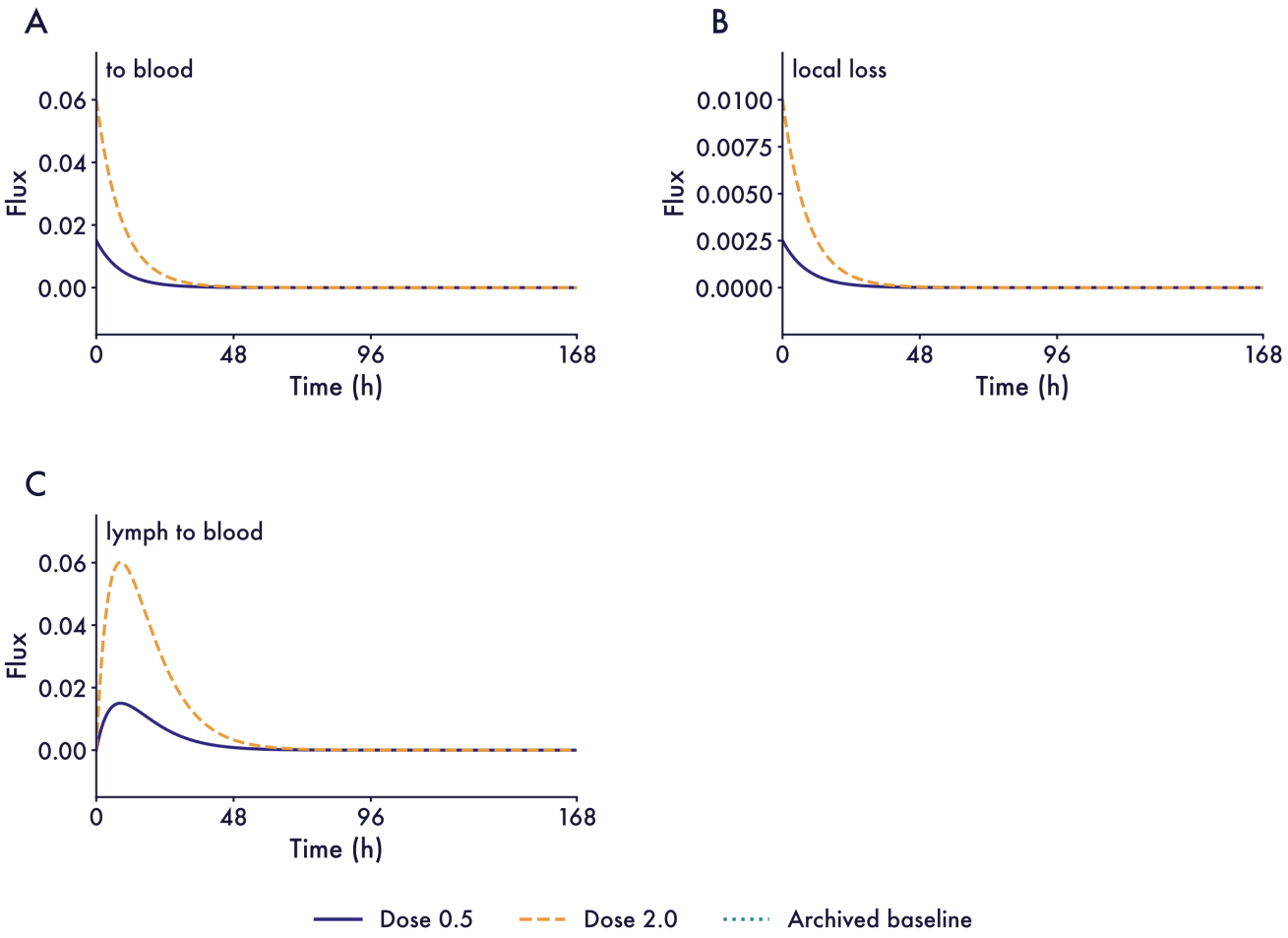


Figure 12. Absorption fluxes. A: direct transfer from depot to blood. B: local loss at the depot. C: lymph-to-blood transfer. Compare the uptake routes with local loss to see where material leaving the depot goes; leaving the depot does not imply that all of it reaches blood.

Delivery Figure 13

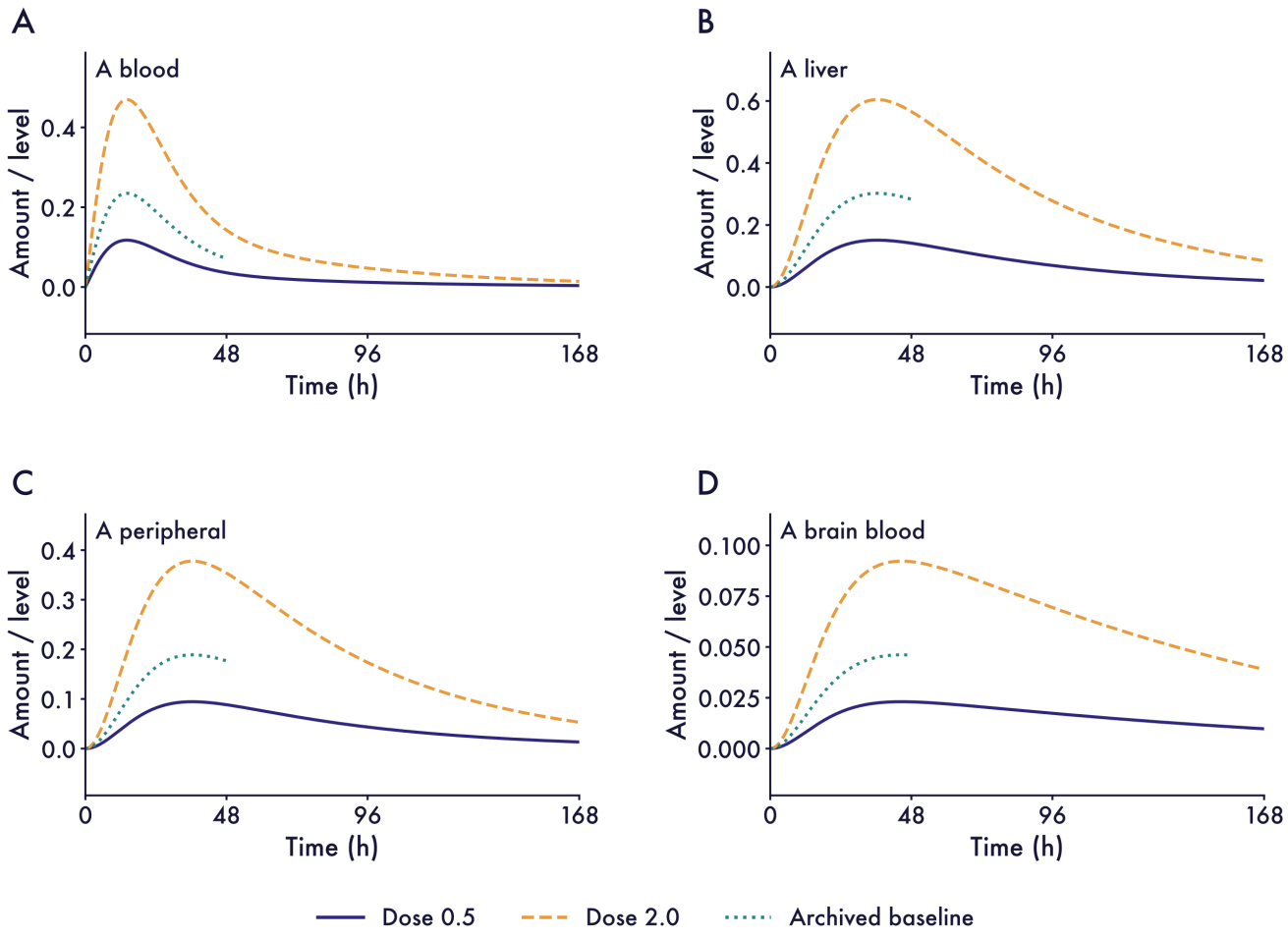


Figure 13. Distribution profiles. A–D: AAV amounts in blood, liver, peripheral tissue and brain circulation. A denotes the amount in each model compartment. Compare peak timing and persistence across compartments to follow redistribution. Brain-circulation material is upstream of barrier crossing and is distinct from delivery into brain cells.

Delivery Figure 14

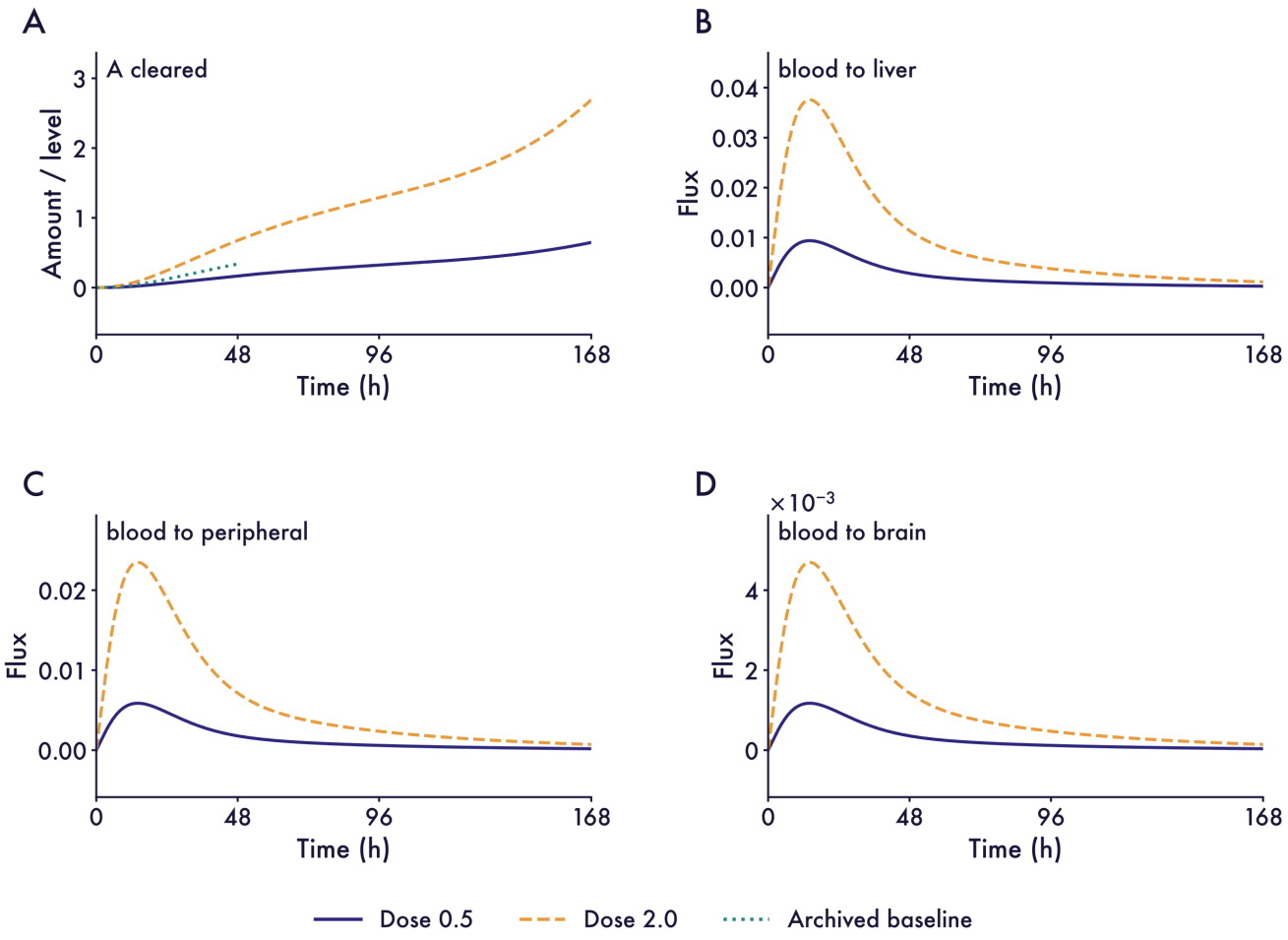


Figure 14. Distribution and clearance. A: cumulative cleared material. B–D: transfer rates from blood into liver, peripheral tissue and brain circulation. The cumulative cleared pool records material lost over time, while B–D show instantaneous transfer rates. Together these separate tissue uptake from material permanently removed from circulation.

Delivery Figure 15

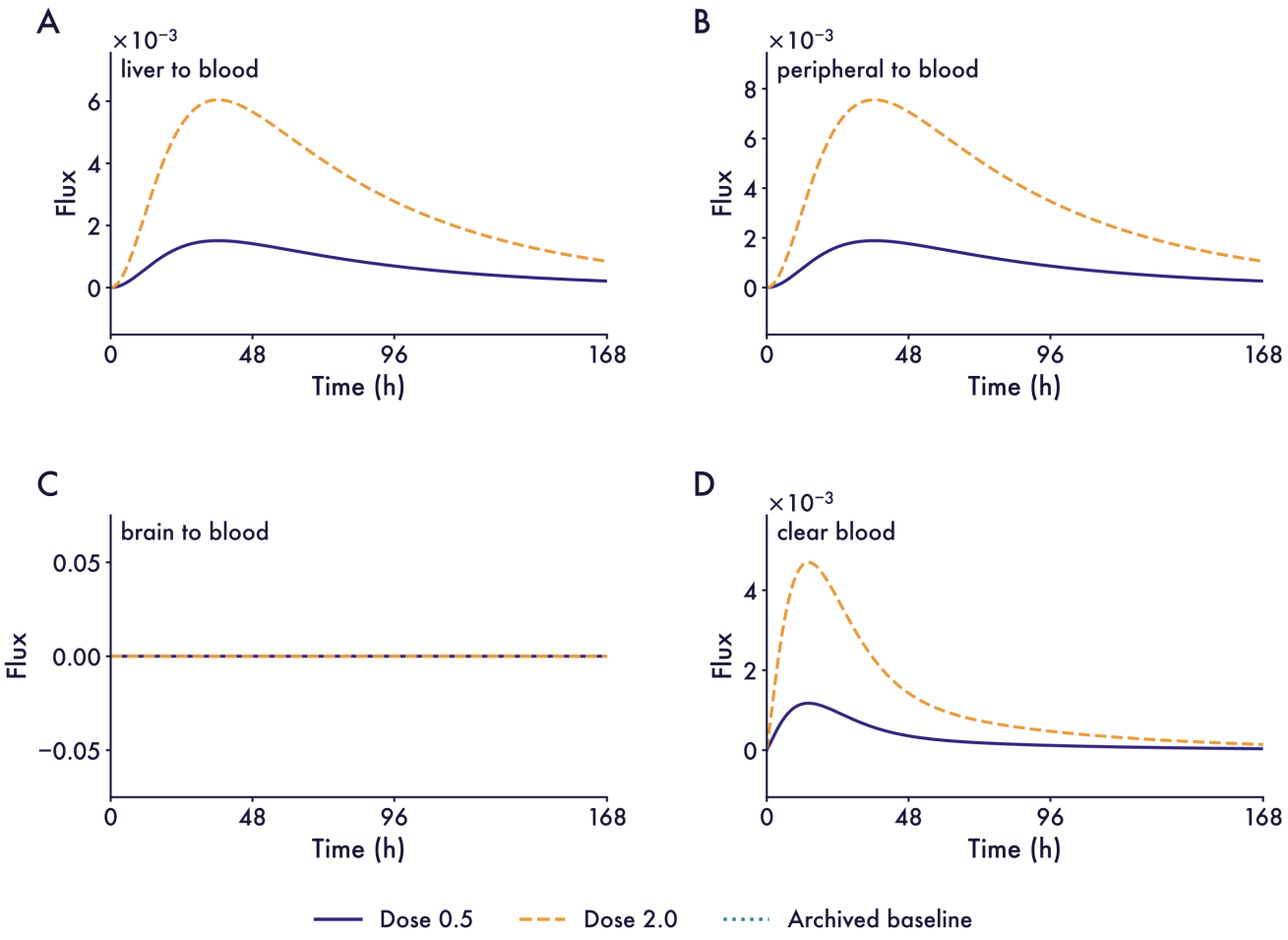


Figure 15. Return and clearance fluxes. A–C: return rates from liver, peripheral tissue and brain circulation to blood. D: clearance from blood. Return fluxes recycle material into circulation, whereas clearance removes it. These competing routes help explain the rise and fall of tissue amounts in Figure 13.

Delivery Figure 16

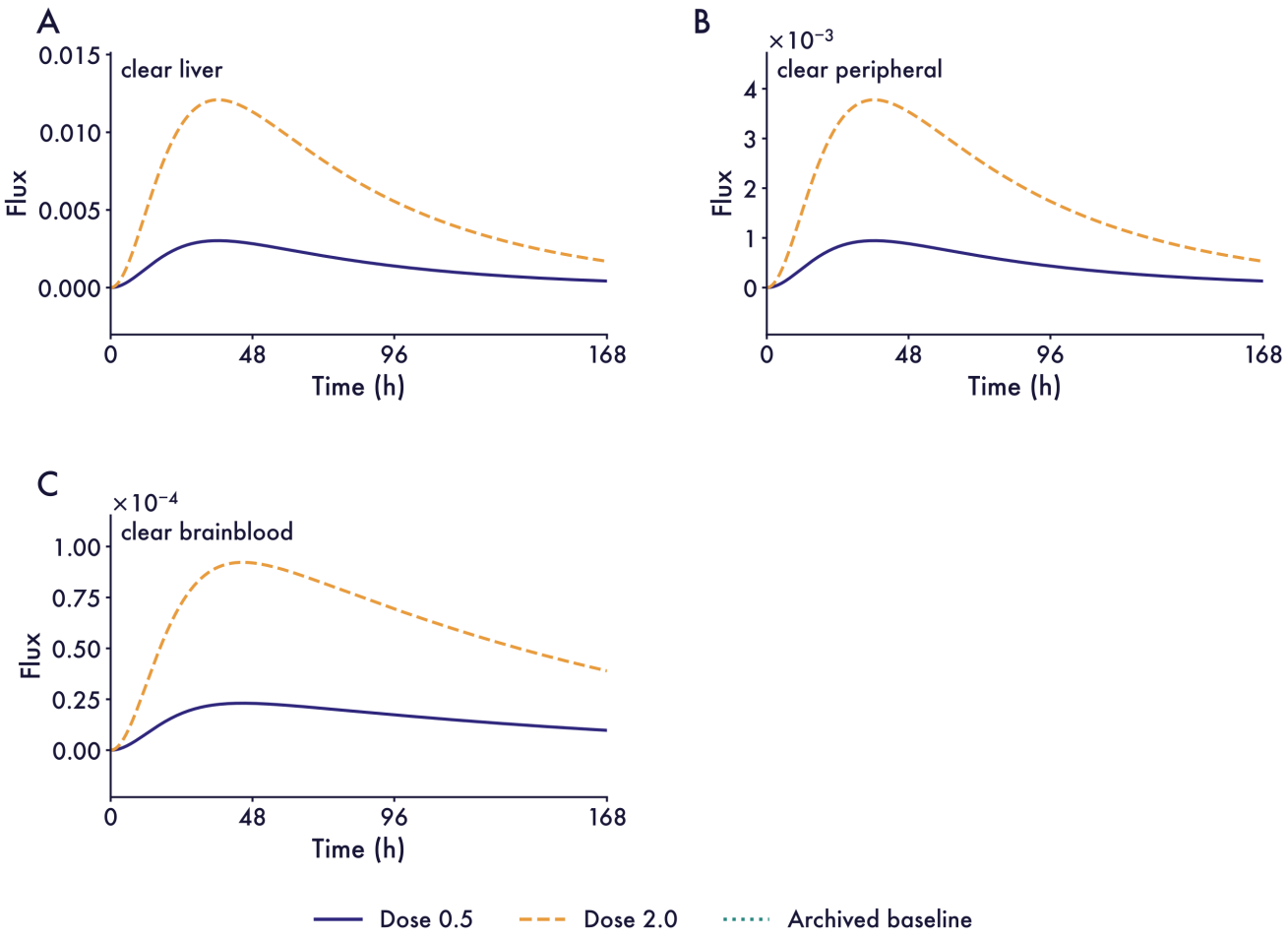


Figure 16. Clearance fluxes. A–C: clearance rates from liver, peripheral tissue and brain circulation. These loss routes contribute to the cumulative cleared pool in Figure 14; their values are rates, rather than amounts currently stored in each tissue.

Delivery Figure 17

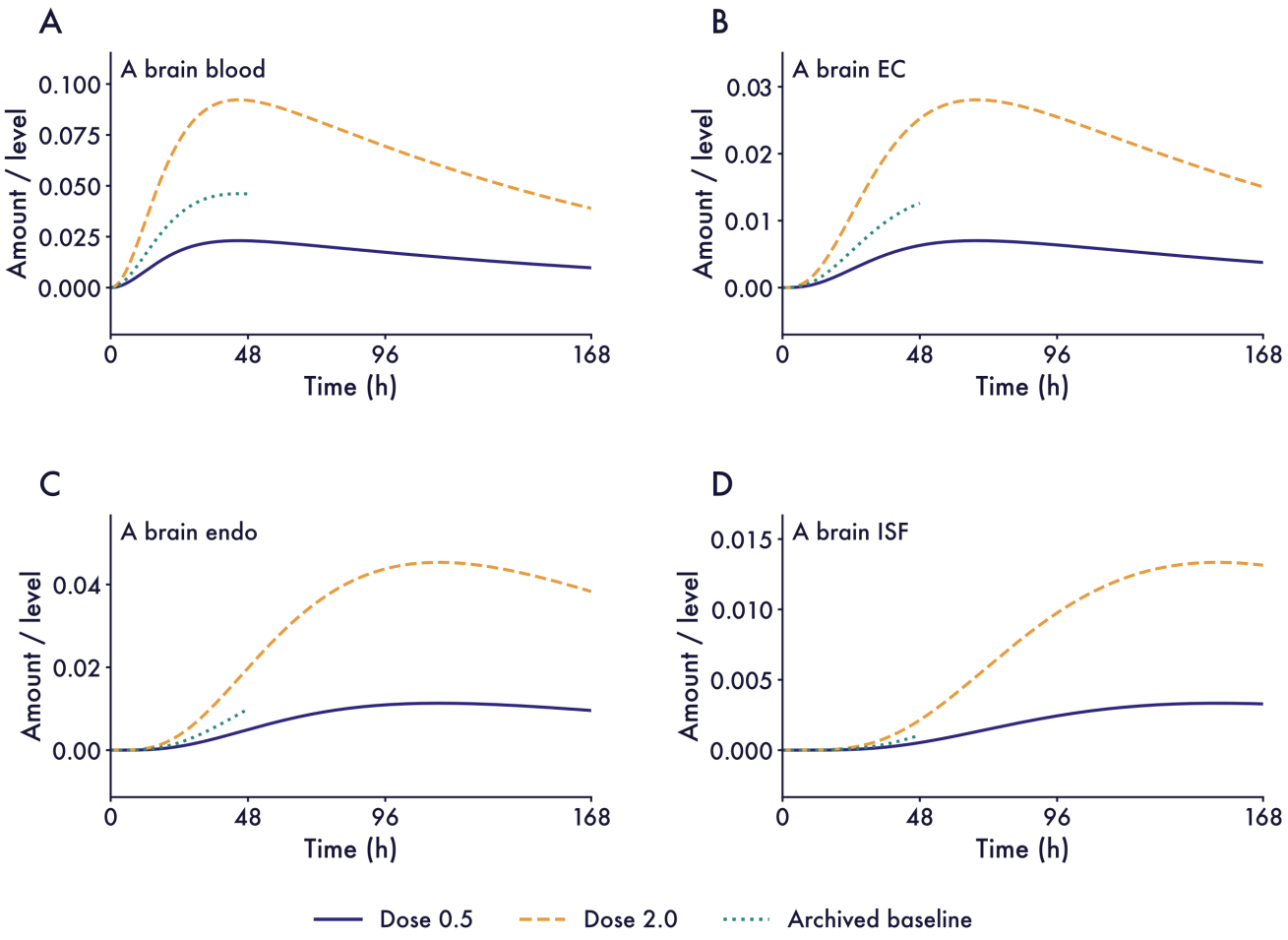


Figure 17. Blood–brain barrier states. A–D: AAV amounts in brain blood, the endothelial compartment (EC), endosomes (endo) and interstitial fluid (ISF), tracing blood–brain barrier transport. Progression through these compartments distinguishes arrival at brain vessels from release beyond the barrier. The ISF pool is the extracellular source available for subsequent brain-cell uptake.

Delivery Figure 18

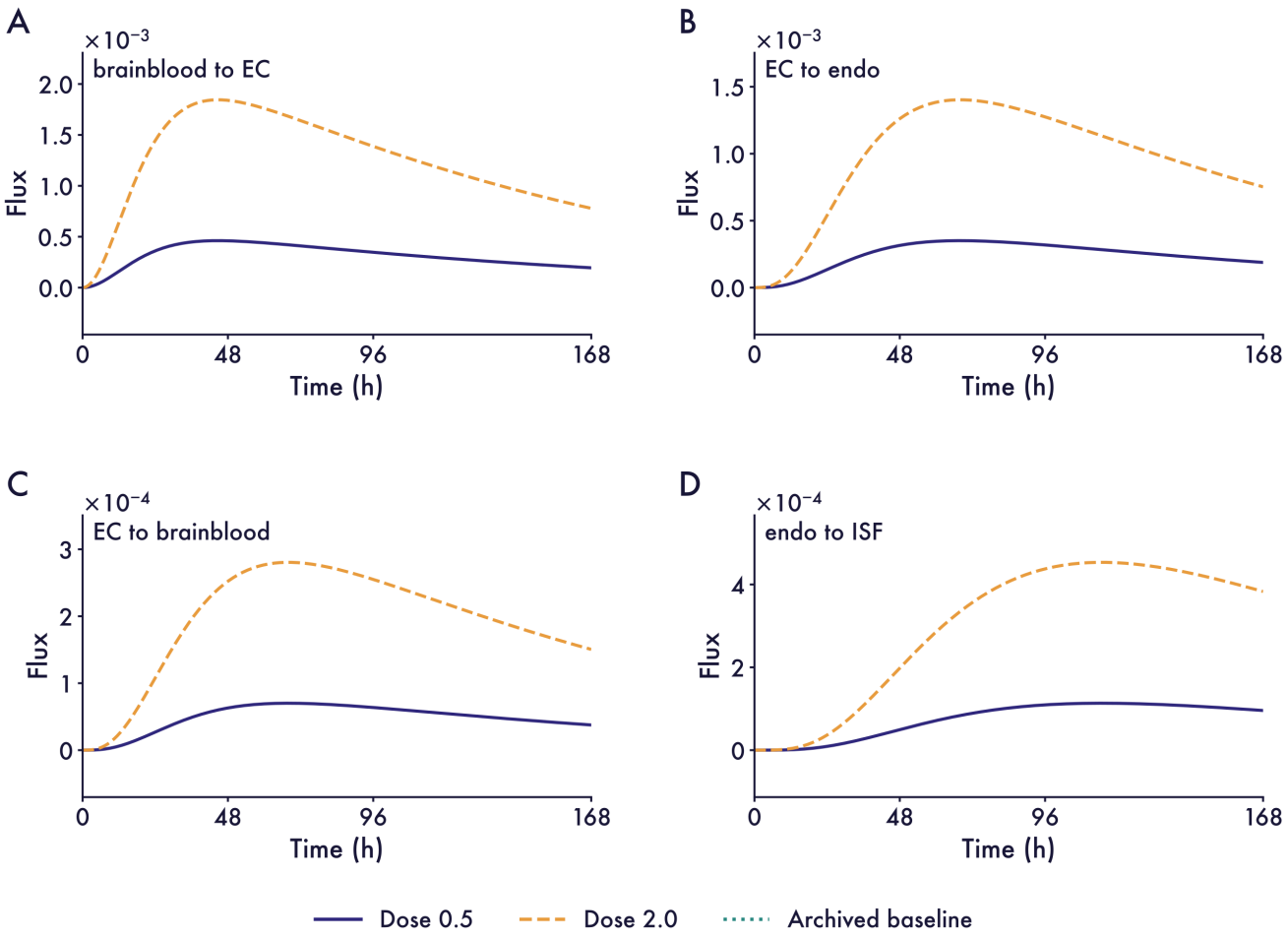


Figure 18. Blood–brain barrier transport. A: uptake from brain blood into the endothelial compartment. B: endosomal entry. C: return to blood. D: endosomal release into interstitial fluid. Compare forward transport with the return route: endothelial uptake can either progress toward tissue release or recycle to blood. Successful uptake is therefore only one step in the delivery chain.

Delivery Figure 19

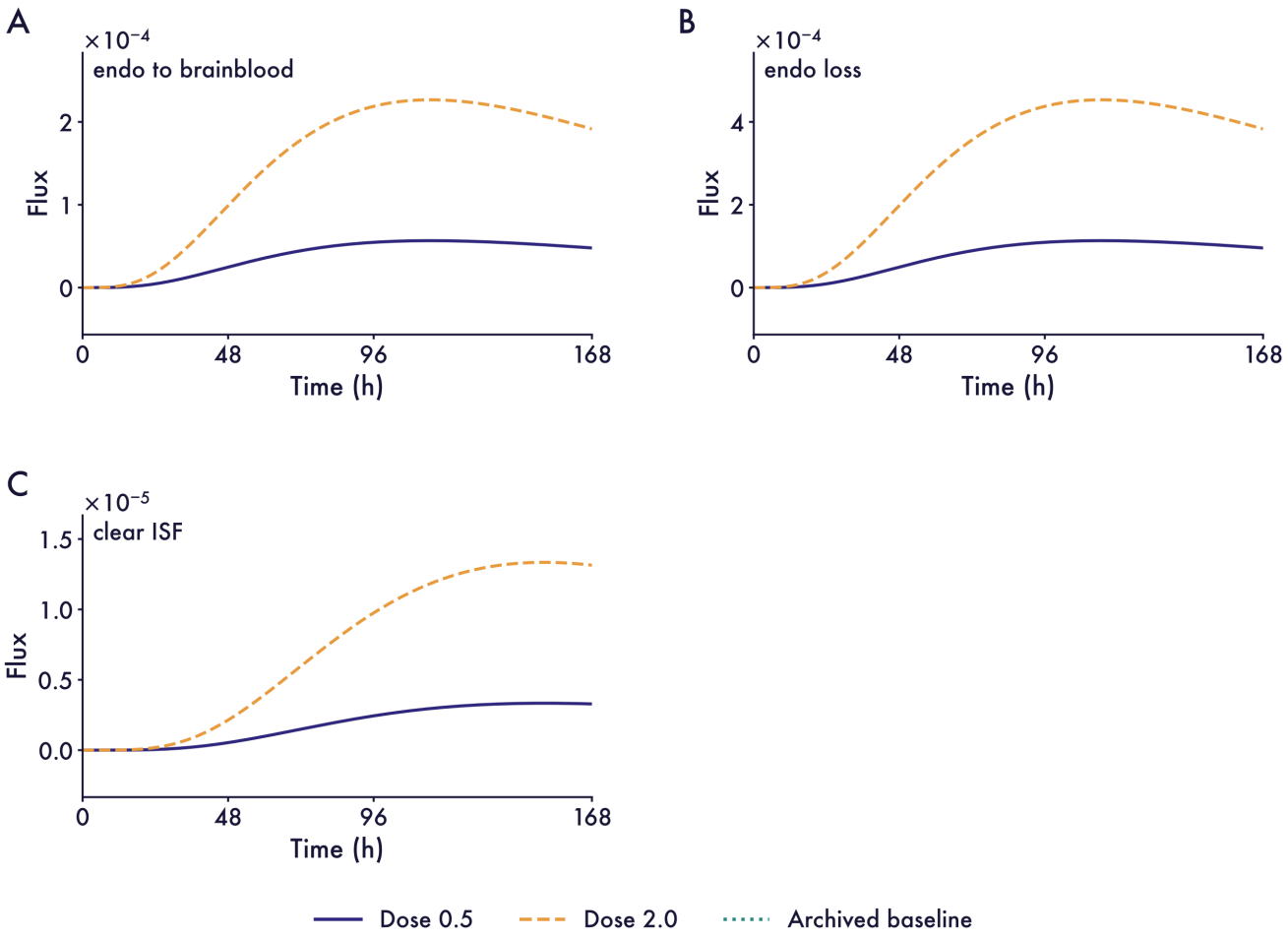


Figure 19. Blood–brain barrier return and loss. A: endosomal return to brain blood. B: endosomal loss. C: clearance from interstitial fluid (ISF). These branches account for material that does not remain available in tissue. Read them alongside forward release in Figure 18 to understand the competing fates at the barrier.

Delivery Figure 20

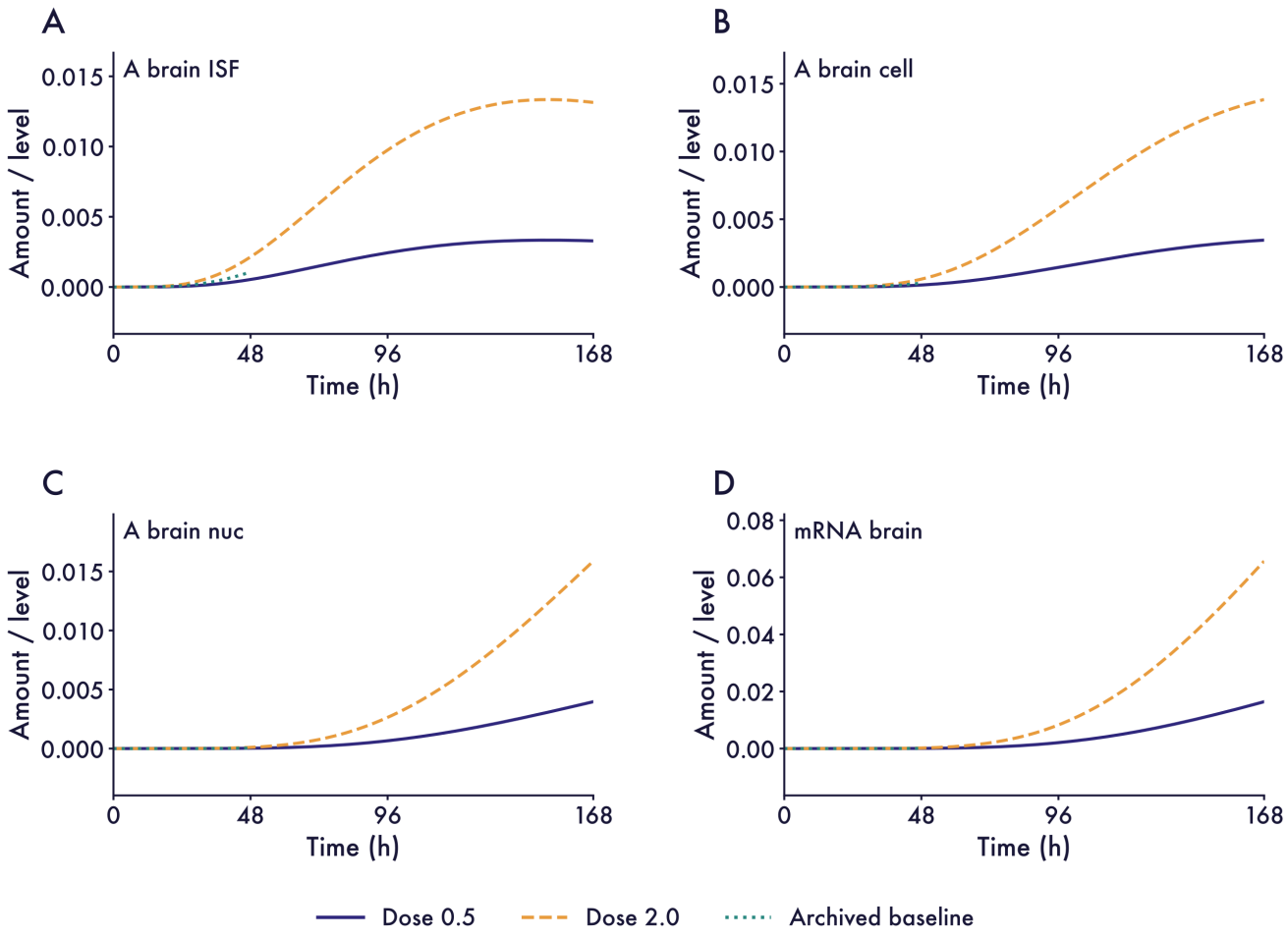


Figure 20. Cellular delivery states. A–C: AAV amounts in interstitial fluid, brain cells and nuclei. D: editor messenger RNA (mRNA_brain). The sequence connects extracellular availability to nuclear delivery and transcription. mRNA is an expression output, so its rise need not coincide with the peak vector amount.

Delivery Figure 21

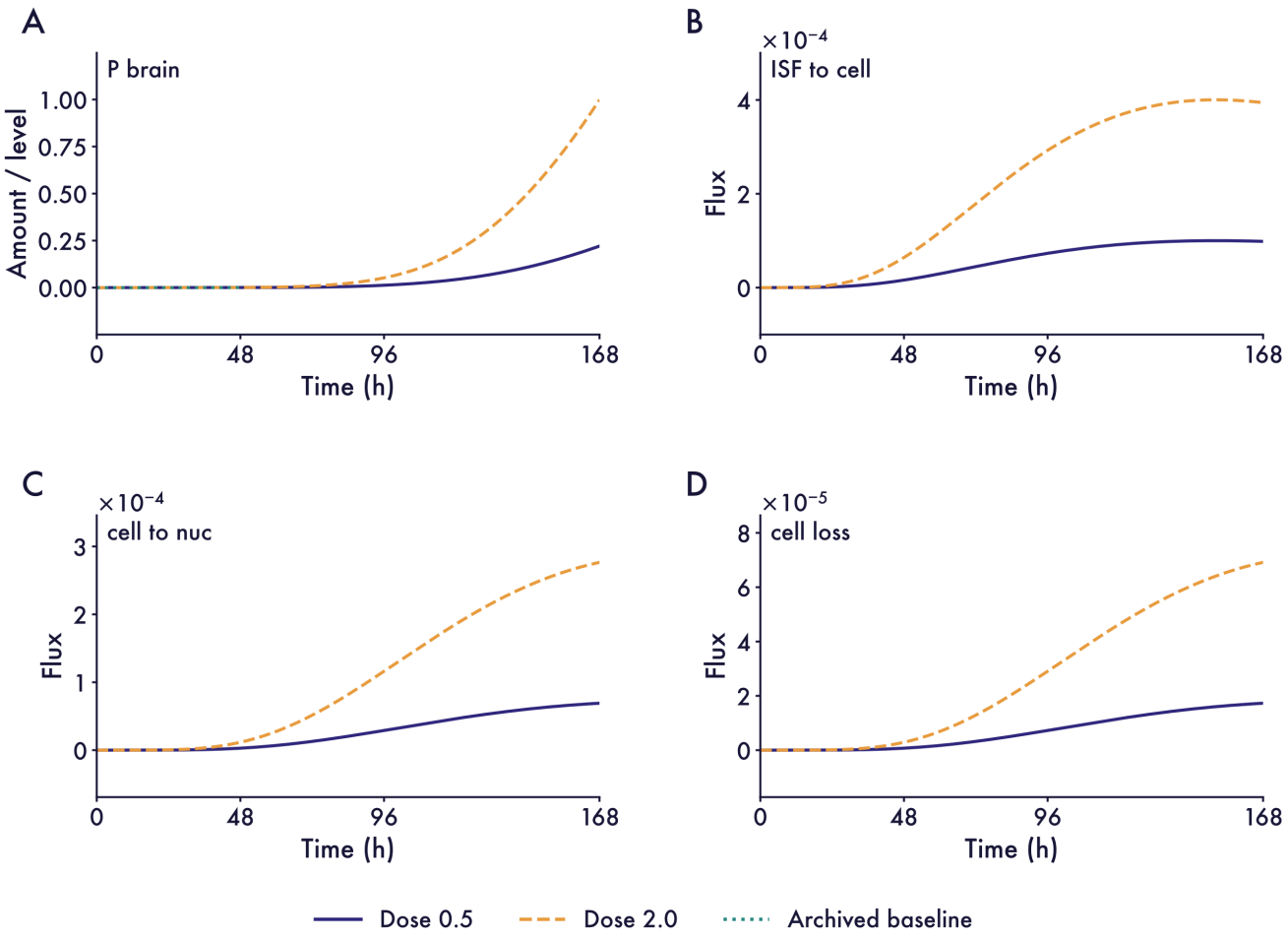


Figure 21. Expression and delivery fluxes. A: active editor protein in brain cells (P_brain). B: uptake from interstitial fluid. C: nuclear entry. D: intracellular vector loss. The protein trace is the editor supply used by the RNA-reaction model. Its timing reflects both upstream delivery and the expression/degradation processes detailed in Figures 22–23.

Delivery Figure 22

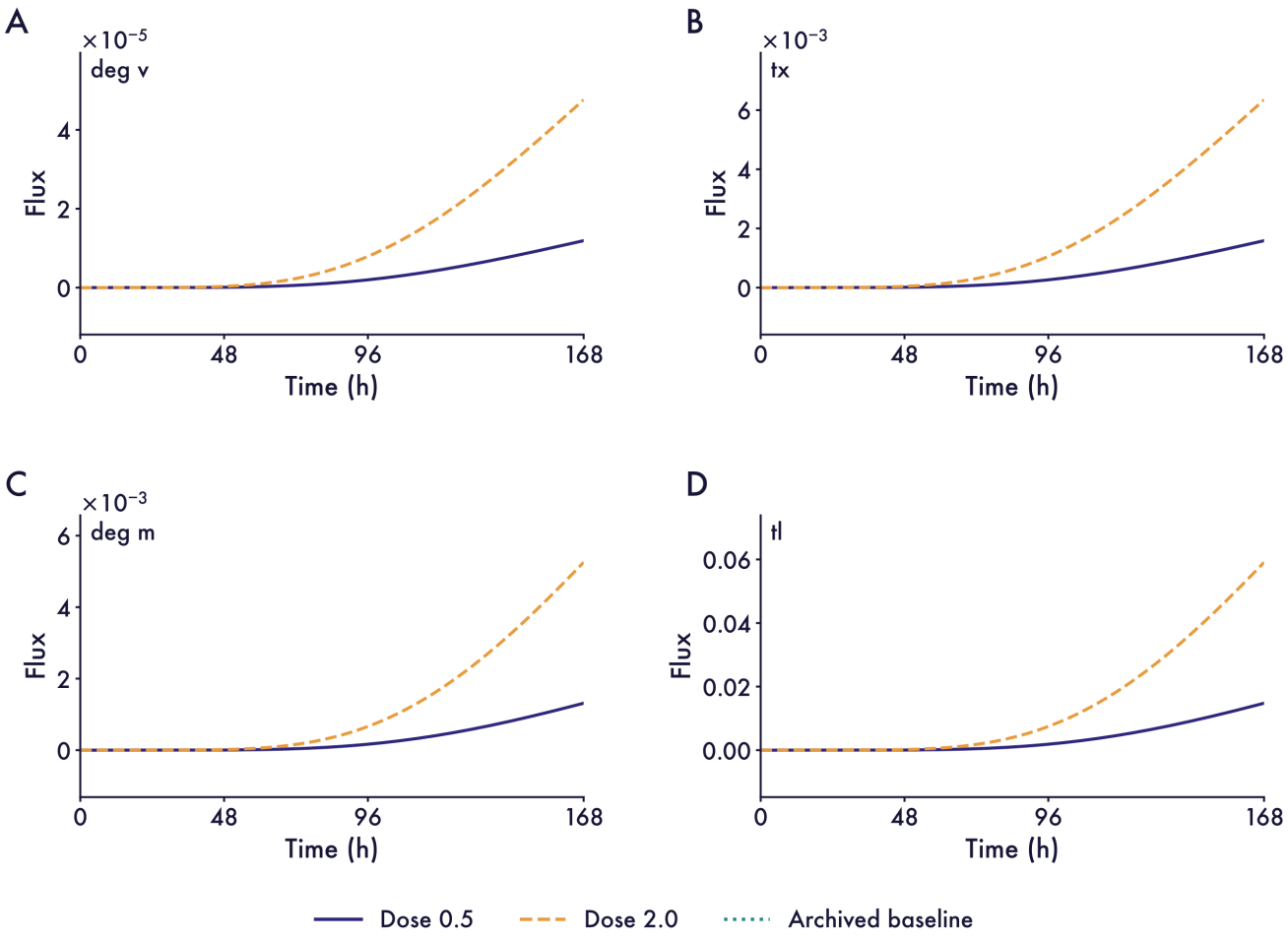


Figure 22. Expression and degradation fluxes. A: nuclear vector degradation (deg_v). B: transcription (tx). C: messenger-RNA degradation (deg_m). D: translation into editor protein (tl). Compare transcription with mRNA loss, and translation with protein loss in Figure 23. Their balance explains persistence of editor expression after the delivery input changes.

Delivery Figure 23

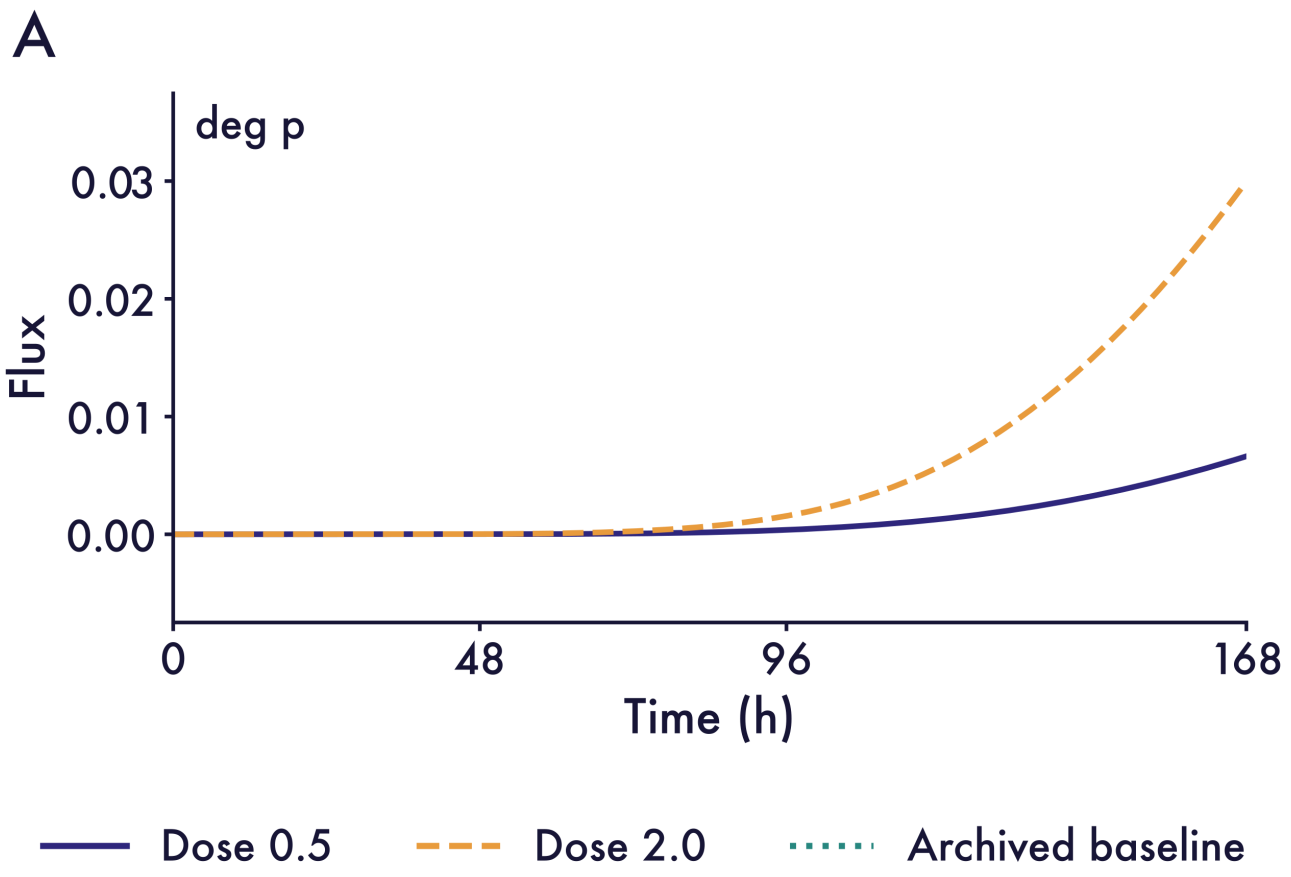


Figure 23. Protein degradation. Editor-protein degradation rate (deg_p), removing active protein from the brain-cell compartment. Read this loss together with translation in Figure 22: protein abundance reflects production minus removal, rather than the translation rate alone.

Delivery Figure 24

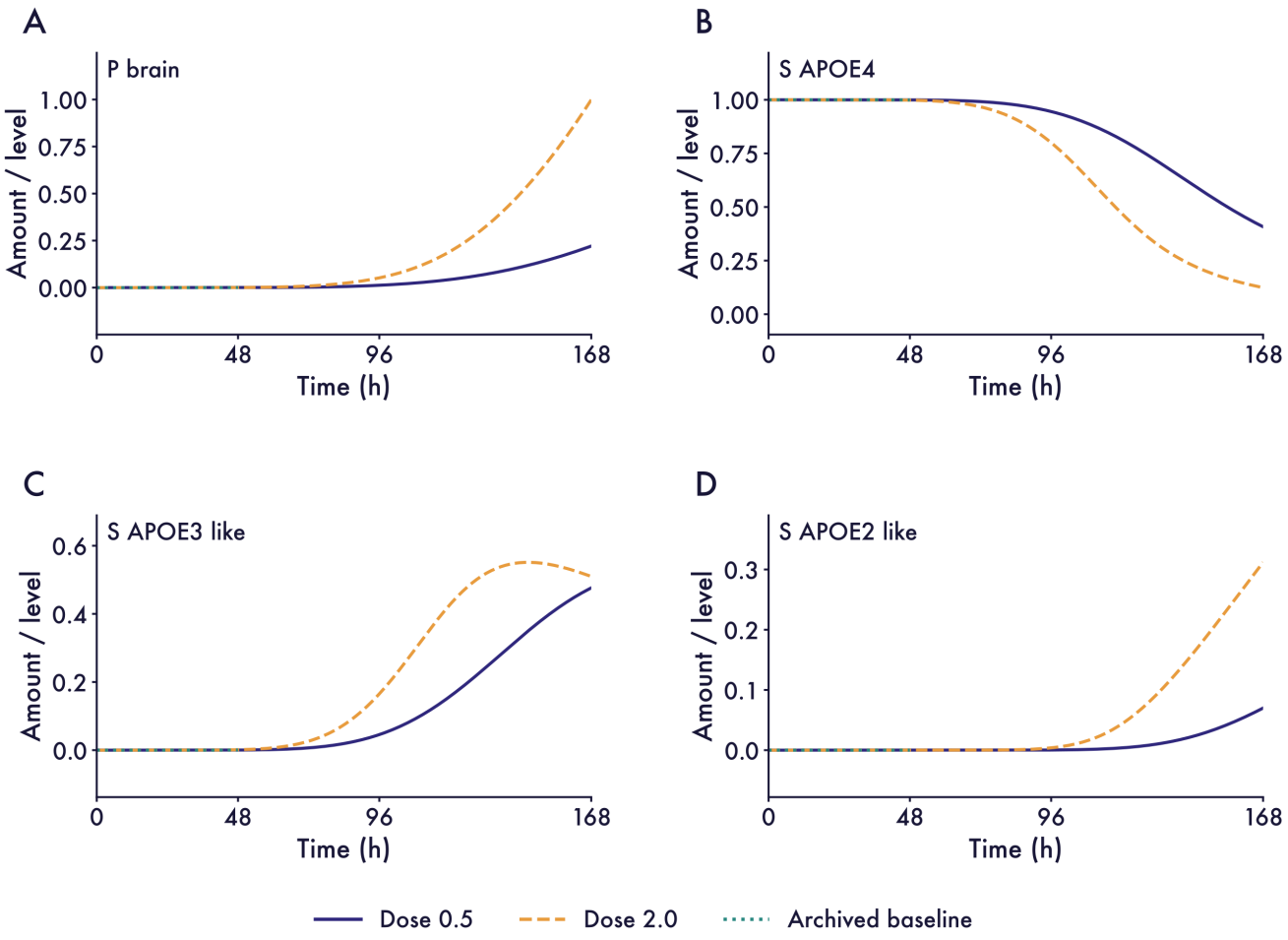


Figure 24. APOE editing states. A: active editor protein. B–D: APOE4, APOE3-like and APOE2-like RNA pools. The latter two correspond to editing at the residue-112 codon and at both residue-112/158 codons. Read the intended APOE3-like pool separately from APOE2-like accumulation. A decline in starting APOE4 RNA can feed either product, so total conversion alone does not specify the outcome.

Delivery Figure 25

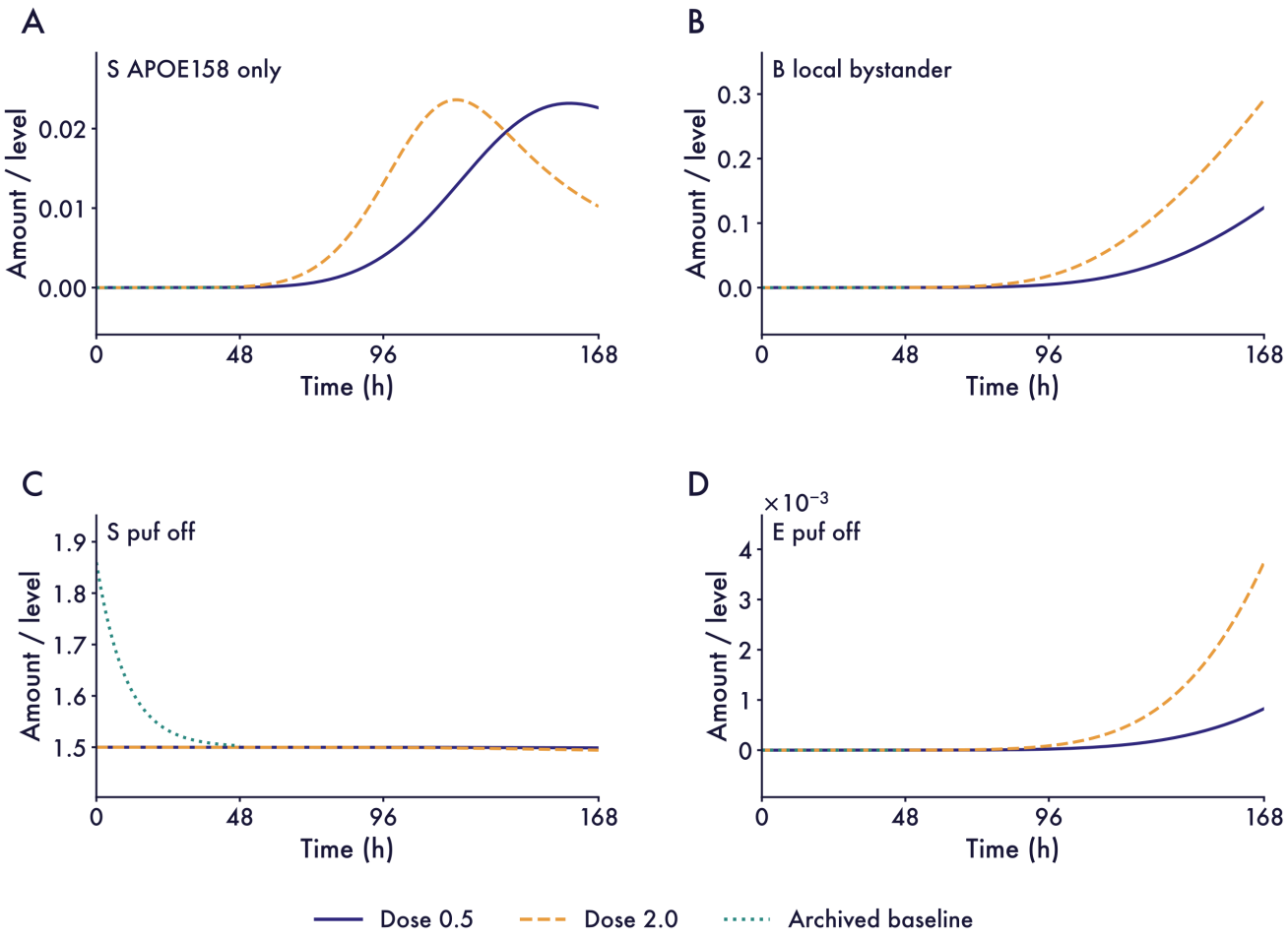


Figure 25. Bystander and background states. A: RNA edited only at the residue-158 codon. B: local bystander burden. C–D: unedited and edited RNA pools associated with mismatched PUF recognition. The panels separate editing of the other APOE site from local bystander events and mismatched-recognition editing. This identifies distinct contributors that would be hidden in one aggregate burden.

Delivery Figure 26

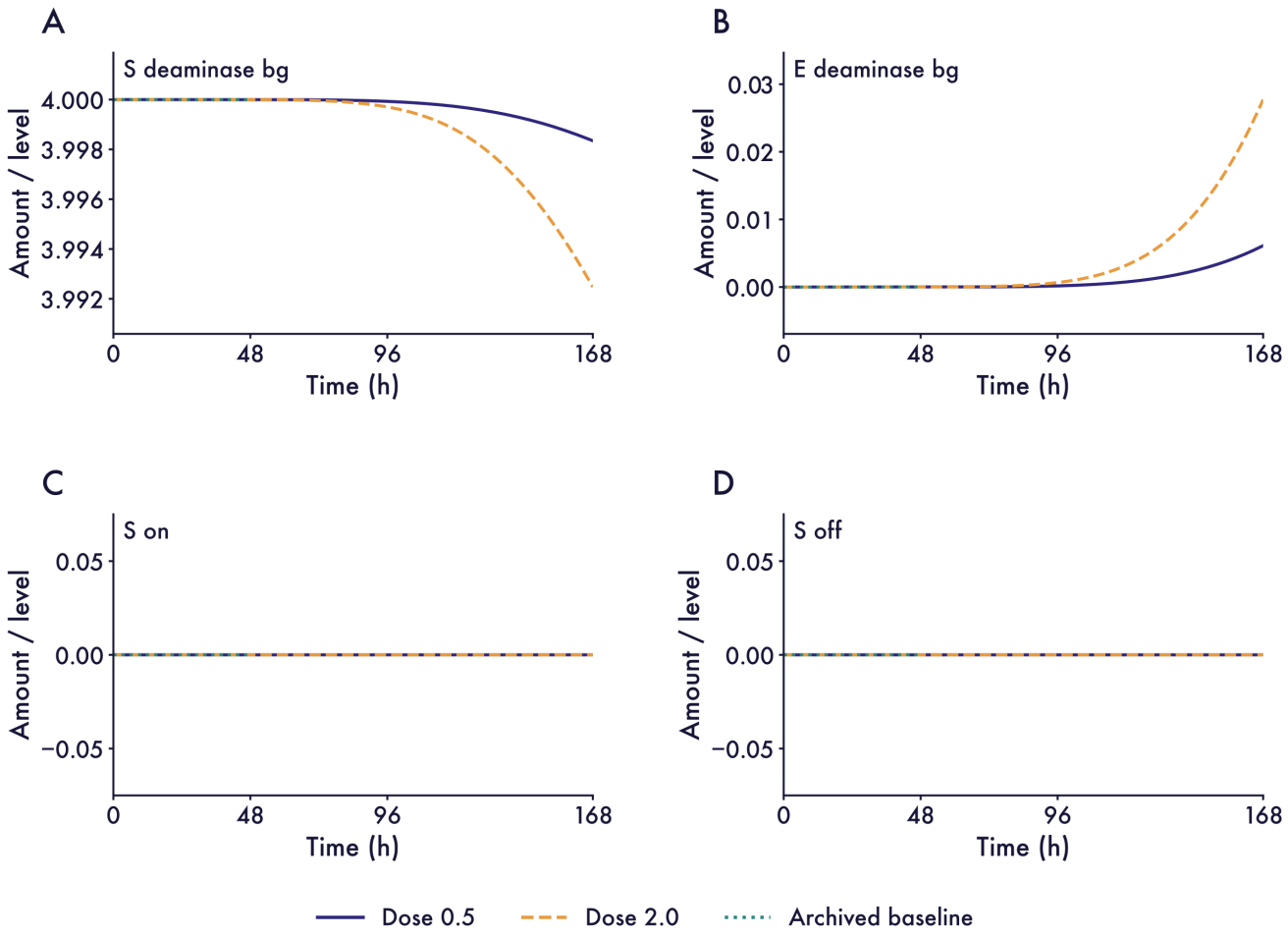


Figure 26. Substrate and complex states. A–B: unedited and edited RNA pools from deaminase-background activity. C–D: unedited on-target (S_on) and off-target (S_off) substrate pools. These pools supply separate reaction branches. Comparing remaining substrate with edited product helps distinguish substrate availability from accumulation of unwanted edits.

Delivery Figure 27

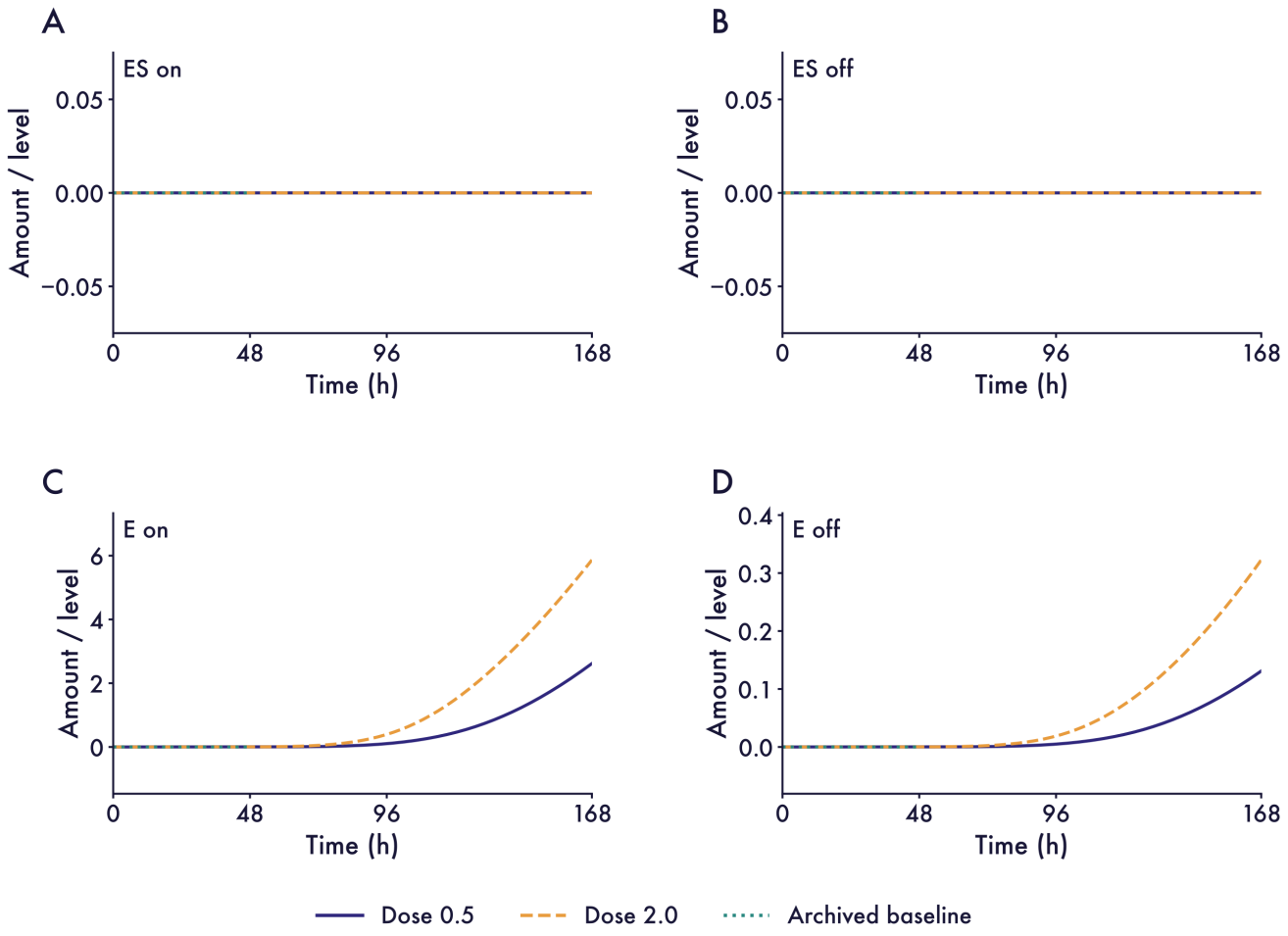


Figure 27. Complex and product states. A–B: editor–RNA complexes for on-target and off-target substrates (ES_on/ES_off). C–D: accumulated edited products (E_on/E_off); E here labels products, not free enzyme. Bound complexes are reaction intermediates; edited products accumulate after catalysis. Their different trajectories connect binding dynamics to the editing summary without treating binding itself as successful editing.

Delivery Figure 28

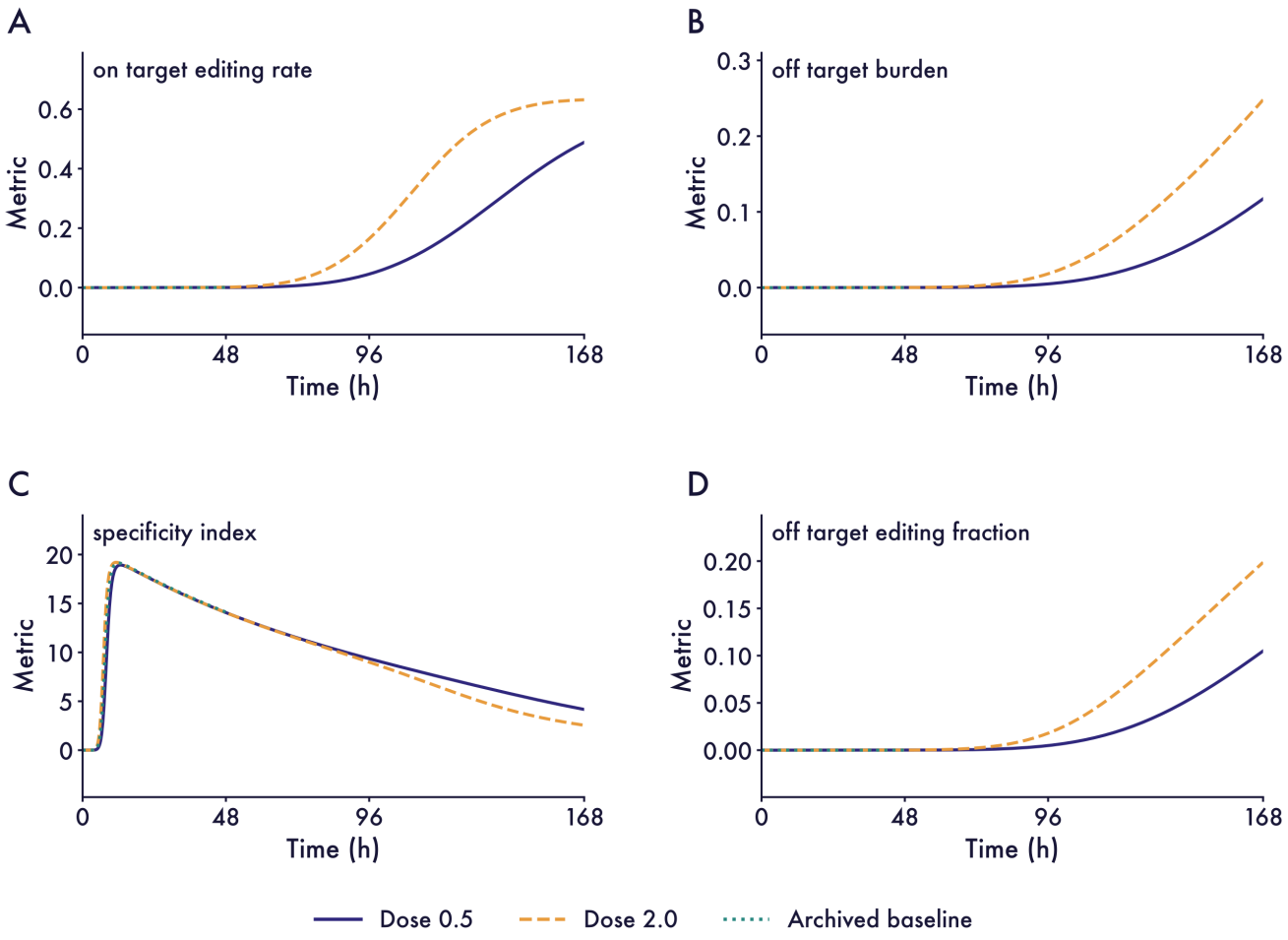


Figure 28. Editing efficiency and specificity. A: on-target editing rate. B: total unwanted-editing burden. C: model specificity index. D: off-target editing fraction. These readouts separate desired yield from unintended editing. Here on-target editing includes both APOE3-like and APOE2-like products. The specificity index divides that combined yield by accumulated unwanted-edit outputs. Read Figure 29 alongside these metrics to identify how much of the yield remains the intended APOE3-like product.

Delivery Figure 29

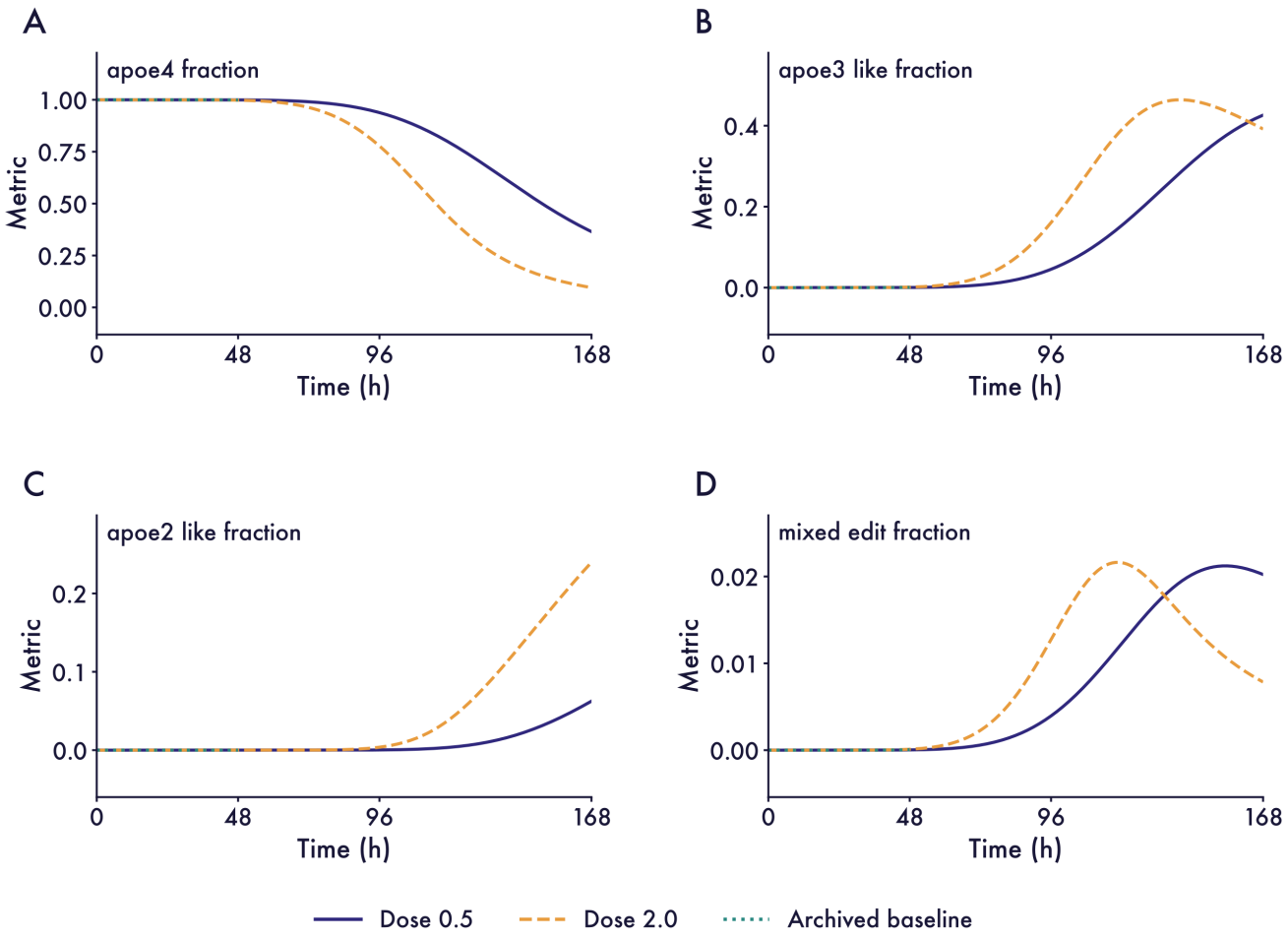


Figure 29. APOE product fractions. A–C: APOE4, APOE3-like and APOE2-like fractions of the RNA pool. D: the mixed-edit fraction, retained as defined in the source model. Compare the intended APOE3-like fraction with APOE2-like accumulation to distinguish the first desired edit from additional conversion. All fractions use the model’s total free and bound APOE RNA pool.

Delivery Figure 30

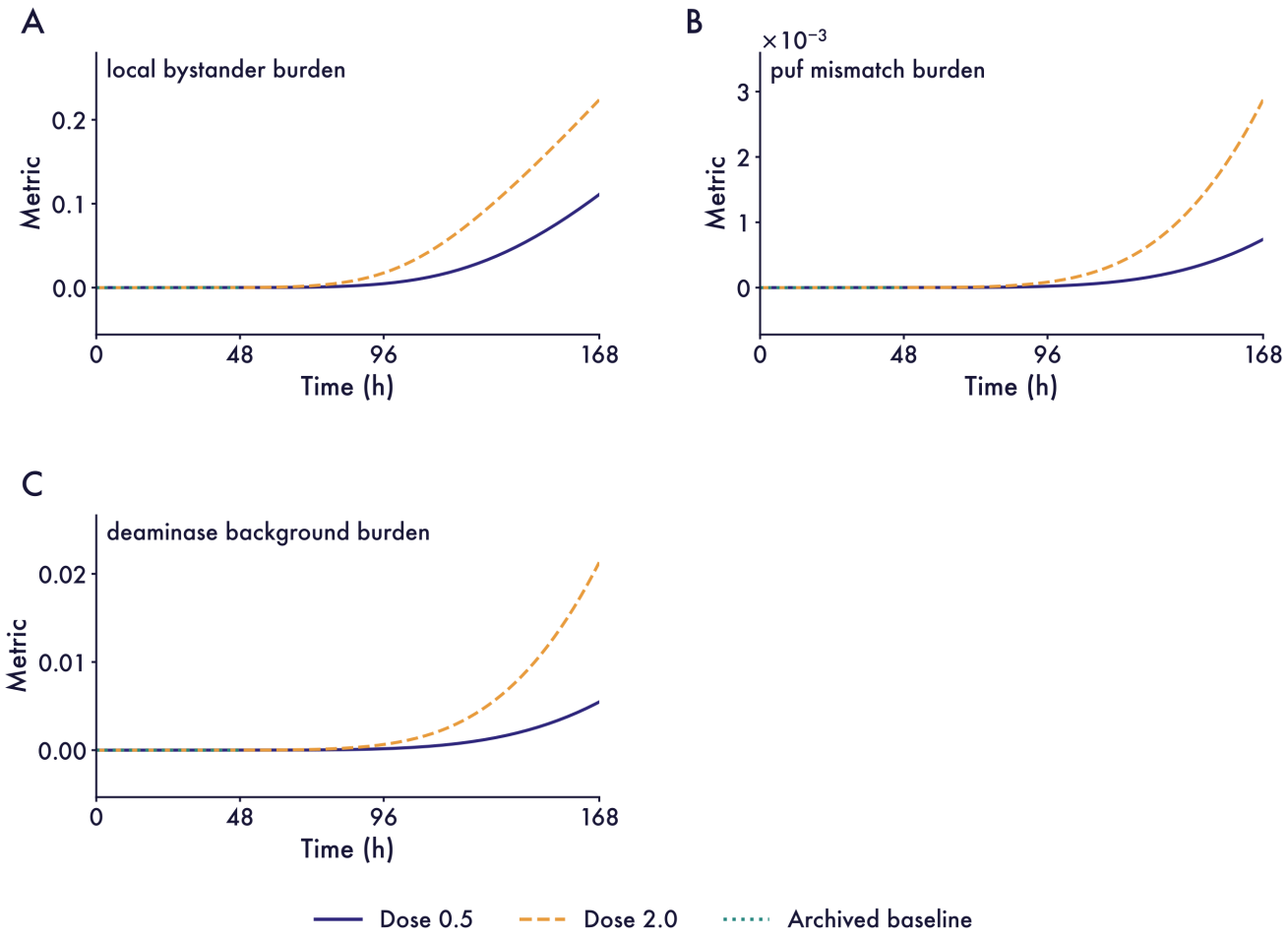


Figure 30. Off-target burden components. A–C: unwanted editing from local bystanders, mismatched PUF recognition and deaminase-background activity. The separate traces identify which unwanted-edit pathway contributes to the aggregate burden. They share the APOE-pool normalization used for the summary metric.

Delivery Figure 31

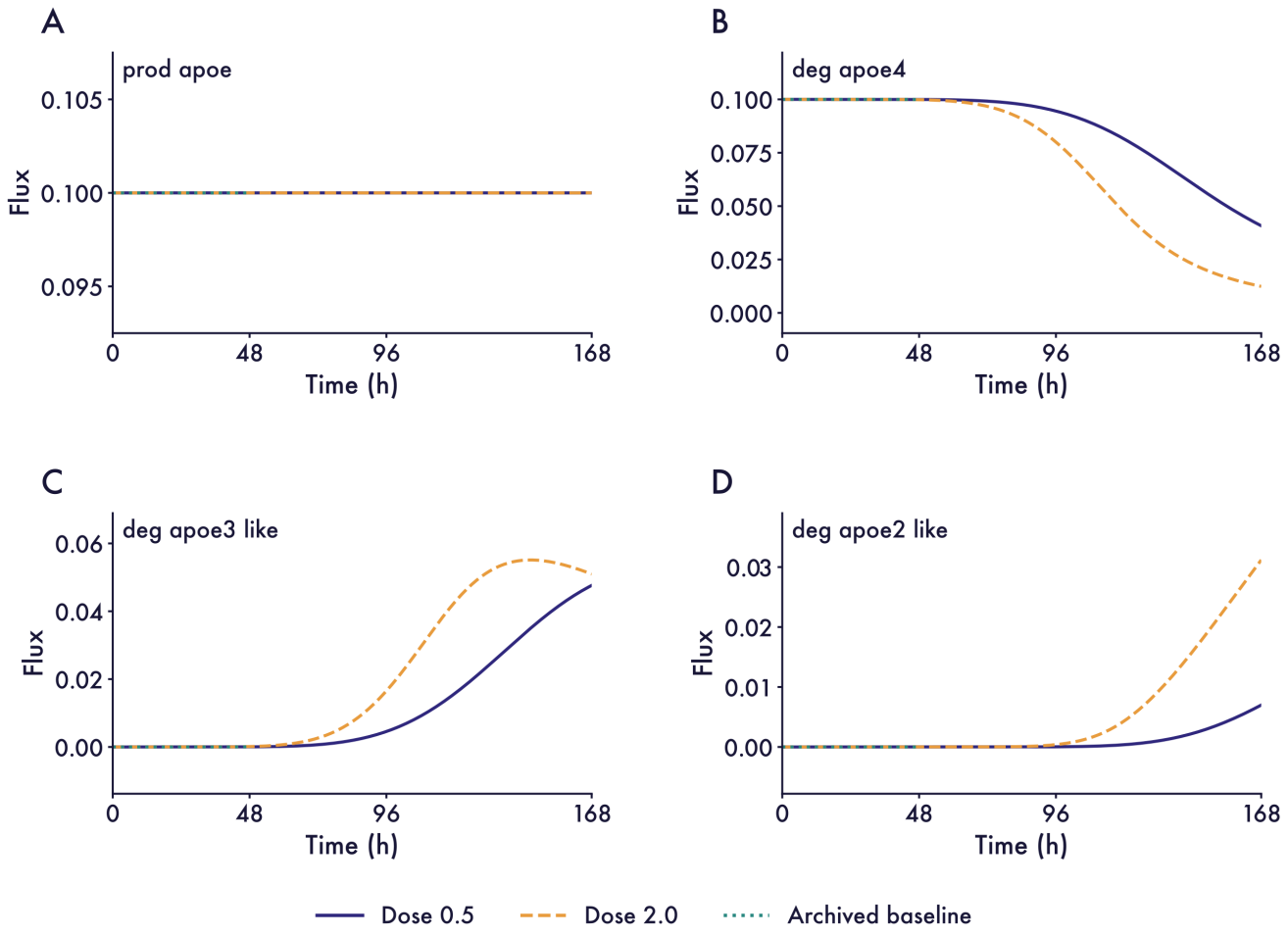


Figure 31. APOE production and degradation. A: production of APOE RNA. B–D: degradation of APOE4, APOE3-like and APOE2-like RNA states. Production replenishes substrate while degradation removes each RNA state. These turnover processes allow product amounts to change even when an editing flux is small.

Delivery Figure 32

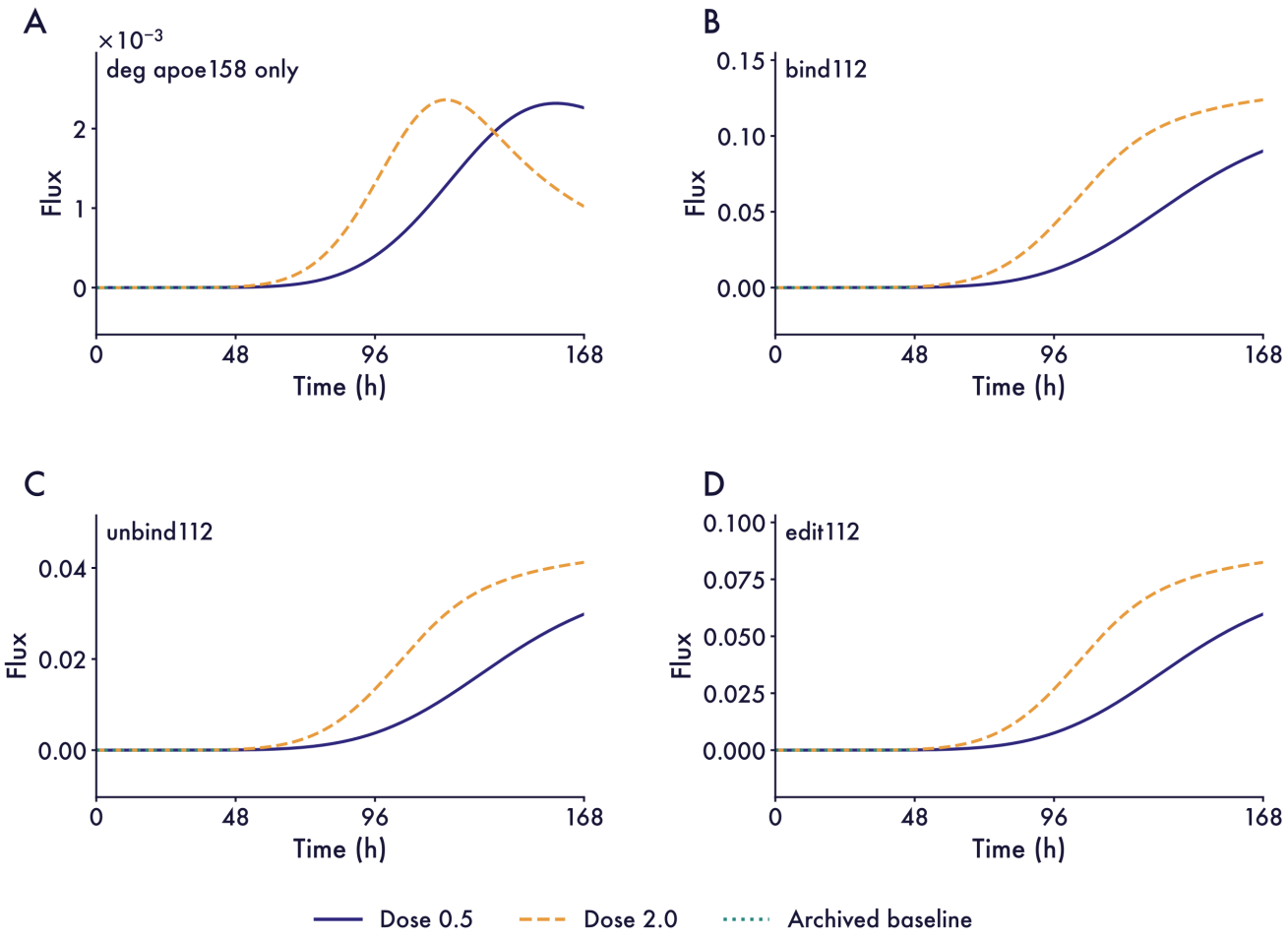


Figure 32. Site 112 editing fluxes. A: degradation of the 158-only RNA state. B–D: editor binding, dissociation and editing at the residue-112 codon. Labels 112/158 refer to APOE amino-acid positions, not RNA nucleotide indices. Binding creates the reaction complex, dissociation returns RNA without editing, and catalysis generates the edited state. Comparing the three fluxes separates encounter frequency from completed conversion.

Delivery Figure 33

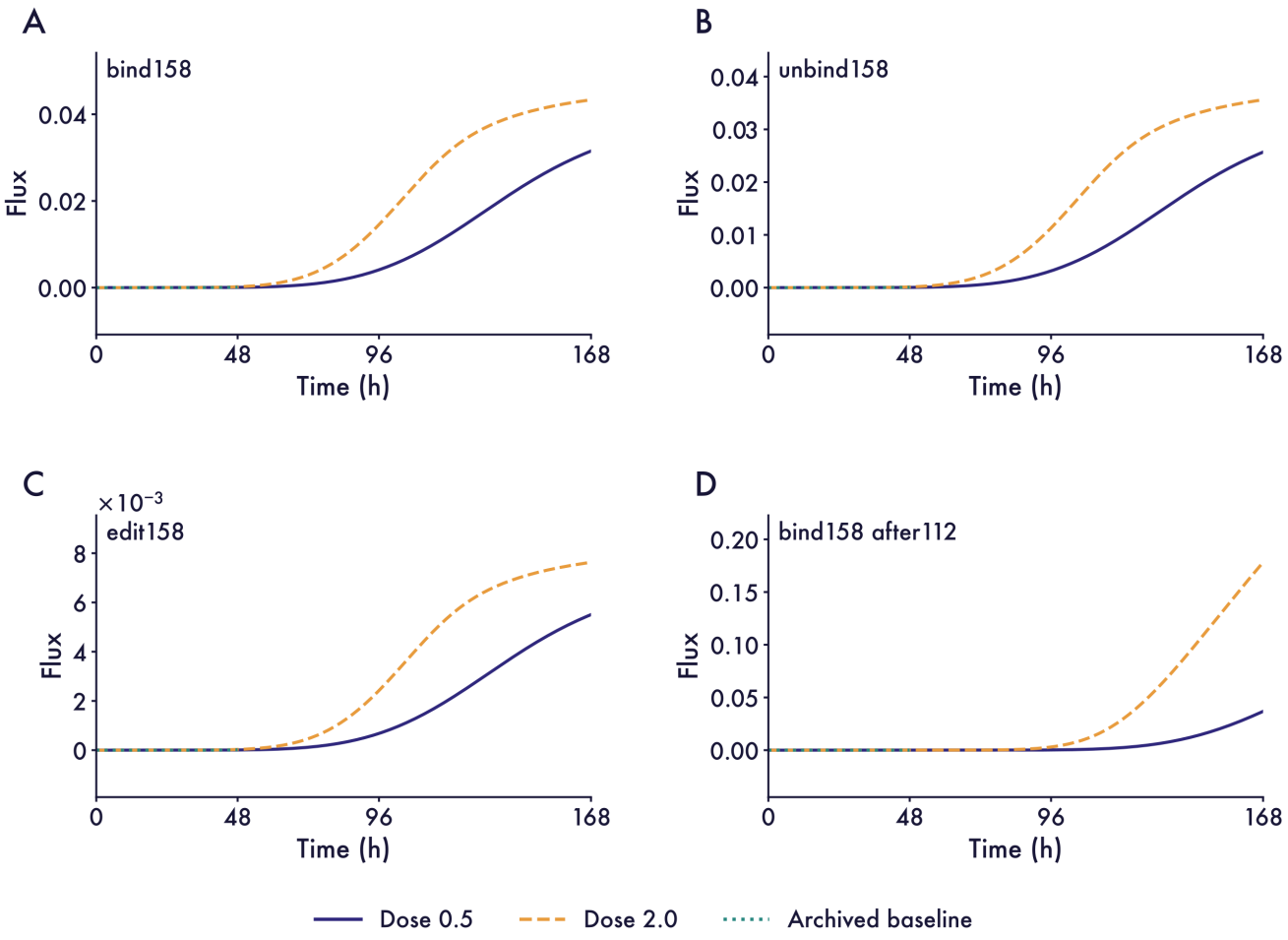


Figure 33. Site 158 editing fluxes. A–C: binding, dissociation and editing at the residue-158 codon. D: binding at 158 after the residue-112 codon has been edited. The final panel starts the second-edit route from the already edited APOE3-like state, linking continued editor exposure to loss of that intended product.

Delivery Figure 34

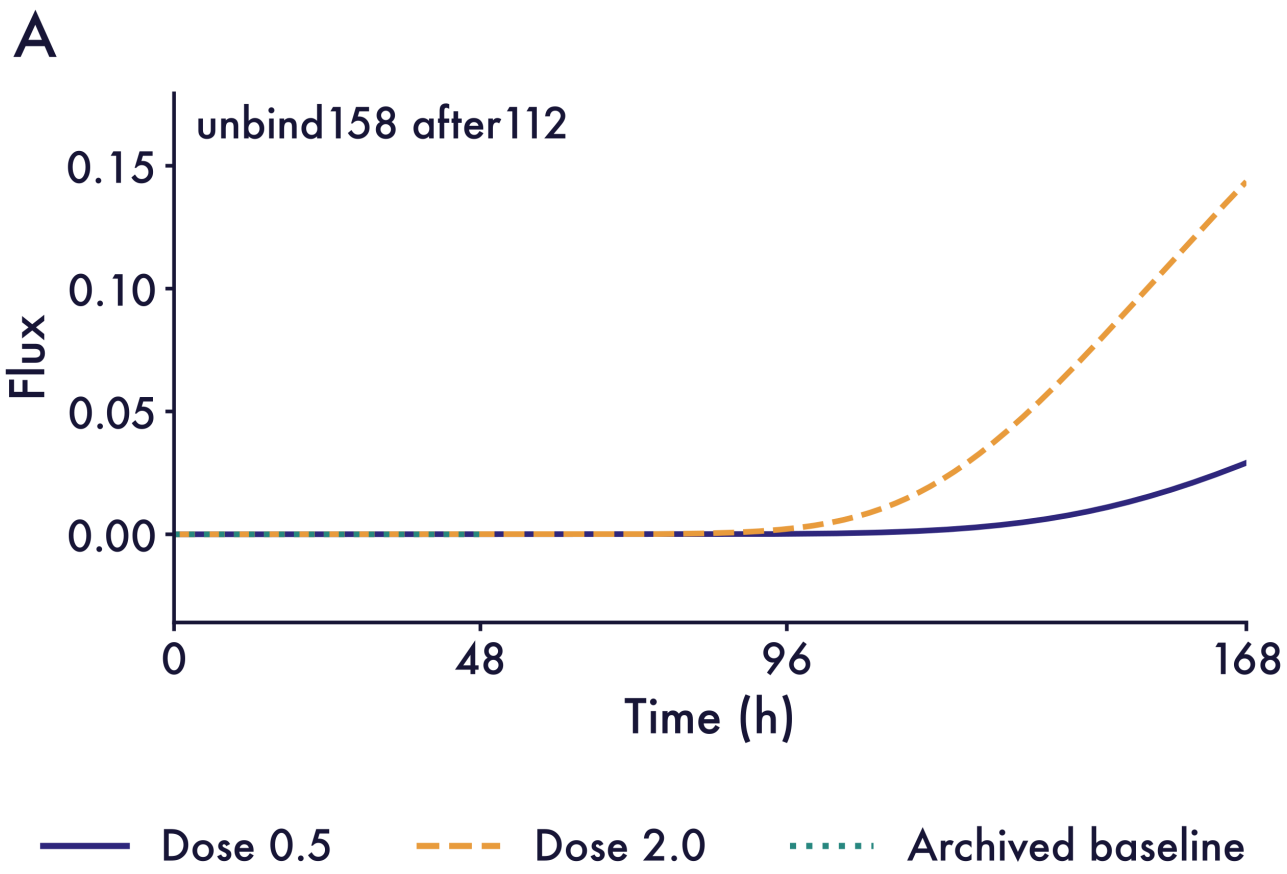


Figure 34. Sequential editing flux. Editor dissociation from the residue-158 site after editing at the residue-112 codon (unbind158_after112). This route competes with the second catalytic edit: dissociation releases the bound RNA without adding the residue-158 change. The corresponding editing flux appears in Figure 35A.

Delivery Figure 35

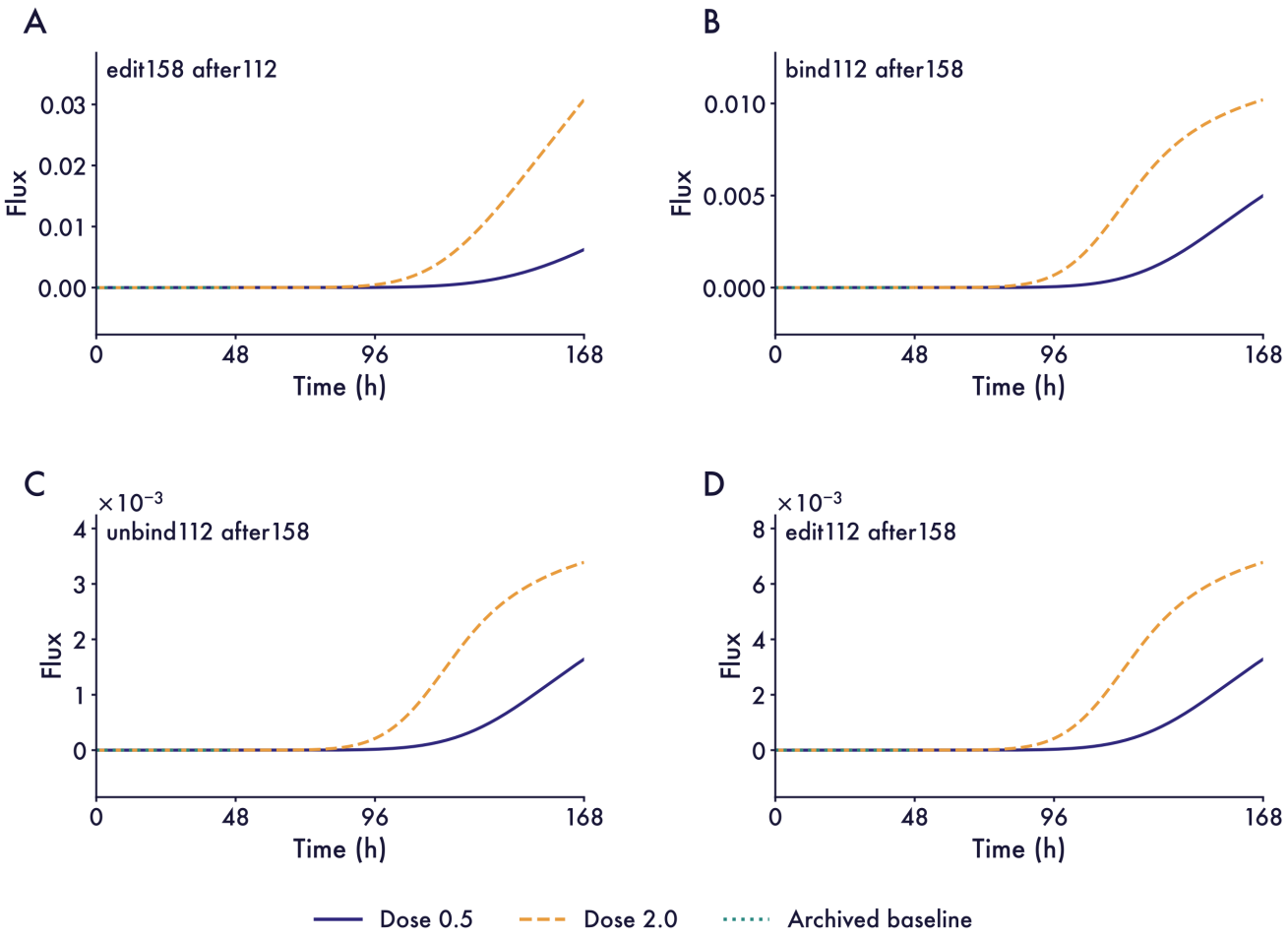


Figure 35. Sequential editing fluxes. A: editing at 158 after editing at 112. B–D: binding, dissociation and editing at 112 after editing at 158. These fluxes distinguish the two editing orders. Both orders can reach the doubly edited APOE2-like state. The separate traces reveal whether that product arises through consumption of APOE3-like RNA or through the 158-only intermediate.

Delivery Figure 36

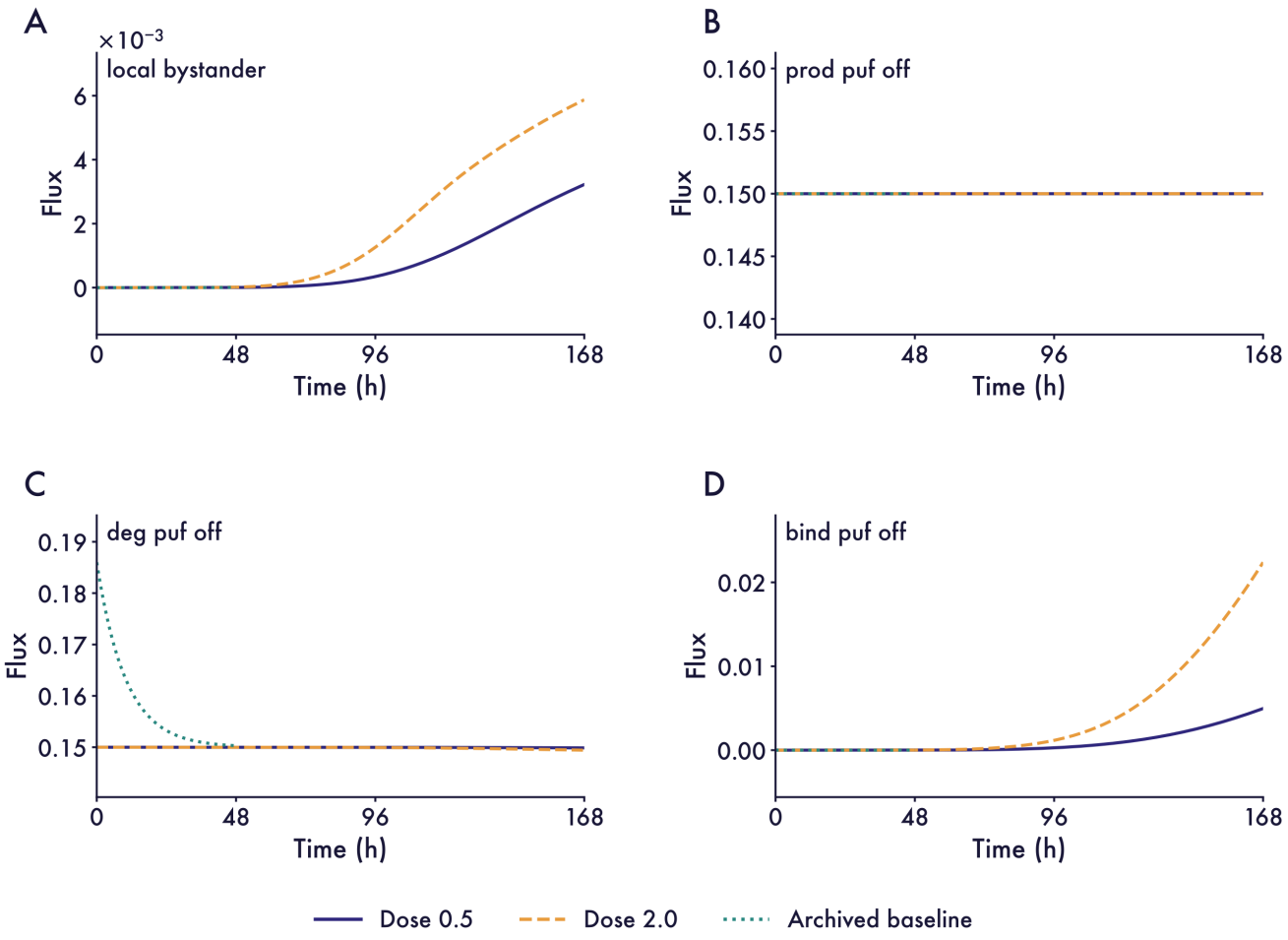


Figure 36. Bystander and mismatch fluxes. A: local bystander editing. B–C: production and degradation of mismatched-PUF RNA substrates. D: editor binding to that off-target pool. These processes distinguish local editing near the intended target from reactions after mismatched PUF recognition. Compare them with their accumulated burdens in Figure 30.

Delivery Figure 37

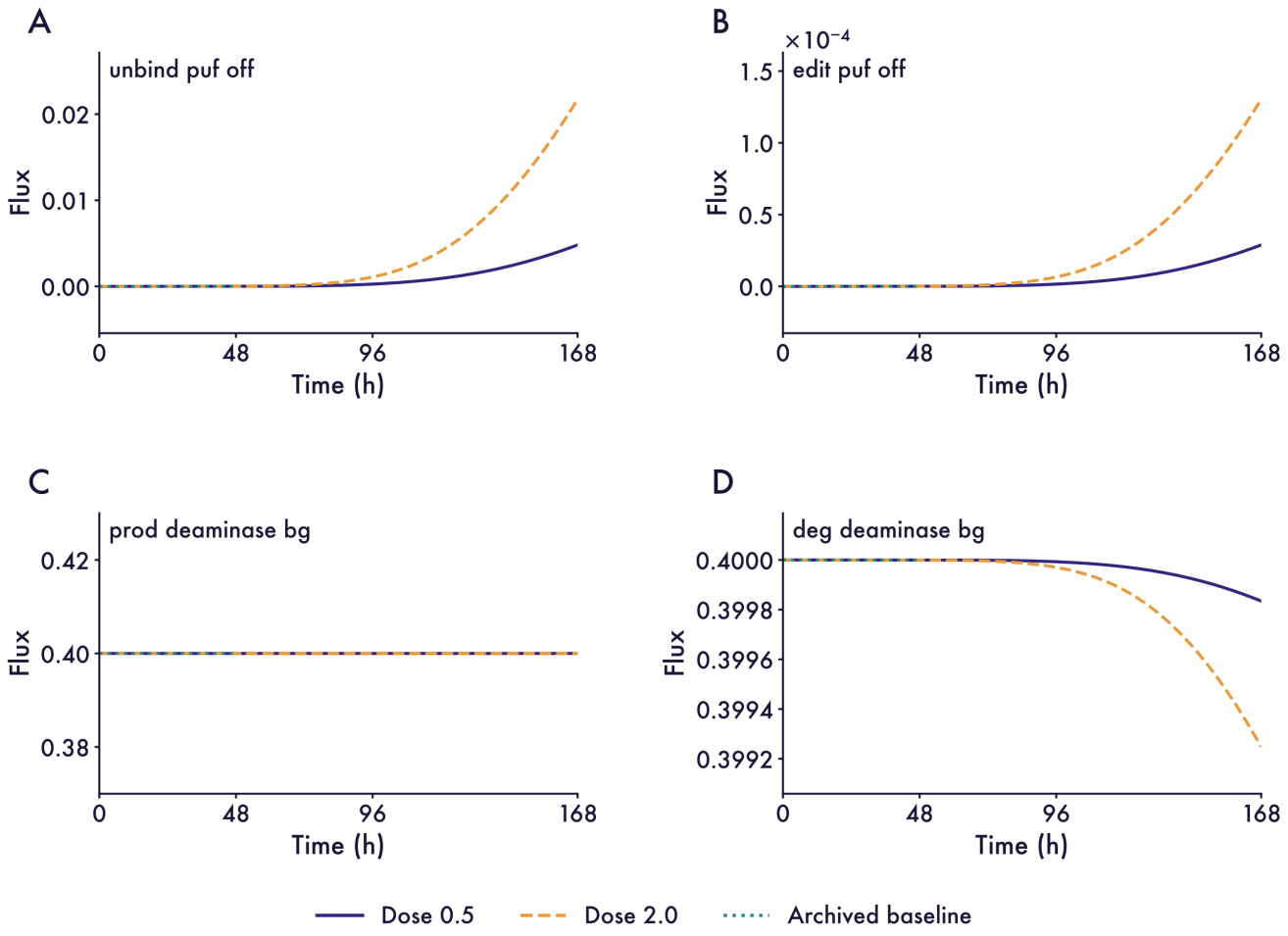


Figure 37. Mismatch and background fluxes. A–B: dissociation and editing of mismatched-PUF substrates. C–D: production and degradation of deaminase-background RNA substrates. Dissociation competes with catalytic conversion in the mismatch branch. The background-substrate production and loss terms support a separate, deaminase-associated source of unwanted edits.

Delivery Figure 38

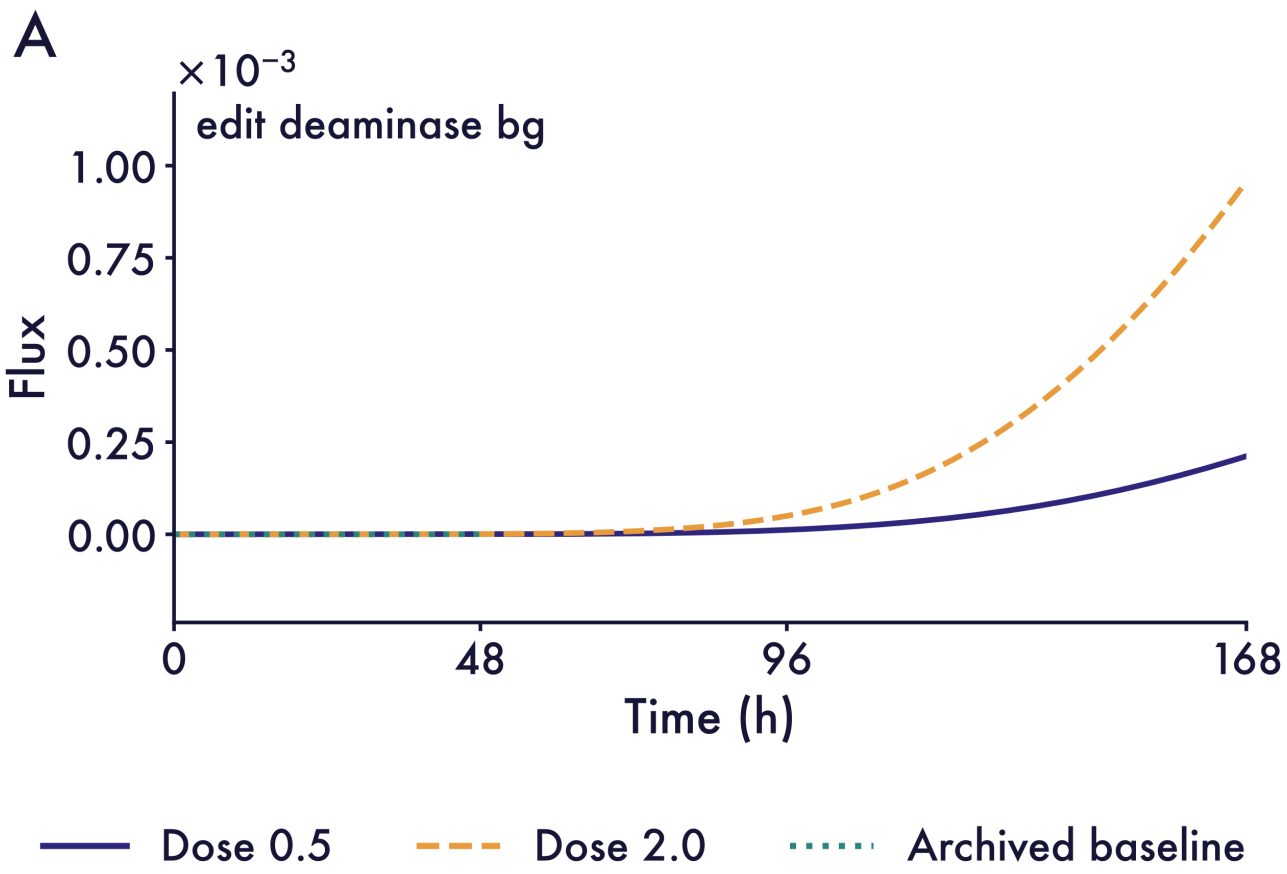


Figure 38. Background editing flux. Editing flux through the deaminase-background pathway (edit_deaminase_bg), tracked separately from PUF-associated off-target editing. Its time integral contributes to accumulated background products; the instantaneous rate and accumulated burden therefore describe different aspects of the same pathway.

Delivery Figure 39

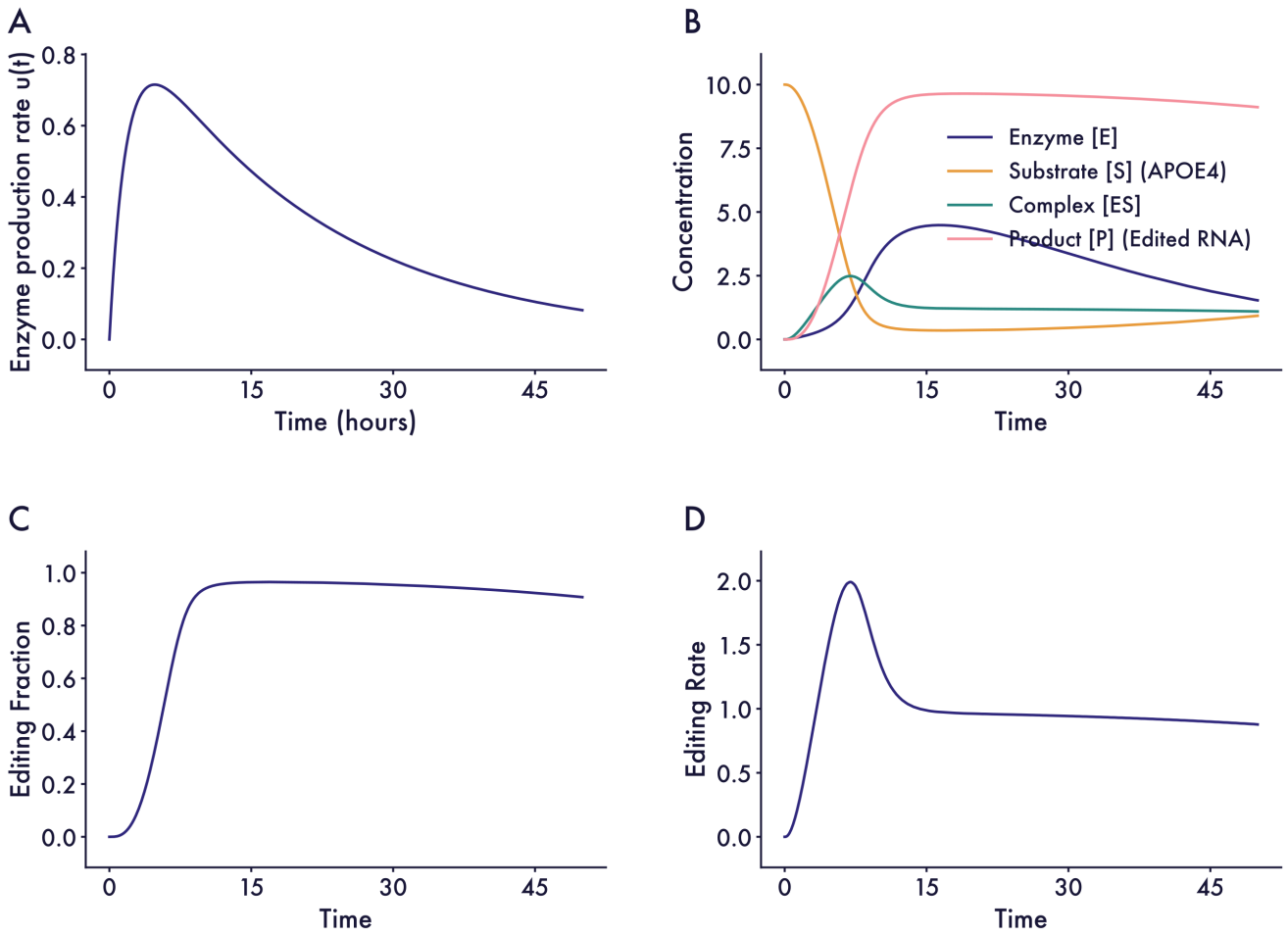


Figure 39. Earlier model trajectories. Four distinct trajectories from the original single-pool notebook model. A shows enzyme production; B follows enzyme, substrate, complex and product concentrations; C shows editing fraction; D shows editing rate. Comparing C and D distinguishes an accumulated fraction from the speed at which editing occurs.

Delivery Figure 40

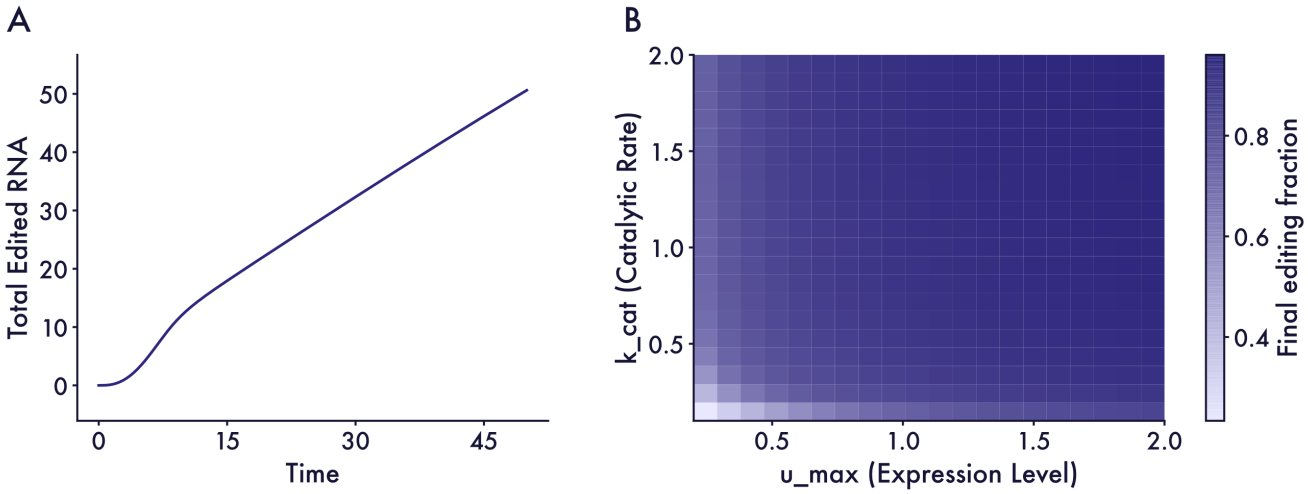


Figure 40. Earlier model accumulation and parameter scan. A, accumulated editing. B, the original 20 × 20 parameter grid. These outputs belong to the earlier model. In B, columns vary maximum expression (u_max), rows vary catalytic rate (k_cat), and color encodes the final editing fraction. The grid separates two parameter effects within the simplified model; it does not include the later multisite product trade-off.

Delivery Table 1

State vector

State group	Variables	Interpretation
Systemic delivery · 7	A_dep, A_lymph, A_blood, A_liver, A_peripheral, A_brain_blood, A_cleared	Vector amount from local depot through circulation and clearance.
Brain transport · 5	A_brain_EC, A_brain_endo, A_brain_ISF, A_brain_cell, A_brain_nuc	Successive barriers from endothelium to the nucleus.
Expression · 2	mRNA_brain, P_brain	Editor transcript and protein abundance.
Binding intermediates · 6	S_on, S_off, ES_on, ES_off, E_on, E_off	Generic target and off-target binding bookkeeping.
APOE editing · 8	S_APOE4, S_APOE3_like, S_APOE2_like, S_APOE158_only, C_APOE112, C_APOE158, C_APOE158_after112, C_APOE112_after158	Codon-resolved substrate, complexes and products.
Off-target editing · 6	B_local_bystander, S_puf_off, C_puf_off, E_puf_off, S_deaminase_bg, E_deaminase_bg	Local bystander, PUF-mismatch and deaminase-background components.

Variables and interpretation reproduced from the dedicated delivery page.

Delivery Table 2

Parameter provenance

Parameter	Nominal	Explored range	Confidence	Source status
k_blood_to_brain	0.01	0.001–0.05	Medium	Literature-informed effective rate
k_brainblood_to_EC	0.02	0.002–0.08	Low	Model prior; experimental priority
k_endo_to_ISF	0.01	0.001–0.04	Low	Model prior; experimental priority
k_tx / k_tl	0.4 / 0.9	0.05–1.20 / 0.10–2.50	Medium	Effective expression rates
k_on_112 / k_cat_112	1.0 / 1.0	0.10–10 / 0.03–3	Low	Normalized editing priors
k_on_puf_off / k_cat_puf_off	0.015 / 0.006	0.001–0.5 / 0.0001–0.2	Low	Normalized off-target priors

Normalized effective-rate convention. Ranges express model uncertainty, not confidence intervals. Consult the source parameter record before reuse.

Delivery Table 3

Sensitivity analysis

Rank	Parameter	Analysis score	Measurement implication
01	k_brainblood_to_EC	2.720	Quantify endothelial uptake.
02	k_blood_to_brain	2.621	Measure brain exposure against blood.
03	k_cat_158	2.323	Resolve codon-158 editing kinetics.
04	k_cat_puf_off	2.310	Measure mismatch-site editing.
05	k_cat_112	2.135	Fit desired-edit kinetics.
08	k_endo_to_ISF	2.024	Quantify trans-endothelial release.

Archived analysis scores within the source model and explored parameter ranges; not universal biological effect sizes.